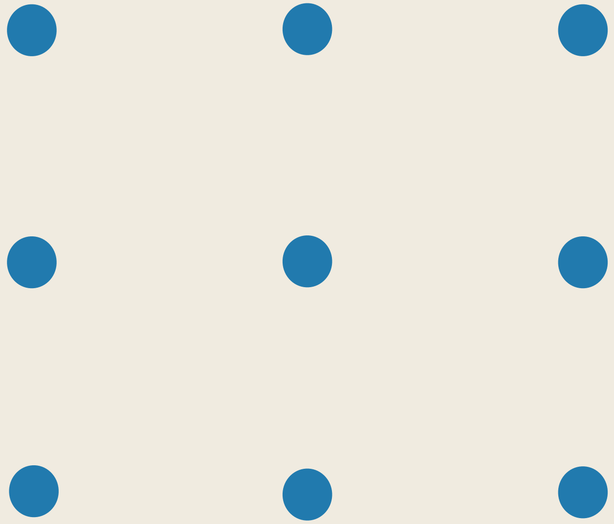
Symmetry and Pseudorandomness in Computation

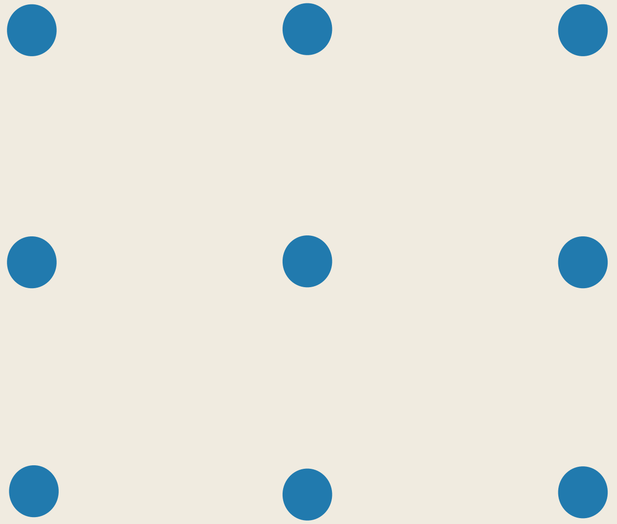
Tushant Mittal
Postdoctoral Researcher, Stanford University

Example 1: Creating Roads



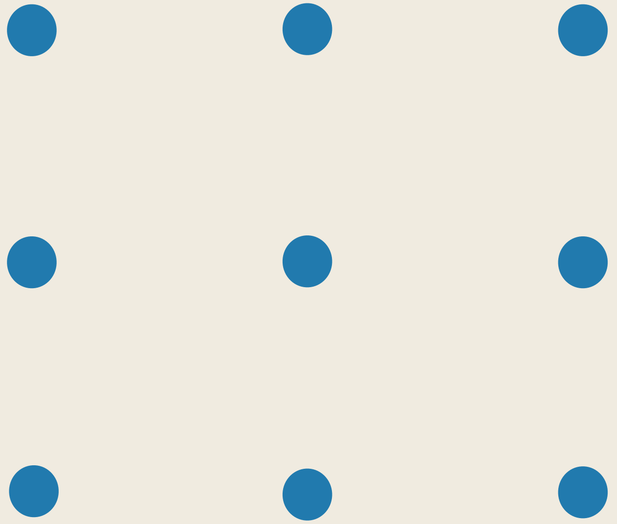
Example 1: Creating Roads

Budget: 4 connections each



Example 1: Creating Roads

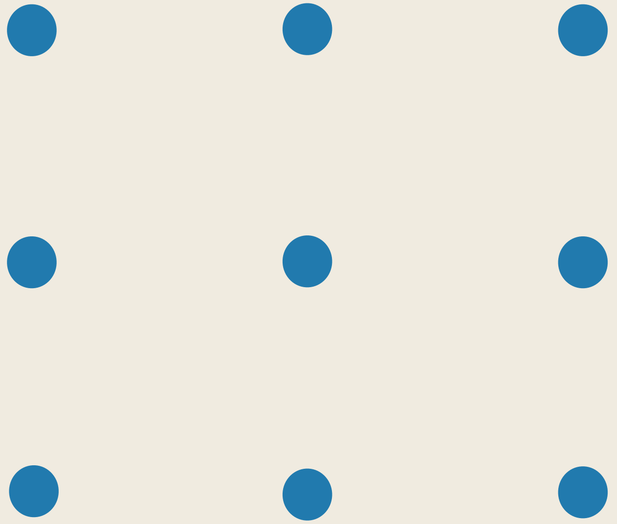
Budget: 4 connections each



Example 1: Creating Roads

Budget: 4 connections each

Objective 1: Max Connectivity

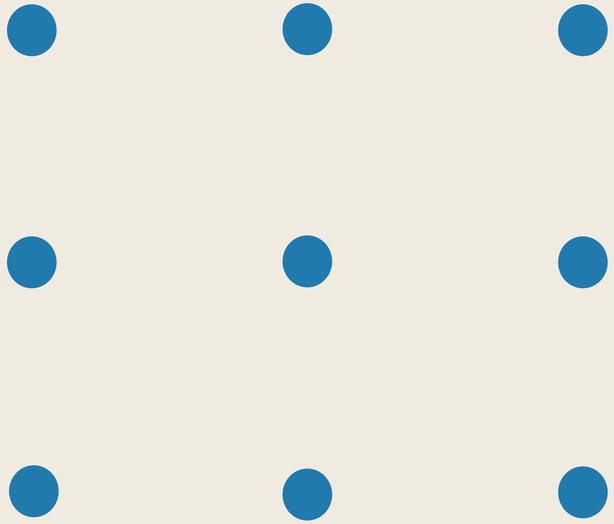


Example 1: Creating Roads

Budget: 4 connections each

Objective 1: Max Connectivity

Objective 2: Easy Navigation



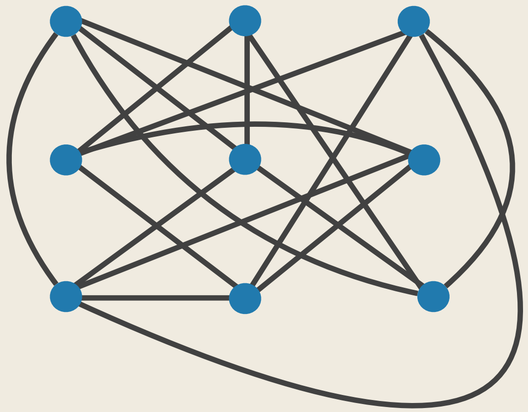
Random vs Symmetric

Goal 1: Max Connectivity

Goal 2: Easy Navigation

Random vs Symmetric

Random Connections

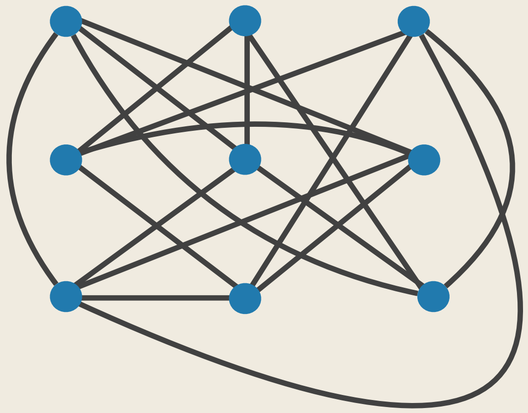


Goal 1: Max Connectivity

Goal 2: Easy Navigation

Random vs Symmetric

Random Connections



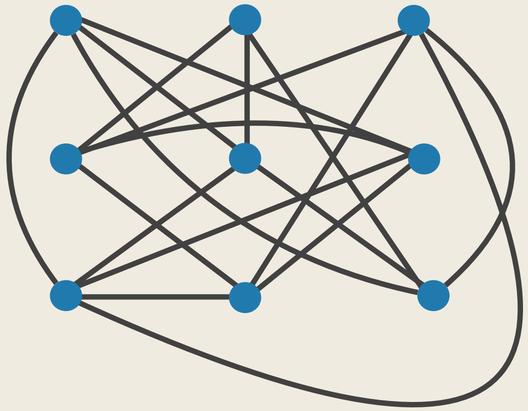
Max Connectivity for the Budget

Goal 1: Max Connectivity

Goal 2: Easy Navigation

Random vs Symmetric

Random Connections



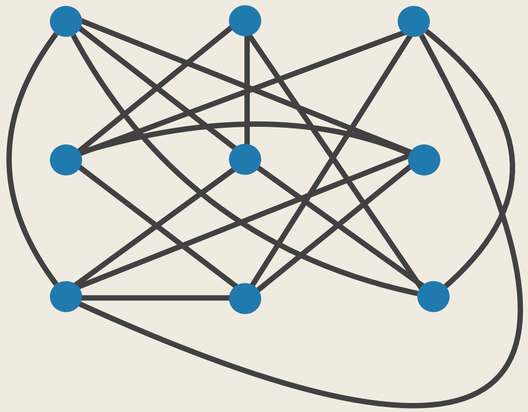
Max Connectivity for the Budget

Goal 1: Max Connectivity

Goal 2: Easy Navigation

Random vs Symmetric

Random Connections

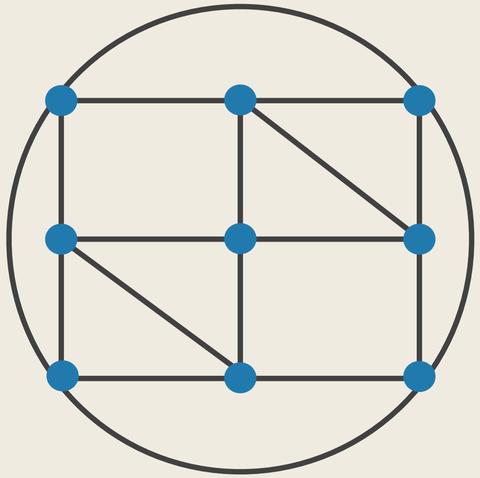


Max Connectivity for the Budget

Goal 1: Max Connectivity

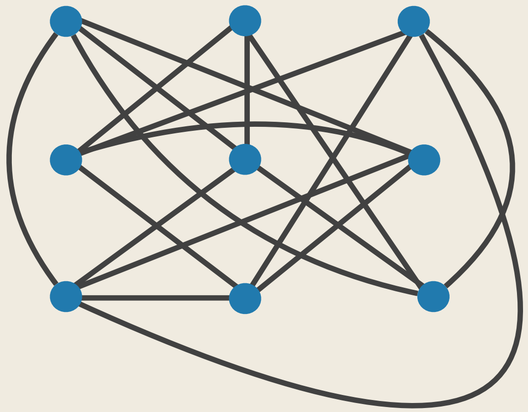
Goal 2: Easy Navigation

Symmetric Connections



Random vs Symmetric

Random Connections

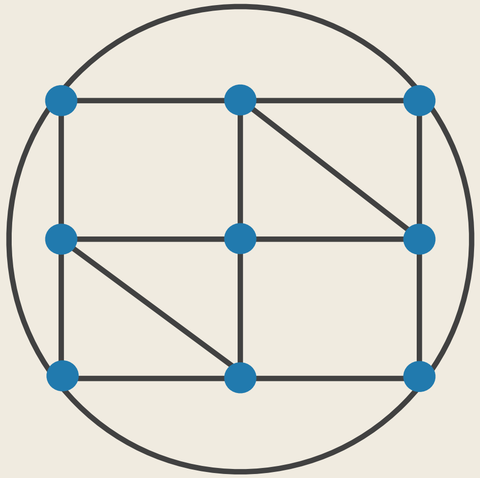


Max Connectivity for the Budget

Goal 1: Max Connectivity

Goal 2: Easy Navigation

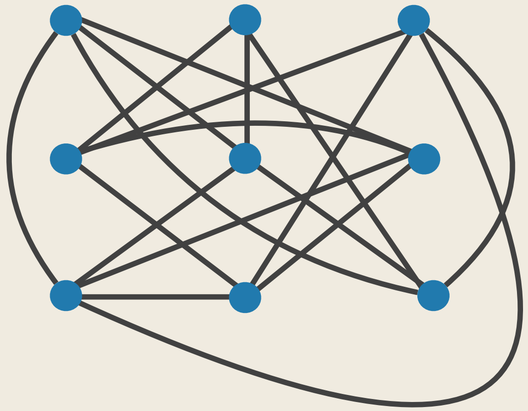
Symmetric Connections



Easy Navigation

Random vs Symmetric

Random Connections

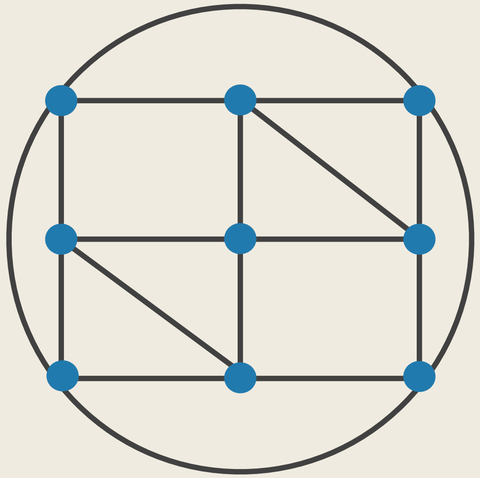


Max Connectivity for the Budget

Goal 1: Max Connectivity

Goal 2: Easy Navigation

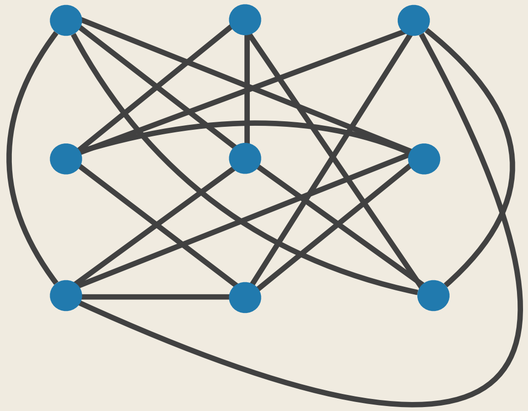
Symmetric Connections



Easy Navigation

Random vs Symmetric

Random Connections



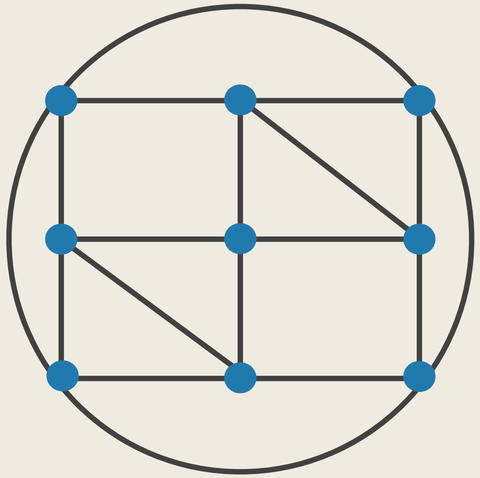
Max Connectivity for the Budget

Goal 1: Max Connectivity

Goal 2: Easy Navigation

BEST OF BOTH?

Symmetric Connections



Easy Navigation

Example 1: Creating Roads

Hyderabad, India

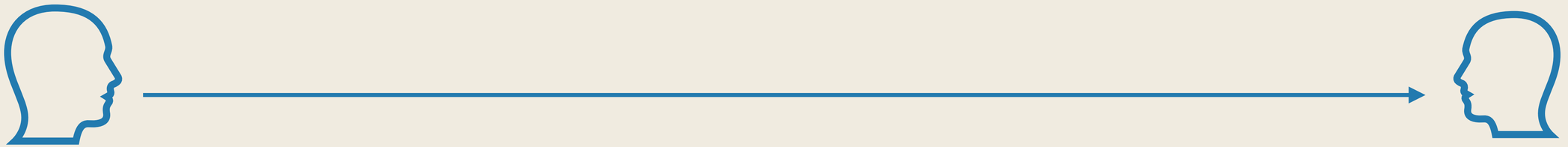
Goal 1: Max Connectivity

Goal 2: Easy Navigation

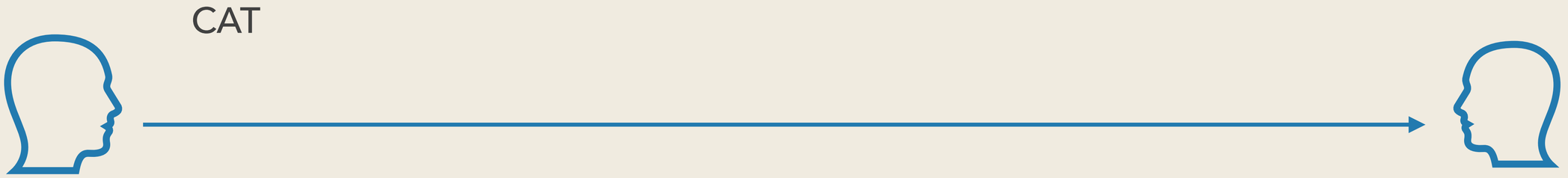
BEST OF BOTH?

Chicago, USA

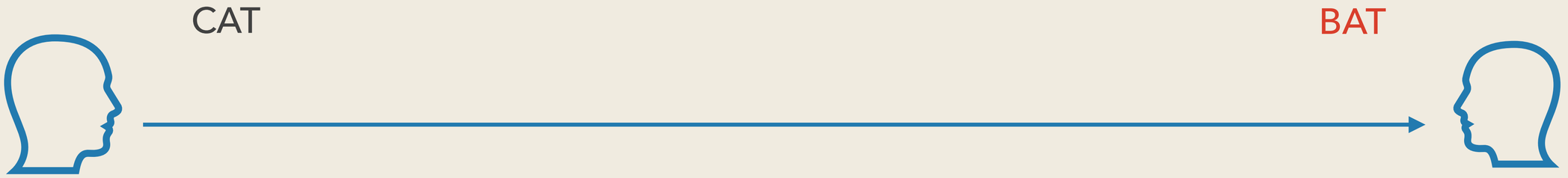
Example 2: Noisy Communication



Example 2: Noisy Communication



Example 2: Noisy Communication



Example 2: Noisy Communication

Example 2: Noisy Communication

Example 2: Noisy Communication

Make Message Robust

Budget: 9-letter words

Random vs Symmetric

Budget: 9-letter words

Goal 1: Max Robustness

Goal 2: Easy Decoding

Random vs Symmetric

Random Encodings

Budget: 9-letter words

Goal 1: Max Robustness

Goal 2: Easy Decoding

Random vs Symmetric

Random Encodings

CAT → SXAKRDOPR

BAT → KRMFLERWP

Budget: 9-letter words

Goal 1: Max Robustness

Goal 2: Easy Decoding

Random vs Symmetric

Random Encodings

Budget: 9-letter words

CAT → SXAKRDOPR

BAT → KRMFLERWP

Goal 1: Max Robustness

Goal 2: Easy Decoding

Max Robustness for the Length

Random vs Symmetric

Random Encodings

Budget: 9-letter words

CAT → SXAKRDOPR

BAT → KRMFLERWP

Goal 1: Max Robustness

Goal 2: Easy Decoding

Max Robustness for the Length

Random vs Symmetric

Random Encodings

Systematic Encodings

Budget: 9-letter words

CAT → SXAKRDOPR

BAT → KRMFLERWP

Goal 1: Max Robustness

Goal 2: Easy Decoding

Max Robustness for the Length

Random vs Symmetric

Random Encodings

CAT → SXAKRDOPR

BAT → KRMFLERWP

Budget: 9-letter words

Goal 1: Max Robustness

Goal 2: Easy Decoding

Systematic Encodings

CAT → CATCATCAT

BAT → BATBATBAT

Max Robustness for the Length

Random vs Symmetric

Random Encodings

CAT → SXAKRDOPR

BAT → KRMFLERWP

Budget: 9-letter words

Goal 1: Max Robustness

Goal 2: Easy Decoding

Systematic Encodings

CAT → CATCATCAT

BAT → BATBATBAT

Max Robustness for the Length

Easy Decoding

Random vs Symmetric

Random Encodings

Budget: 9-letter words

Systematic Encodings

CAT → SXAKRDOPR

BAT → KRMFLERWP

CAT → CATCATCAT

BAT → BATBATBAT

Max Robustness for the Length

Easy Decoding

Random vs Symmetric

Random Encodings

Budget: 9-letter words

Systematic Encodings

CAT → SXAKRDOPR

BAT → KRMFLERWP

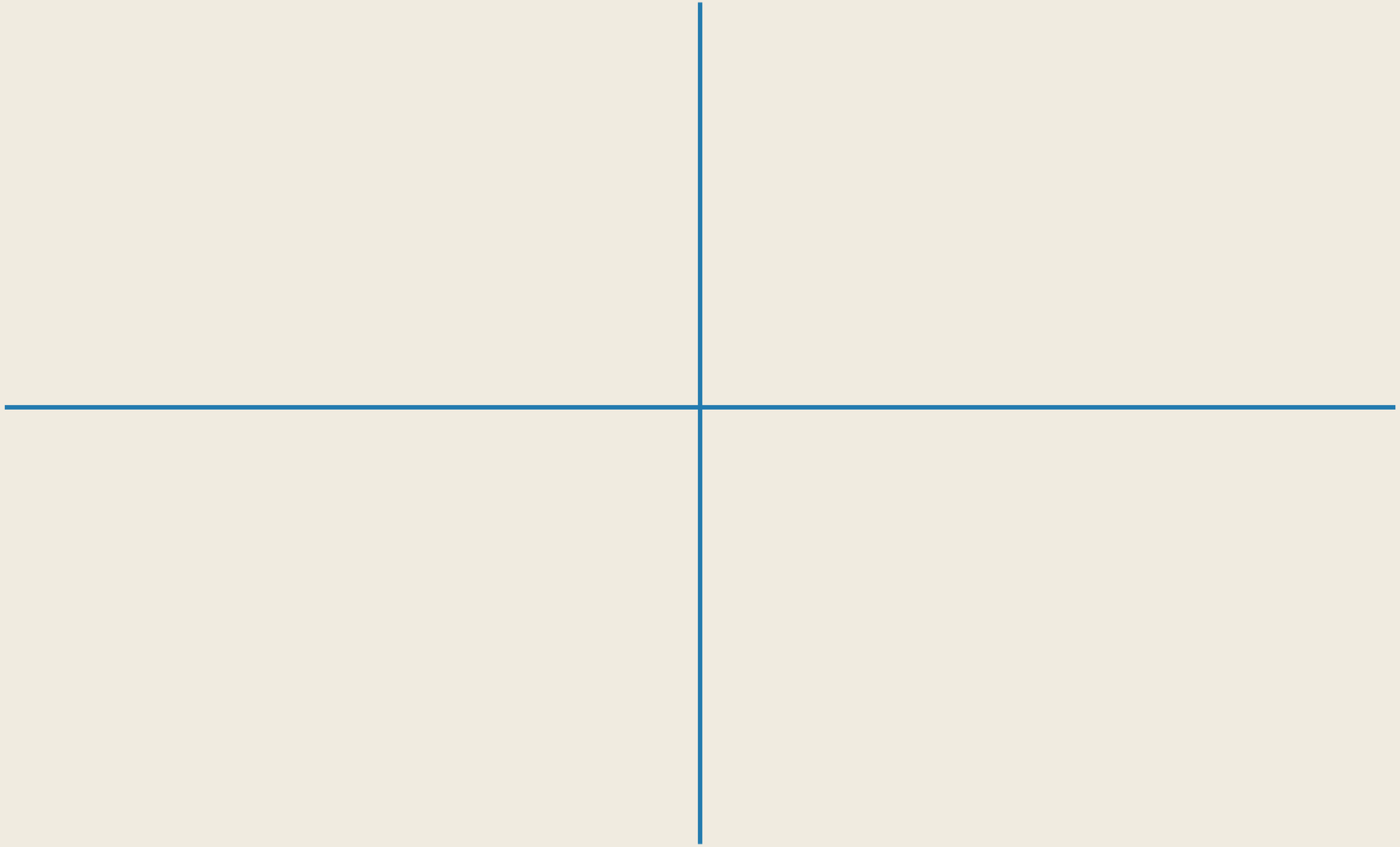
CAT → CATCATCAT

BAT → BATBATBAT

BEST OF BOTH?

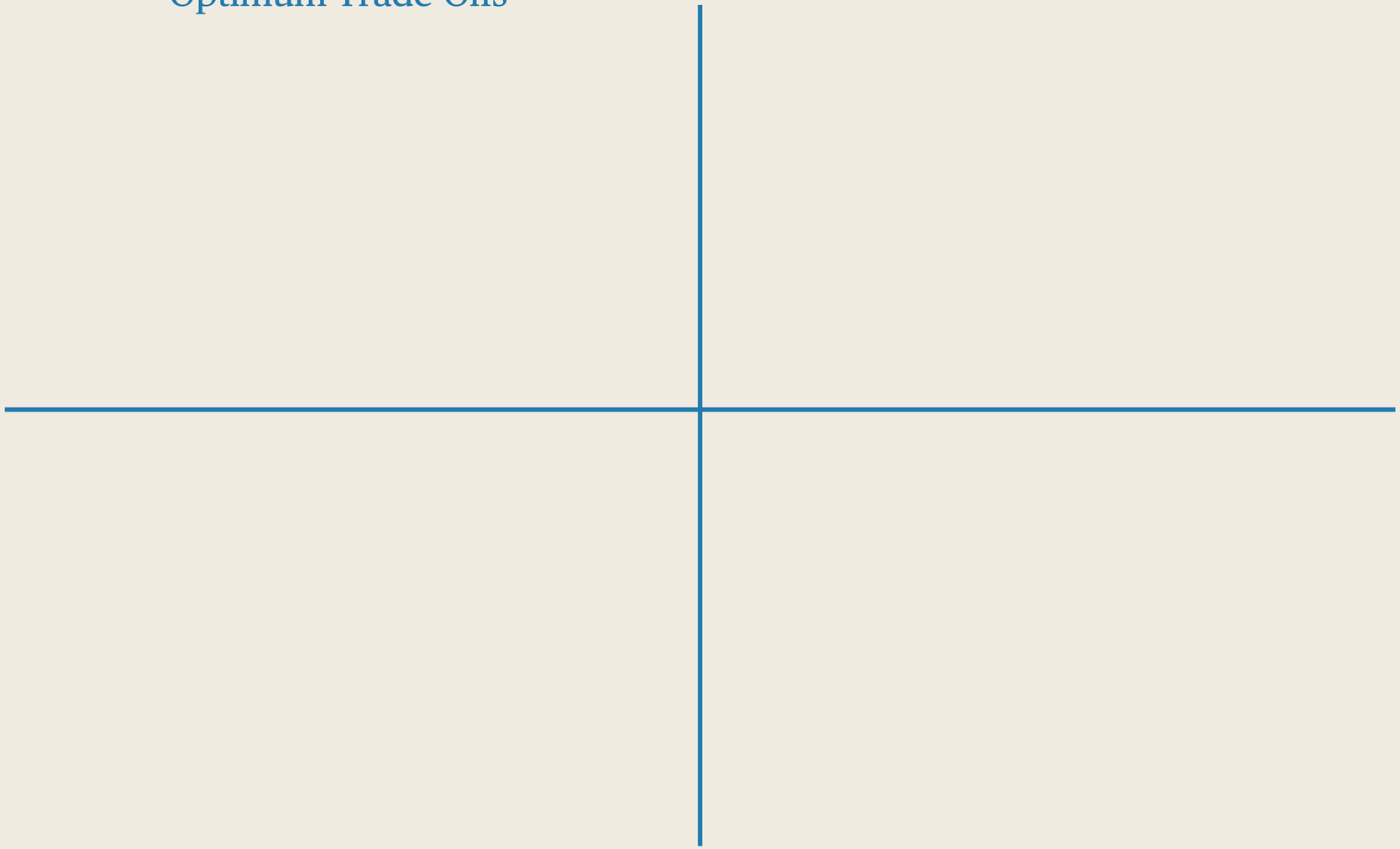
Max Robustness for the Length

Easy Decoding



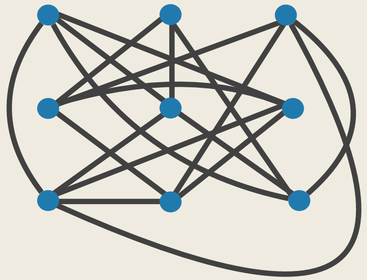
Randomness

Optimum Trade-Offs



Randomness

Optimum Trade-Offs



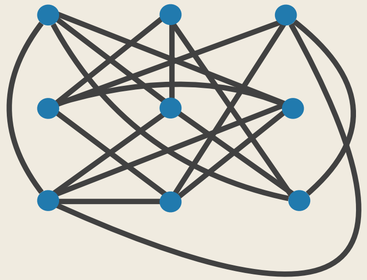
Sparsity vs Connectivity

Networks

Randomness

Networks

Optimum Trade-Offs



Sparsity vs Connectivity

Fault-Tolerance

CAT → SXAKRDOPR

BAT → KRMFLERWP

Information vs Error Tolerance

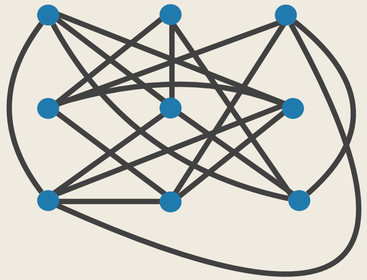
Randomness

Optimum Trade-Offs

Symmetry

Efficiency

Networks



Sparsity vs Connectivity

Fault-Tolerance

CAT → SXAKRDOPR

BAT → KRMFLERWP

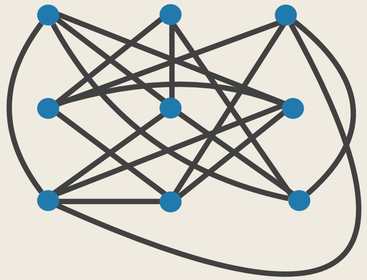
Information vs Error Tolerance

Randomness

Symmetry

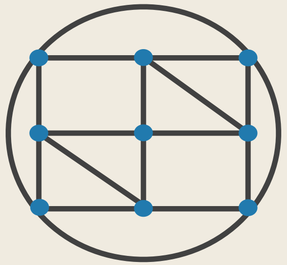
Networks

Optimum Trade-Offs



Sparsity vs Connectivity

Efficiency



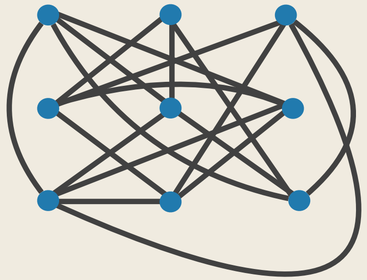
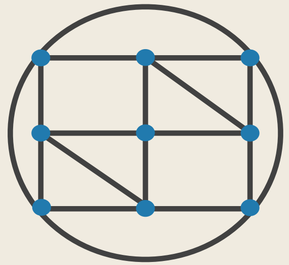
List of Neighbors, Random Walks

Fault-Tolerance

CAT → SXAKRDOPR

BAT → KRMFLERWP

Information vs Error Tolerance

	Randomness	Symmetry
	Optimum Trade-Offs	Efficiency
Networks		
	Sparsity vs Connectivity	List of Neighbors, Random Walks
Fault-Tolerance	CAT → SXAKRDOPR BAT → KRMFLERWP	CAT → CATCATCAT BAT → BATBATBAT
	Information vs Error Tolerance	Encoding, Checking, Storage

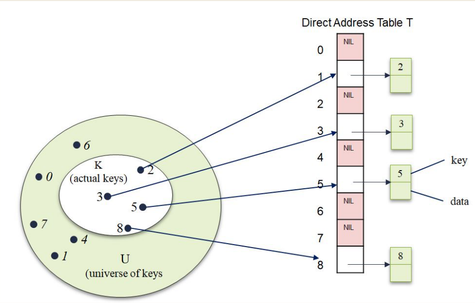
This is a general phenomenon

This is a general phenomenon

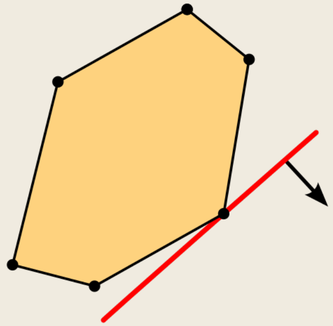
RANDOM CHOICES PERFORM WELL

This is a general phenomenon

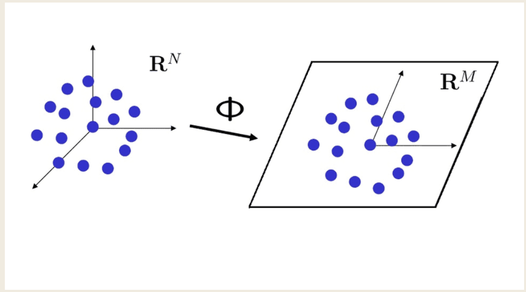
RANDOM CHOICES PERFORM WELL



Universal Hashing



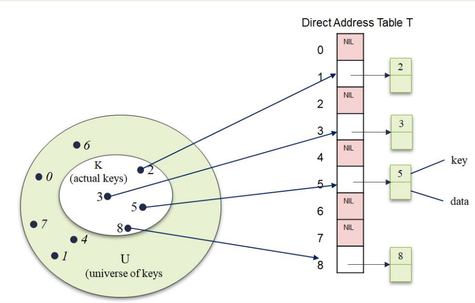
Linear Programming



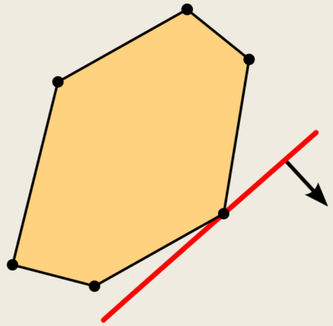
Dimensionality Reduction

This is a general phenomenon

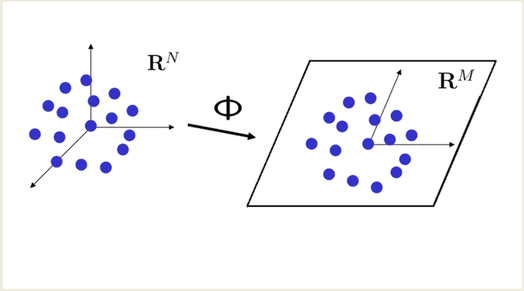
RANDOM CHOICES PERFORM WELL



Universal Hashing



Linear Programming

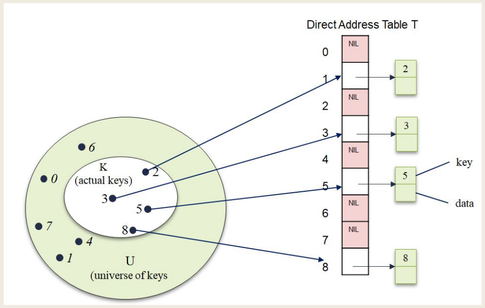


Dimensionality Reduction

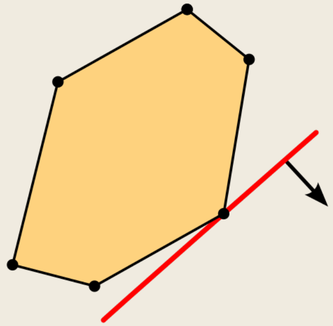
SYMMETRY HELPS WITH EFFICIENCY

This is a general phenomenon

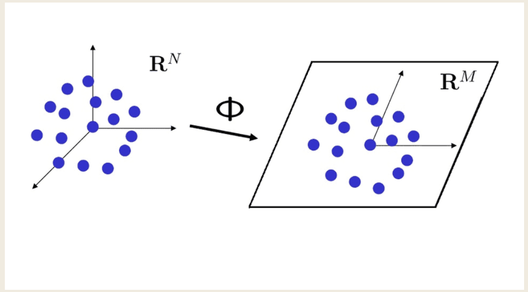
RANDOM CHOICES PERFORM WELL



Universal Hashing

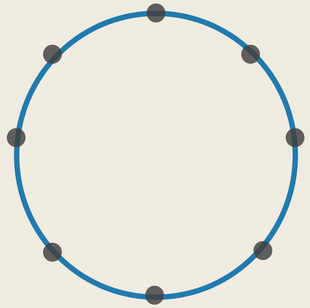


Linear Programming

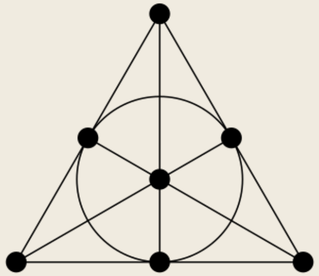


Dimensionality Reduction

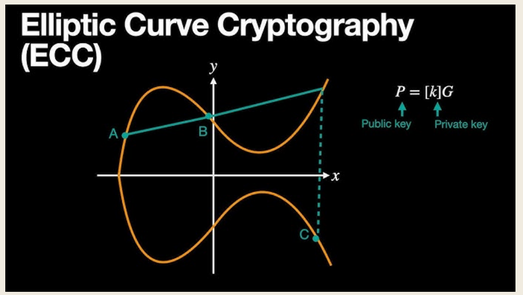
SYMMETRY HELPS WITH EFFICIENCY



Low Overhead



Exact Solutions



Useful Operations

They are at odds

RANDOMNESS

SYMMETRY

They are at odds

RANDOMNESS

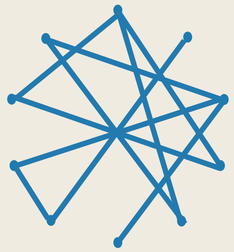
Good Robustness

SYMMETRY

They are at odds

RANDOMNESS

Good Robustness

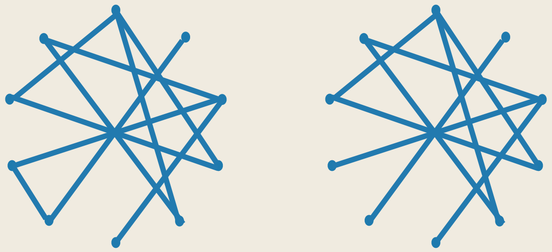


SYMMETRY

They are at odds

RANDOMNESS

Good Robustness

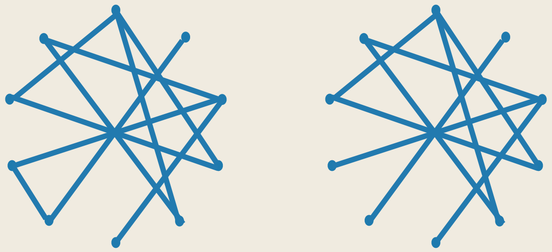


SYMMETRY

They are at odds

RANDOMNESS

Good Robustness



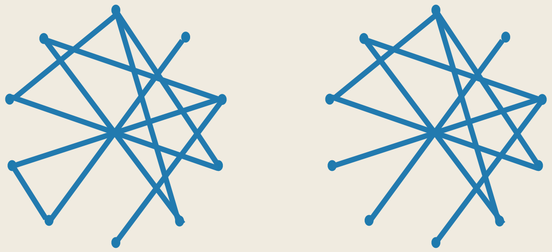
SYMMETRY

Fragile

They are at odds

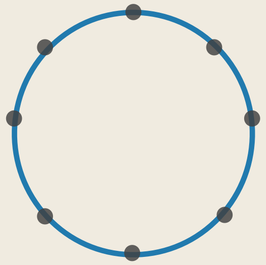
RANDOMNESS

Good Robustness



SYMMETRY

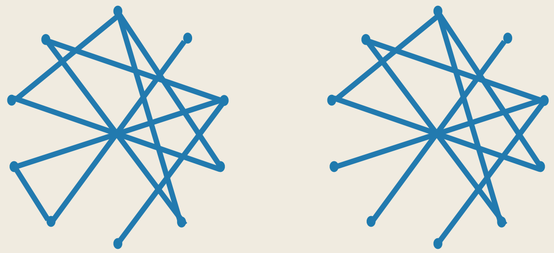
Fragile



They are at odds

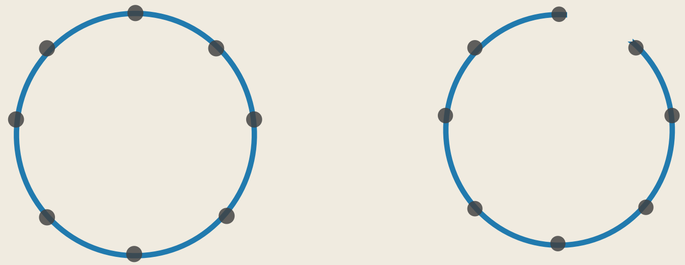
RANDOMNESS

Good Robustness



SYMMETRY

Fragile

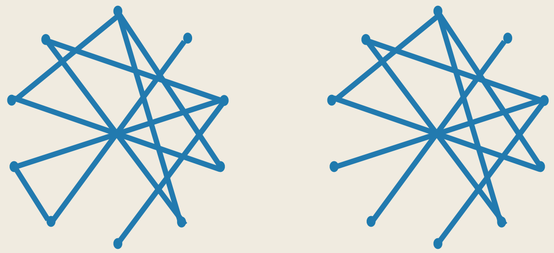


They are at odds

RANDOMNESS

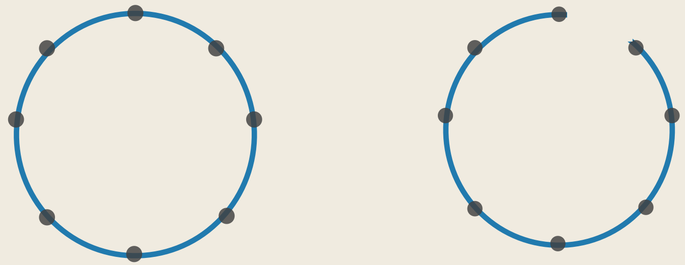
Good Robustness

No Structure



SYMMETRY

Fragile

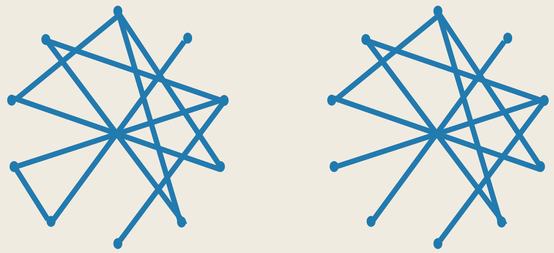


They are at odds

RANDOMNESS

Good Robustness

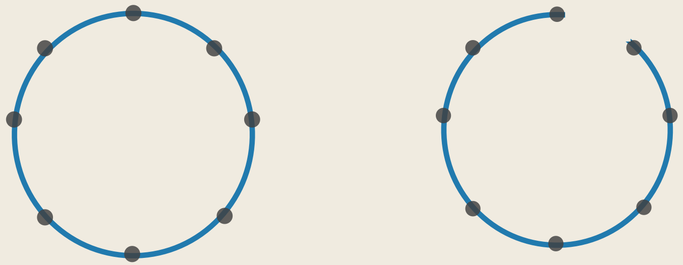
No Structure



SYMMETRY

Fragile

Nice Structure $\Longrightarrow$ algorithms

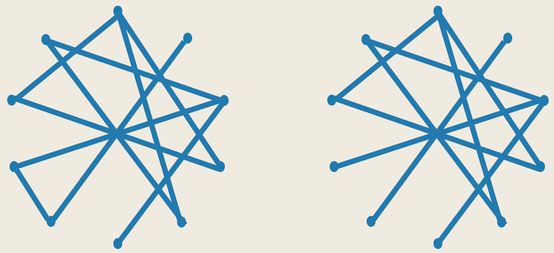


They are at odds

RANDOMNESS

Good Robustness

No Structure

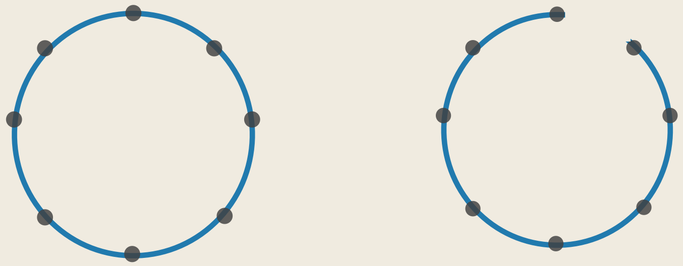


Analytic, Combinatorial

SYMMETRY

Fragile

Nice Structure $\Longrightarrow$ algorithms

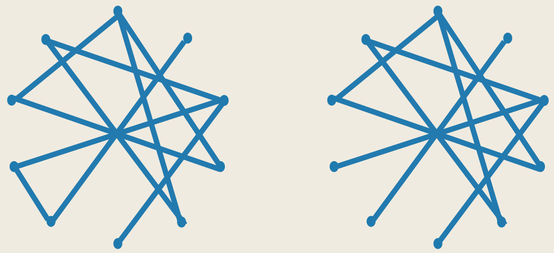


They are at odds

RANDOMNESS

Good Robustness

No Structure

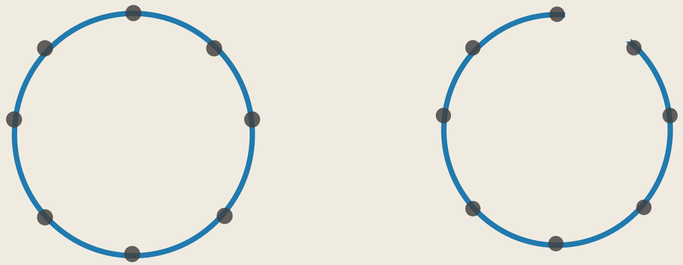


Analytic, Combinatorial

SYMMETRY

Fragile

Nice Structure $\Longrightarrow$ algorithms



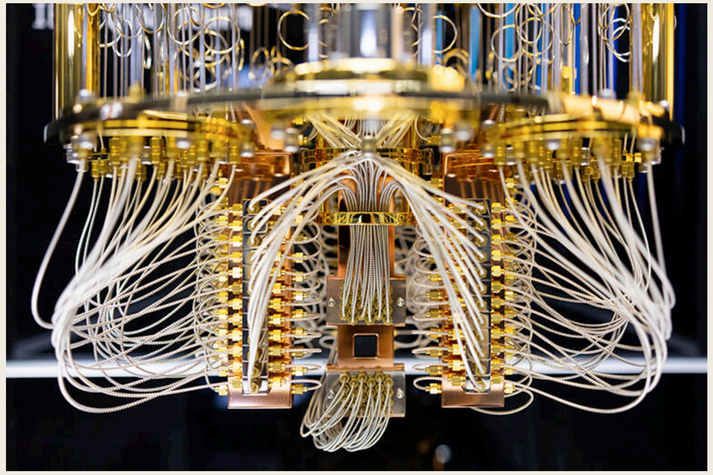
Algebraic

But sometimes we need both

10^{-6}

ERROR RATES

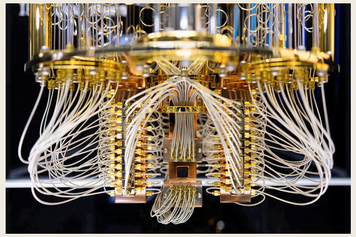
10^{-18}



Quantum Computer

Classical Computer

But sometimes we need both



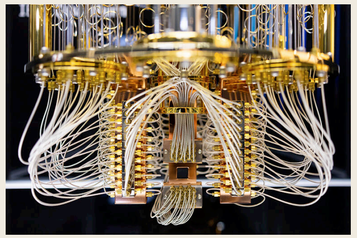
Quantum Computer

10^{-6}

Classical Computer

10^{-18}

But sometimes we need both



Quantum Computer

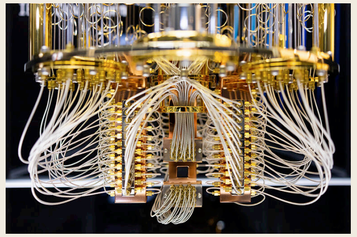
10^{-6}

Classical Computer

10^{-18}

Need Max Robustness

But sometimes we need both



Quantum Computer

10^{-6}

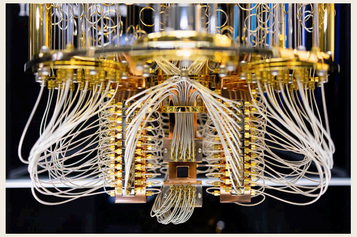
Goal 1: Max Robustness

Classical Computer

10^{-18}

Need Max Robustness

But sometimes we need both



Quantum Computer

10^{-6}

Goal 1: Max Robustness

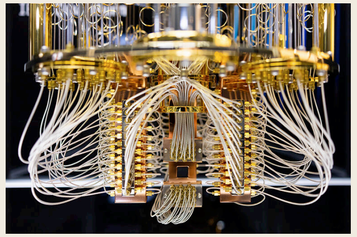
Classical Computer

10^{-18}

Need Max Robustness

Very Efficient Operations

But sometimes we need both



Quantum Computer

10^{-6}

Classical Computer

10^{-18}

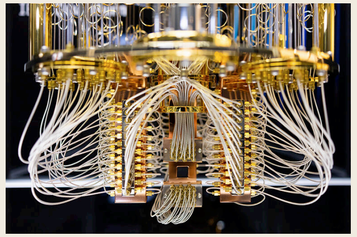
Goal 1: Max Robustness

Goal 2: Efficient Operations

Need Max Robustness

Very Efficient Operations

But sometimes we need both



Quantum Computer

10^{-6}

Goal 1: Max Robustness

Goal 2: Efficient Operations

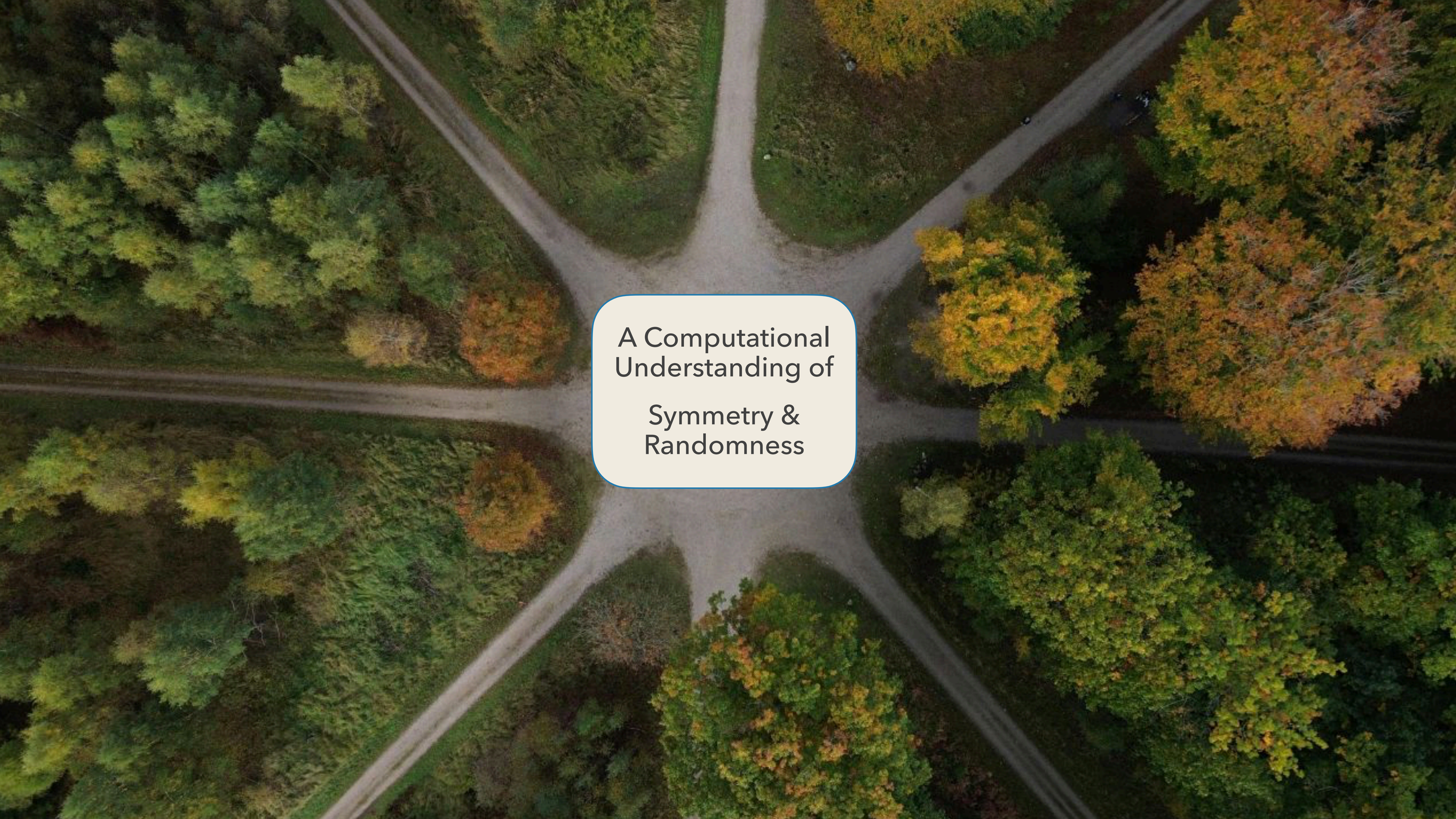
Classical Computer

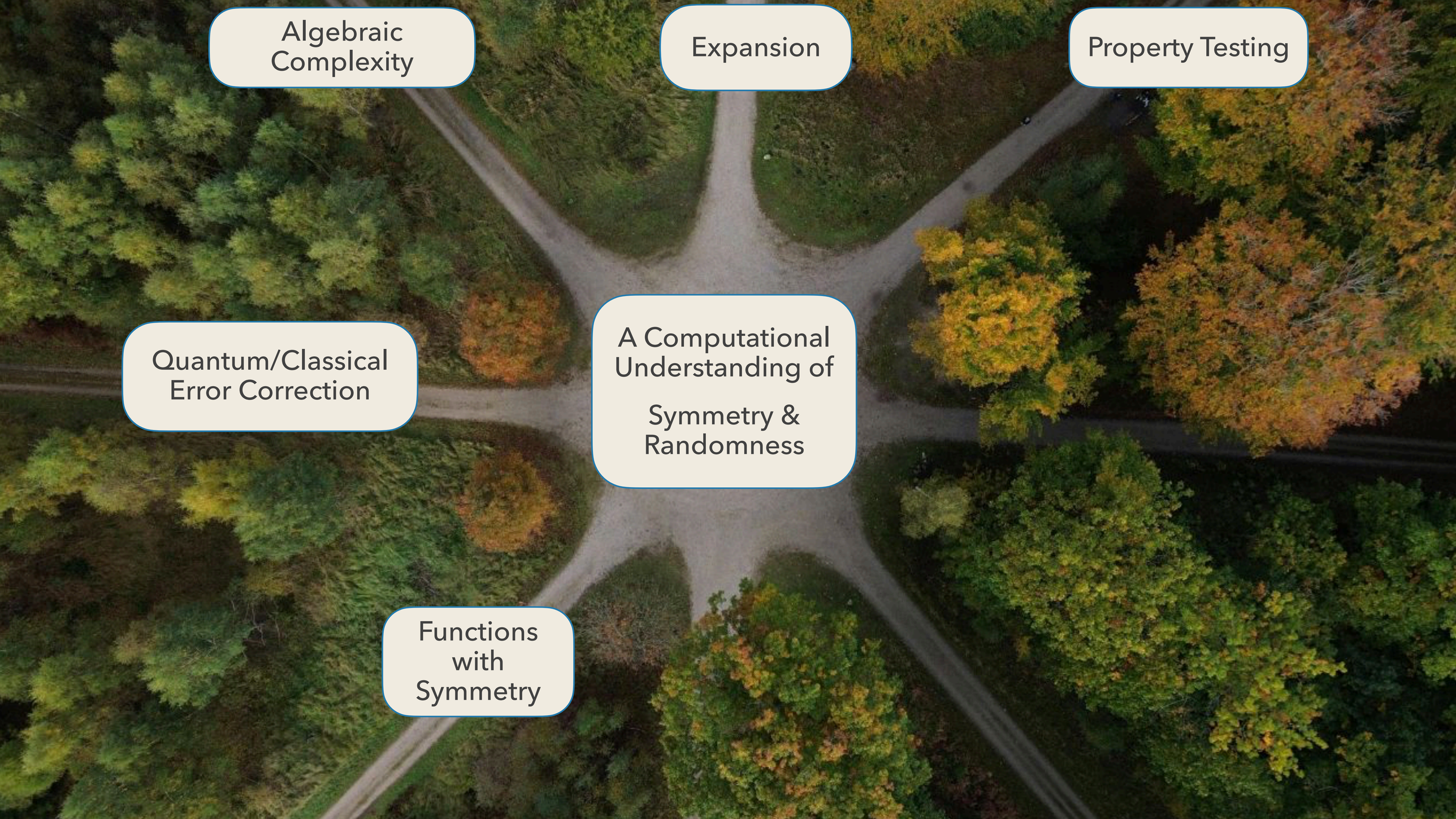
10^{-18}

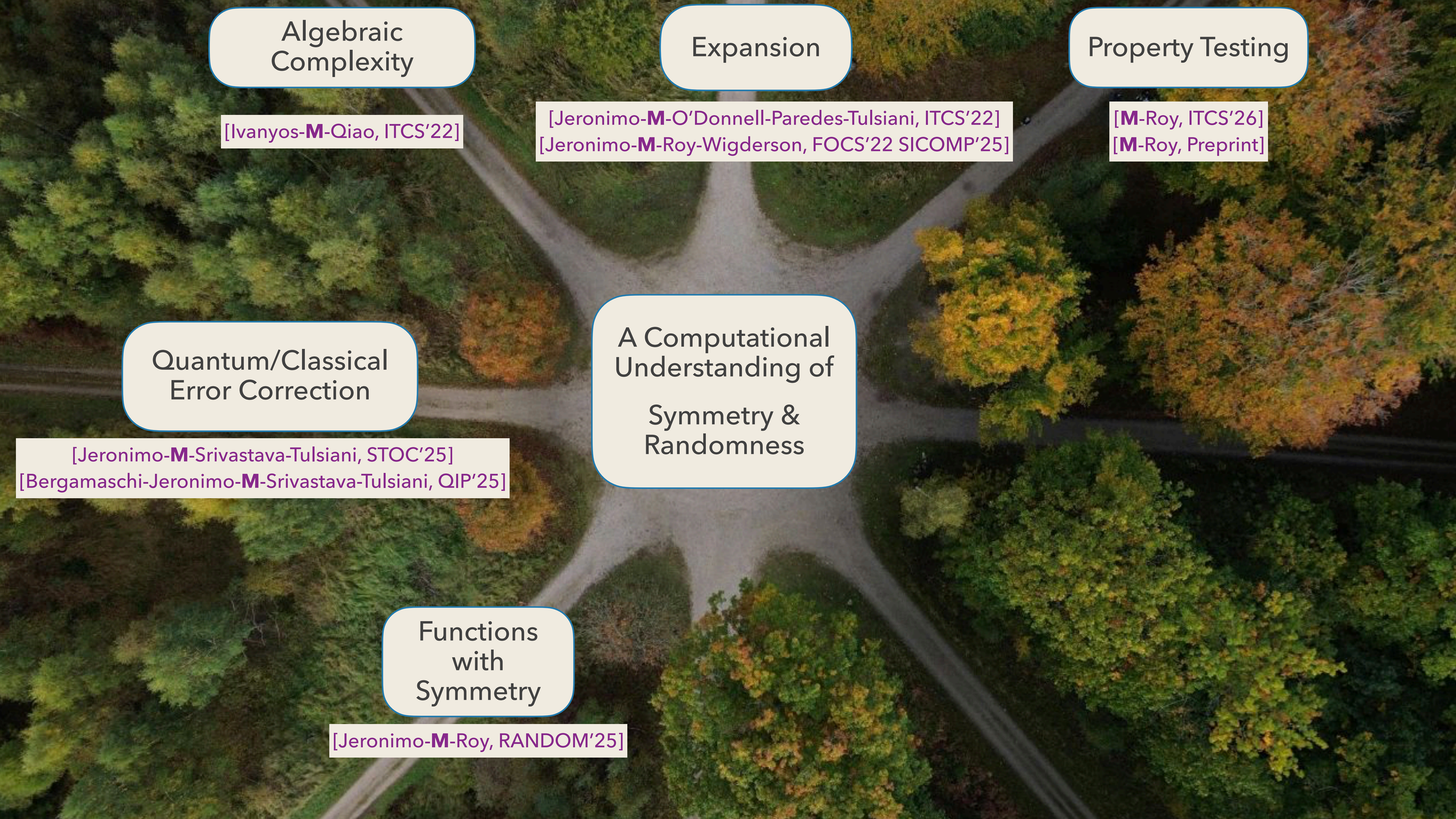
REALLY NEED BOTH!

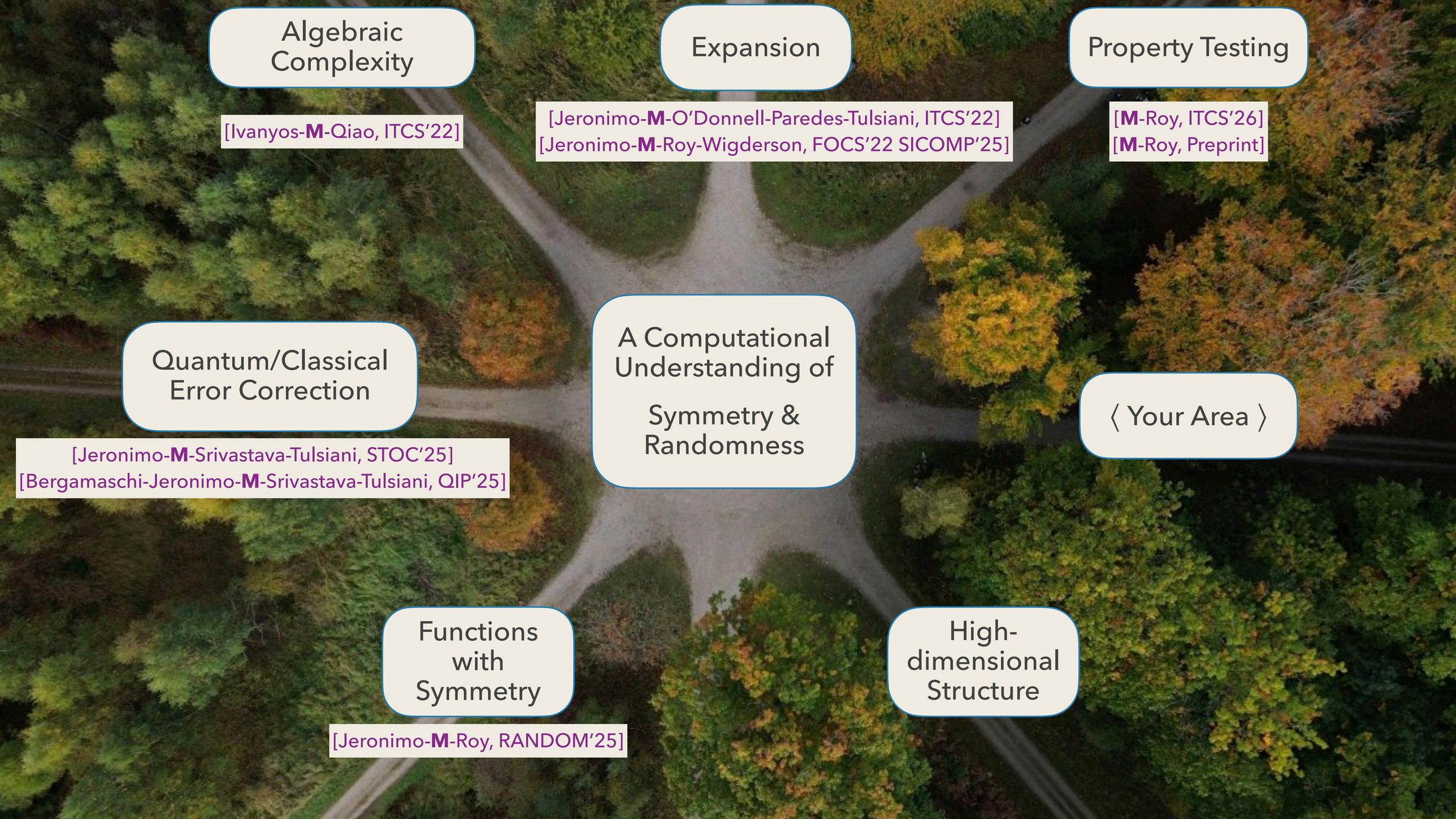
Need Max Robustness

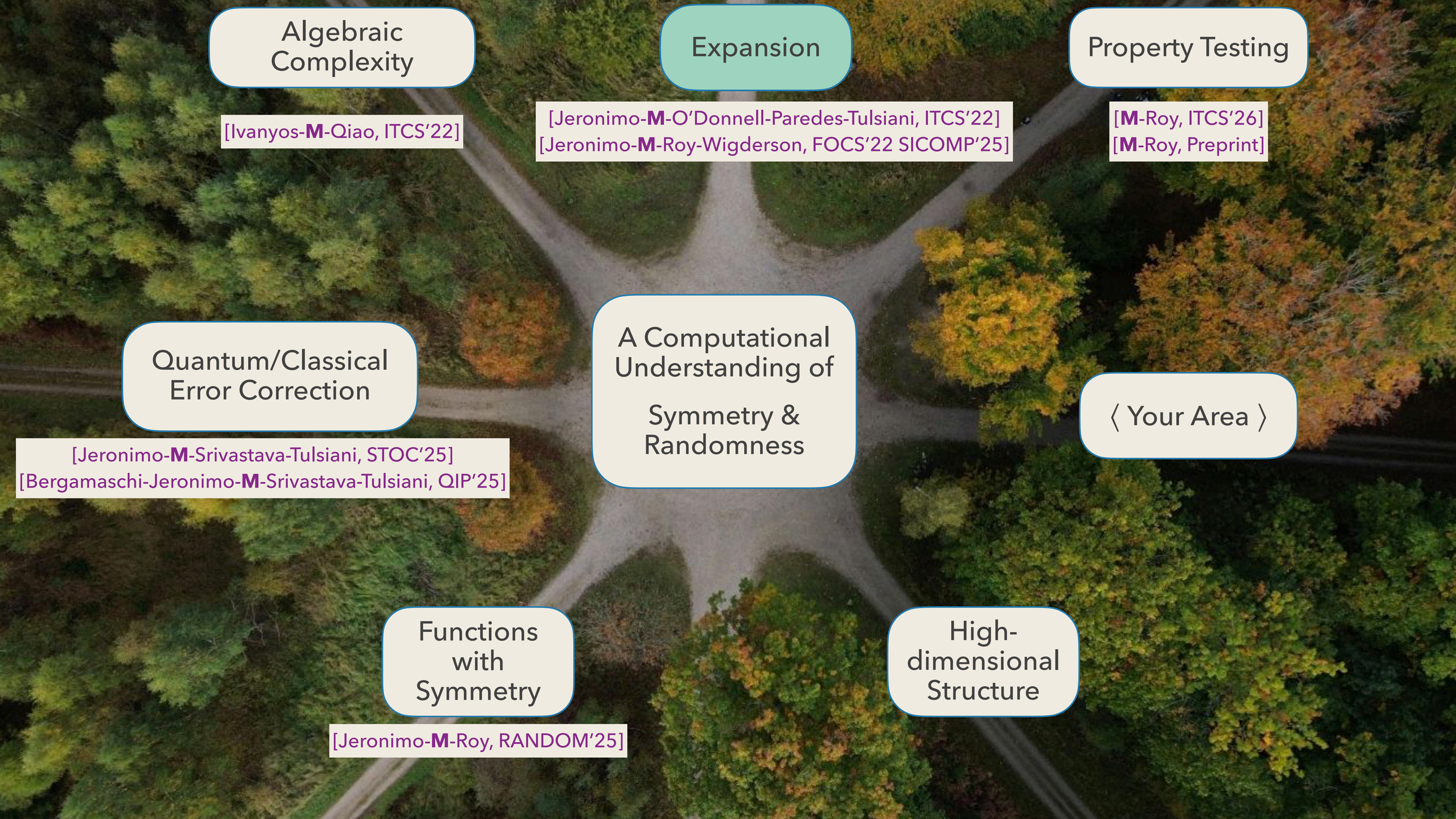
Very Efficient Operations


A Computational
Understanding of
Symmetry &
Randomness

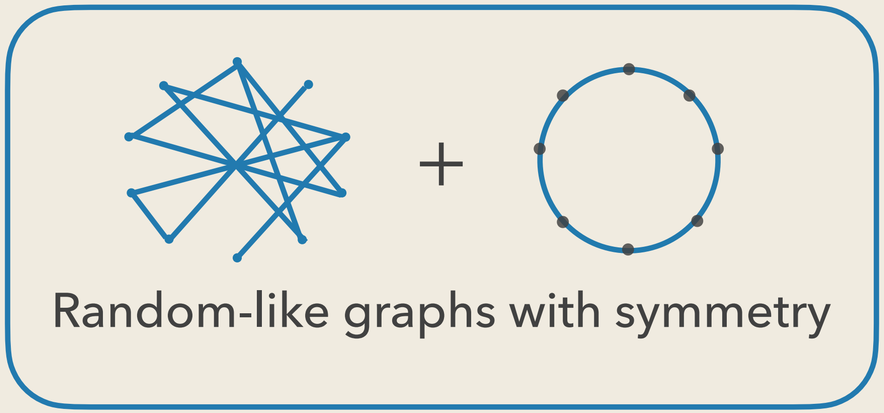

Algebraic Complexity
Expansion
Property Testing
Quantum/Classical Error Correction
A Computational Understanding of
Symmetry & Randomness
Functions with Symmetry


Algebraic Complexity
[Ivanyos-**M**-Qiao, ITCS'22]
Expansion
[Jeronimo-**M**-O'Donnell-Paredes-Tulsiani, ITCS'22]
[Jeronimo-**M**-Roy-Wigderson, FOCS'22 SICOMP'25]
Property Testing
[**M**-Roy, ITCS'26]
[**M**-Roy, Preprint]
Quantum/Classical Error Correction
[Jeronimo-**M**-Srivastava-Tulsiani, STOC'25]
[Bergamaschi-Jeronimo-**M**-Srivastava-Tulsiani, QIP'25]
A Computational Understanding of
Symmetry & Randomness
Functions with Symmetry
[Jeronimo-**M**-Roy, RANDOM'25]

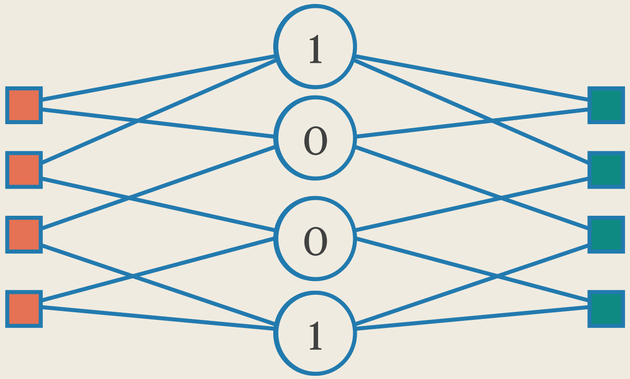

Algebraic Complexity
[Ivanyos-**M**-Qiao, ITCS'22]
Expansion
[Jeronimo-**M**-O'Donnell-Paredes-Tulsiani, ITCS'22]
[Jeronimo-**M**-Roy-Wigderson, FOCS'22 SICOMP'25]
Property Testing
[**M**-Roy, ITCS'26]
[**M**-Roy, Preprint]
Quantum/Classical Error Correction
[Jeronimo-**M**-Srivastava-Tulsiani, STOC'25]
[Bergamaschi-Jeronimo-**M**-Srivastava-Tulsiani, QIP'25]
A Computational Understanding of
Symmetry & Randomness
⟨ Your Area ⟩
Functions with Symmetry
[Jeronimo-**M**-Roy, RANDOM'25]
High-dimensional Structure


Algebraic Complexity
[Ivanyos-**M**-Qiao, ITCS'22]
Expansion
[Jeronimo-**M**-O'Donnell-Paredes-Tulsiani, ITCS'22]
[Jeronimo-**M**-Roy-Wigderson, FOCS'22 SICOMP'25]
Property Testing
[**M**-Roy, ITCS'26]
[**M**-Roy, Preprint]
Quantum/Classical Error Correction
[Jeronimo-**M**-Srivastava-Tulsiani, STOC'25]
[Bergamaschi-Jeronimo-**M**-Srivastava-Tulsiani, QIP'25]
A Computational Understanding of
Symmetry & Randomness
⟨ Your Area ⟩
Functions with Symmetry
[Jeronimo-**M**-Roy, RANDOM'25]
High-dimensional Structure

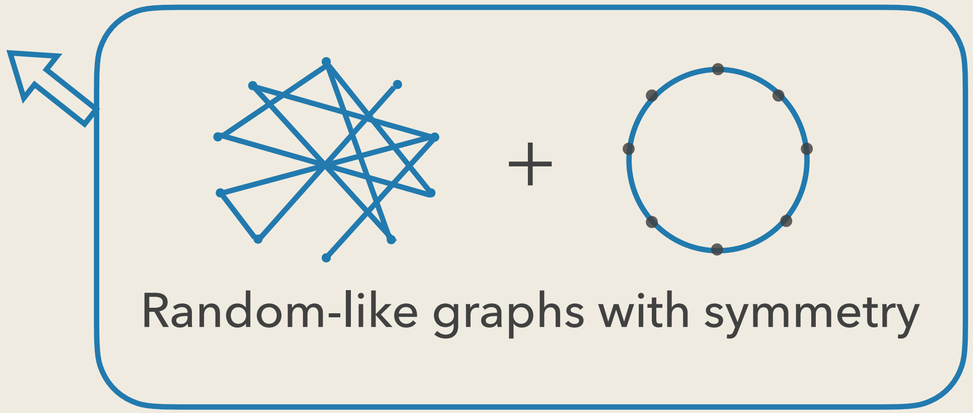
What we will see today



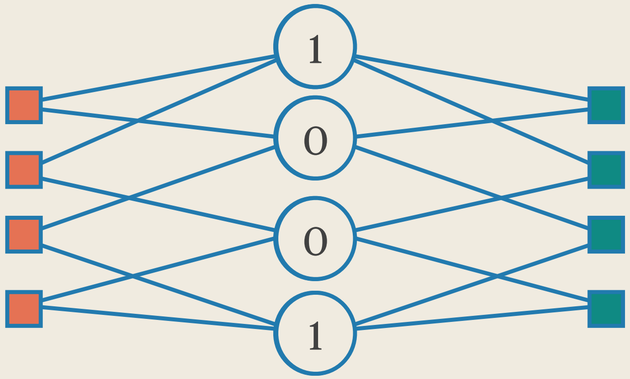
What we will see today



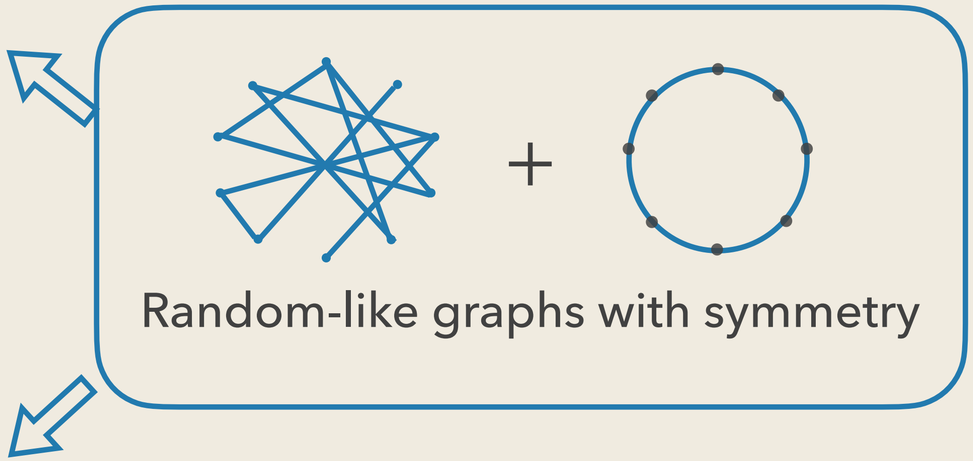
Quantum LDPC Codes

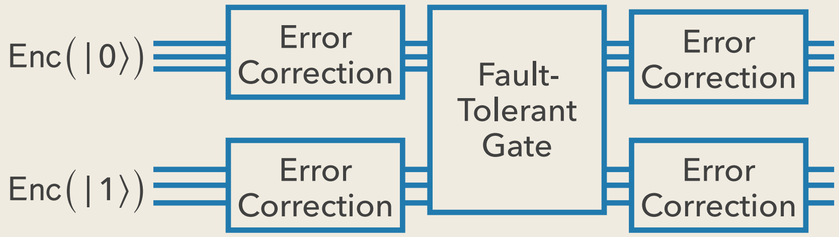


What we will see today



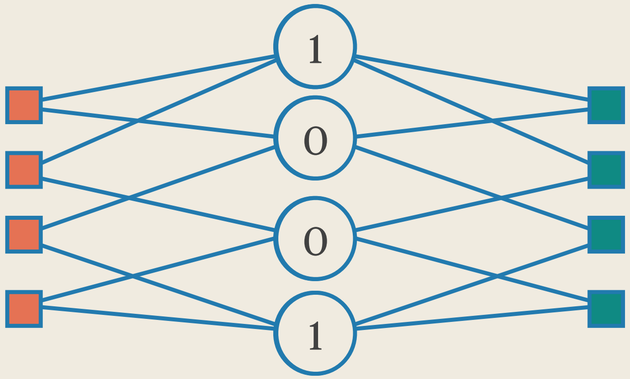
Quantum LDPC Codes



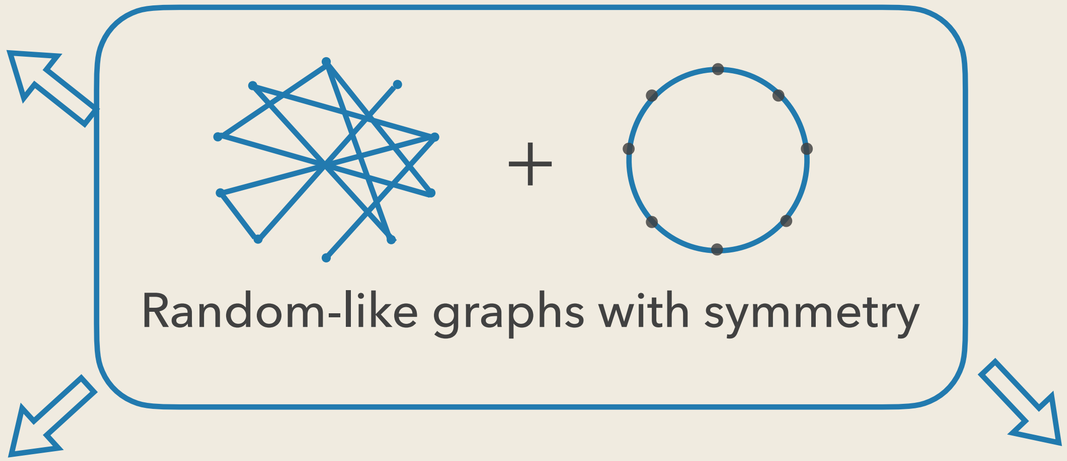


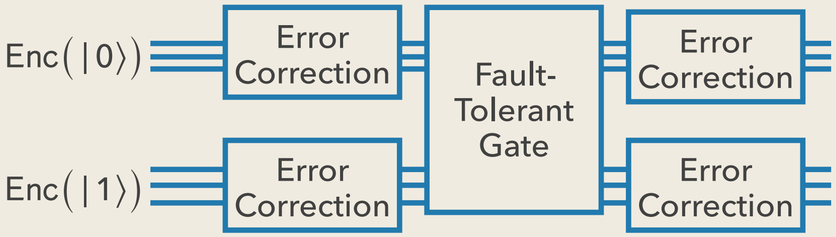
Fault-tolerant Quantum Gadgets

What we will see today

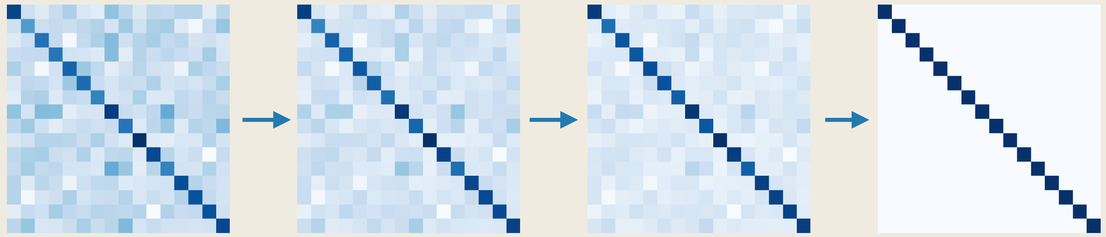


Quantum LDPC Codes



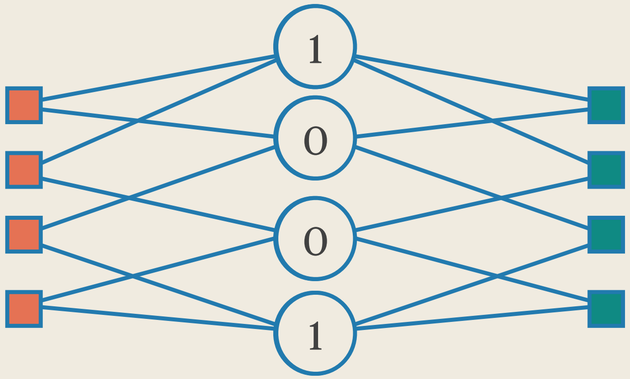


Fault-tolerant Quantum Gadgets



Near-Optimal Quantum Expanders (Fast Mixing Channel)

What we will see today

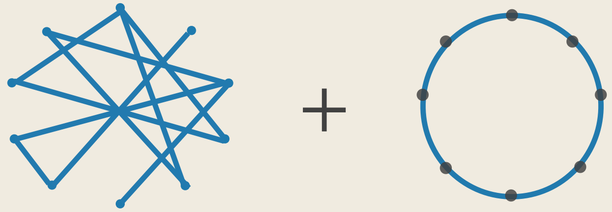


Quantum LDPC Codes

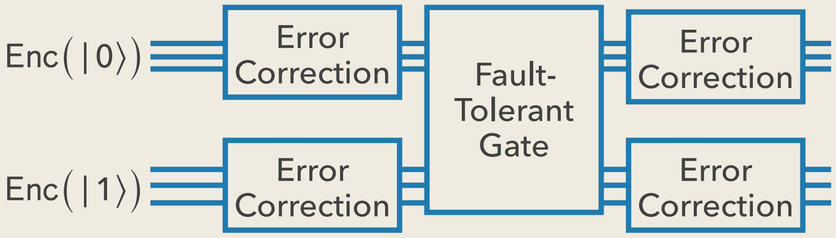
Weak Expanders

Generic Amplification Machinery

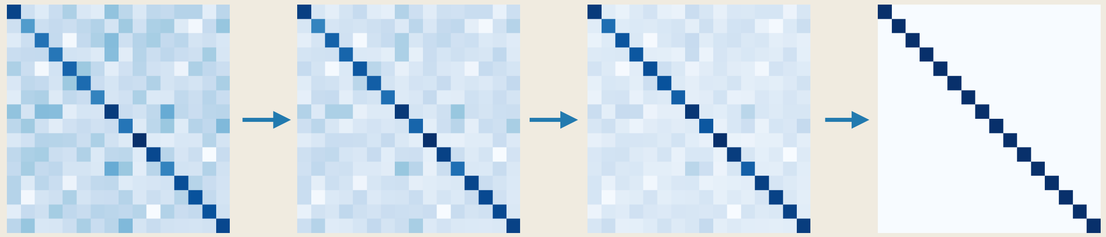
Near-Optimal Expanders



Random-like graphs with symmetry



Fault-tolerant Quantum Gadgets



Near-Optimal Quantum Expanders (Fast Mixing Channel)

Today’s Agenda

Today's Agenda

Definitions

Today's Agenda

Definitions

Symmetry and Expansion for Quantum Error Correction

Today's Agenda

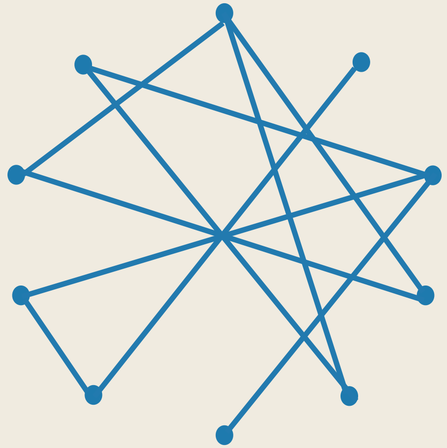
Definitions ▶ Symmetry and Expansion for Quantum Error Correction ▶ Symmetry and Expansion for Other Pseudorandom Objects

Today's Agenda

Definitions ▶ Symmetry and Expansion for Quantum Error Correction ▶ Symmetry and Expansion for Other Pseudorandom Objects ▶ Future Plans

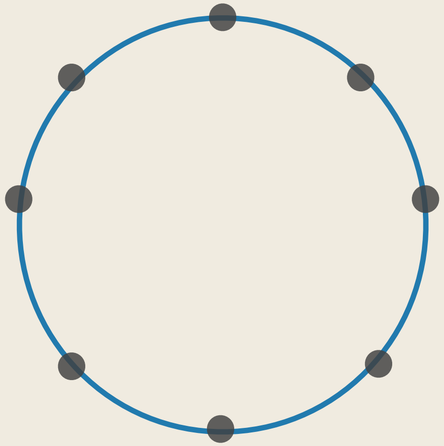
Definitions: Let’s get started

Constructing Expanders with Symmetry



(Near)-Optimal Expanders

+



Symmetry

Sparse yet well-connected graphs

Image Credits – Merill Sherman (Quanta)

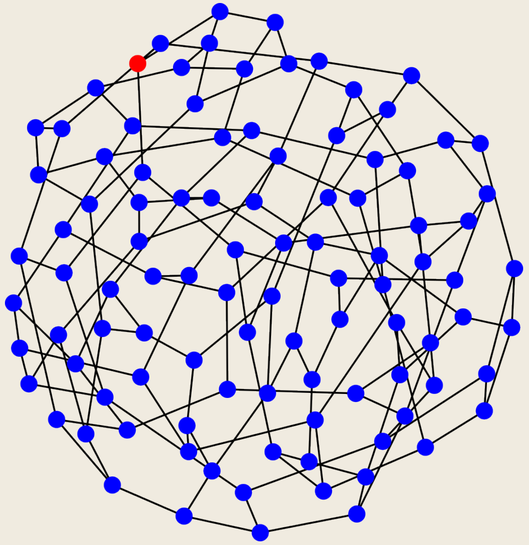
Sparse yet well-connected graphs

Budget: 3 connections each

Objective: Max Connectivity

Image Credits – Merill Sherman (Quanta)

Sparse yet well-connected graphs

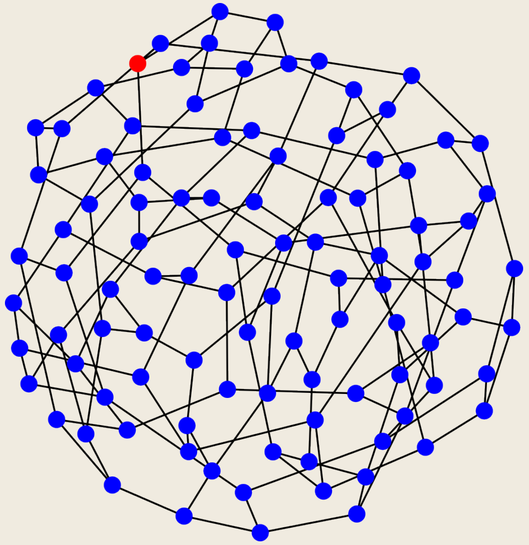


Budget: 3 connections each

Objective: Max Connectivity

Image Credits – Merill Sherman (Quanta)

Sparse yet well-connected graphs



Budget: 3 connections each

Objective: Max Connectivity

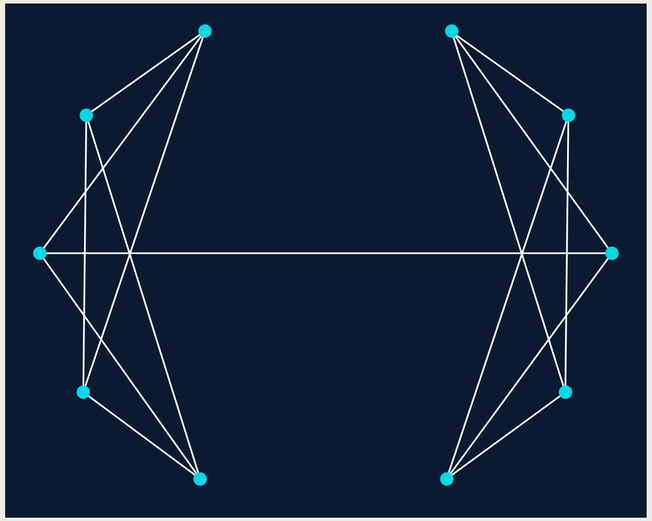
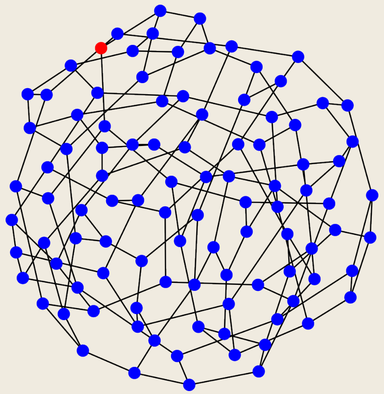
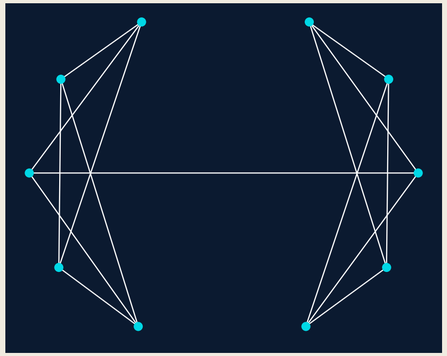


Image Credits – Merill Sherman (Quanta)

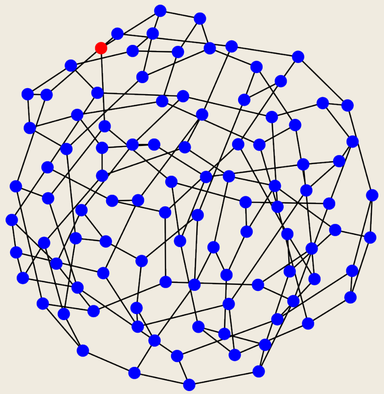
Expansion: Analytic Proxy for Connectivity

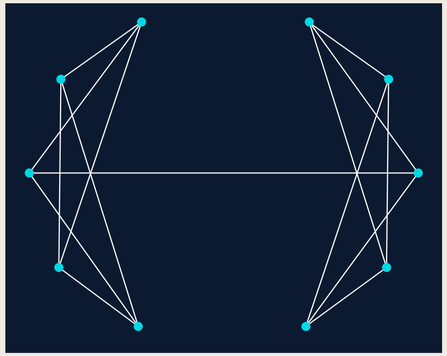




Expansion: Analytic Proxy for Connectivity

Degree – $\deg(X)$

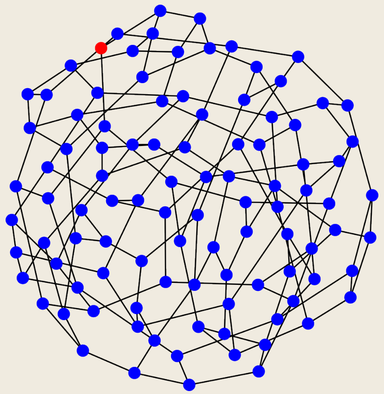


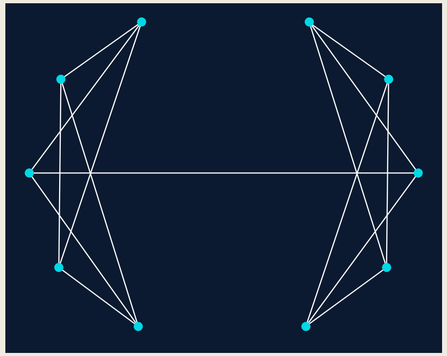


Expansion: Analytic Proxy for Connectivity

Degree – $\deg(X)$

Expansion – $\lambda(X)$



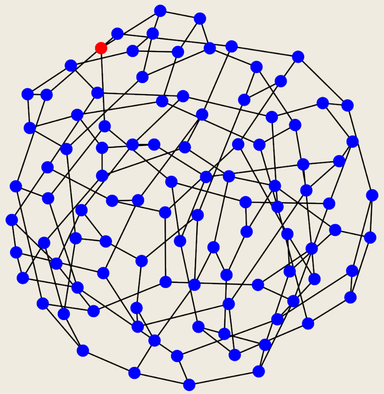


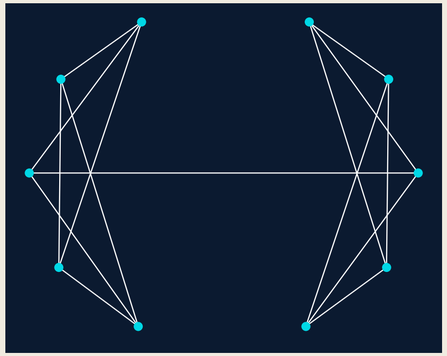
Expansion: Analytic Proxy for Connectivity

Degree – $\deg(X)$

Expansion – $\lambda(X)$

Second Largest Eigenvalue of the Graph Adjacency Matrix



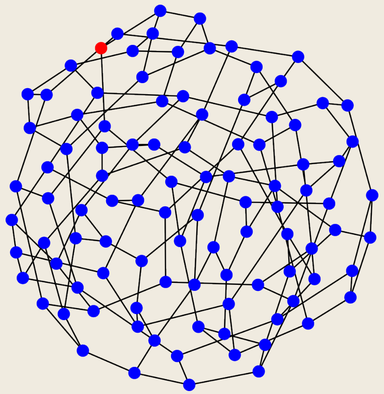


Expansion: Analytic Proxy for Connectivity

Degree – $\deg(X)$

Expansion – $\lambda(X)$

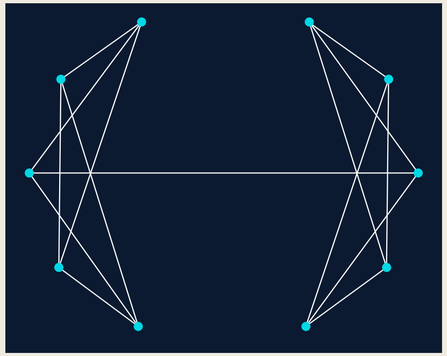
Second Largest Eigenvalue of the Graph Adjacency Matrix



Expander Quality (λ vs deg)

Optimal (Ramanujan) — Okay Expander — Not Expander

$2\sqrt{\deg(X)-1}$ — $0.9\deg(X)$ — $\deg(X)$

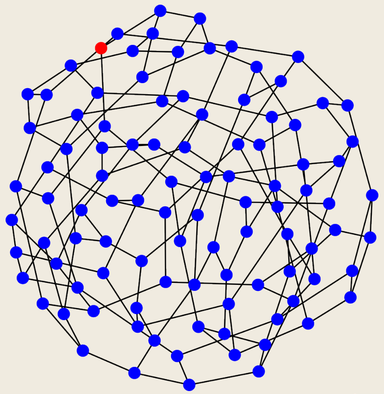


Expansion: Analytic Proxy for Connectivity

Degree – $\deg(X)$

Expansion – $\lambda(X)$

Second Largest Eigenvalue of the Graph Adjacency Matrix

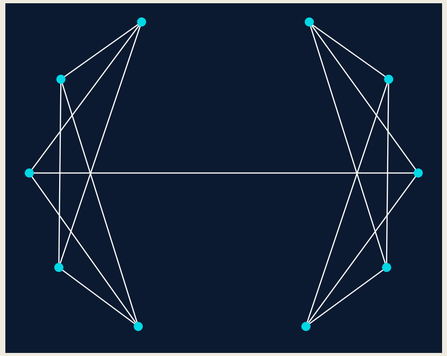


Expander Quality (λ vs deg)

Optimal (Ramanujan) — Okay Expander — Not Expander

$2\sqrt{\deg(X)-1}$ — $0.9\deg(X)$ — $\deg(X)$

Random Graphs (almost) attain this bound!

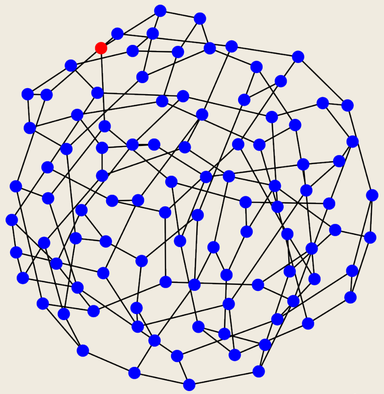


Expansion: Analytic Proxy for Connectivity

Degree – $\deg(X)$

Expansion – $\lambda(X)$

Second Largest Eigenvalue of the Graph Adjacency Matrix

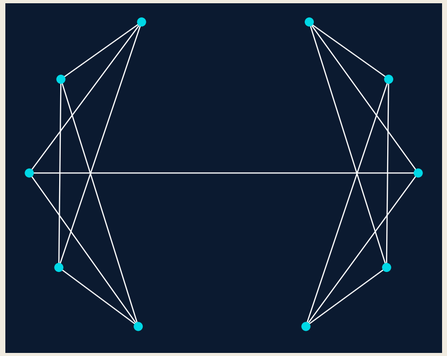


Expander Quality (λ vs $\deg$)

Optimal (Ramanujan) — Okay Expander — Not Expander

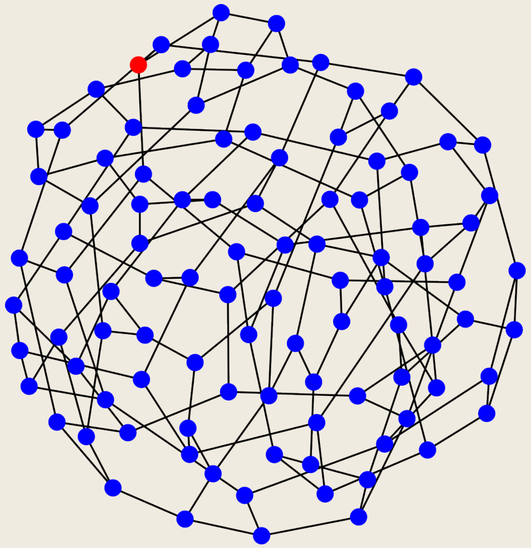
$2\sqrt{\deg(X)-1}$ — $0.9\deg(X)$ — $\deg(X)$

Random Graphs (almost) attain this bound!



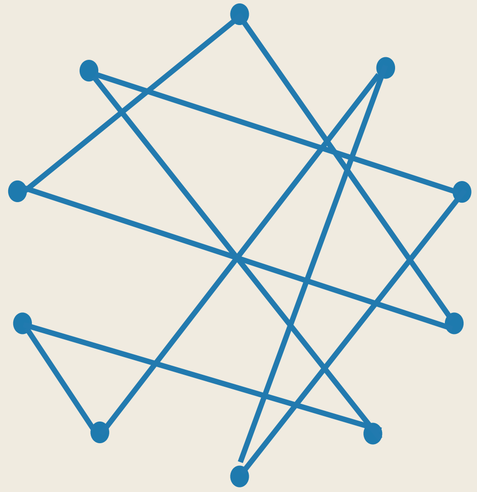
Can we do this without randomness?

Random-like graphs without randomness



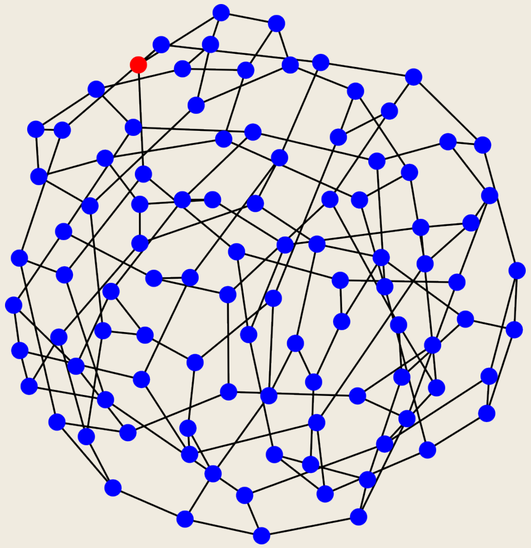
≈

Same Expansion



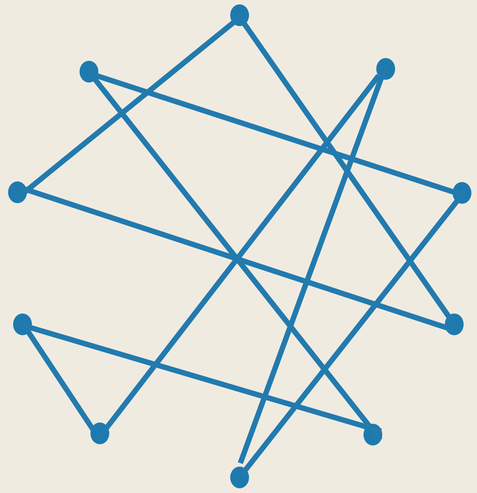
No Randomness

Random-like graphs without randomness



≈

Same Expansion

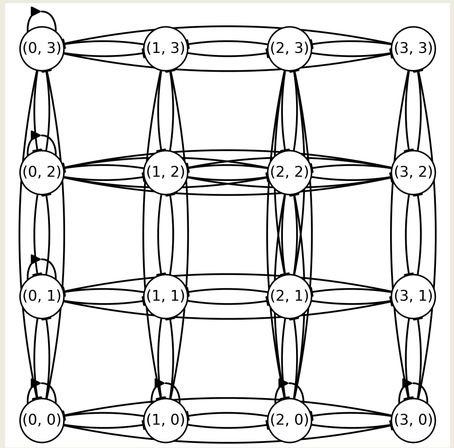


No Randomness

Challenge: Explicit Constructions (Pseudorandom)

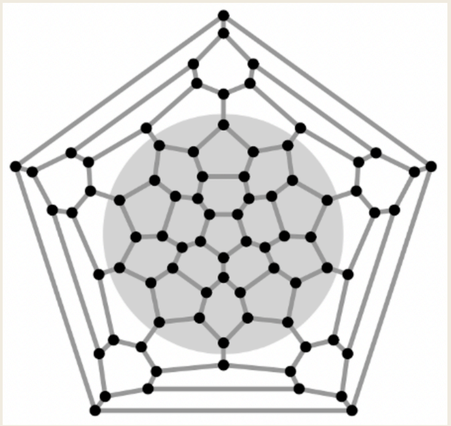
Expanders can be explicitly constructed!

Analytic



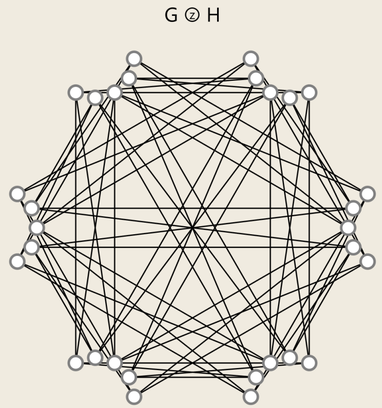
Margulis ’73

Number-Theoretic



Lubotzky–Phillips–Sarnak ’88

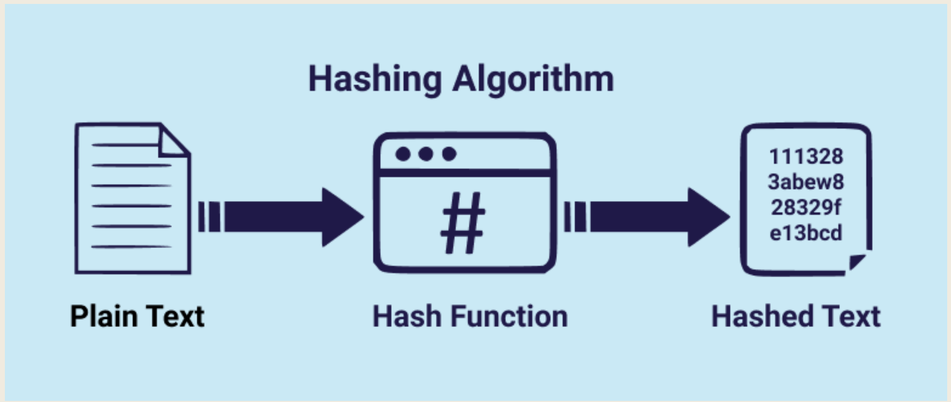
Combinatorial



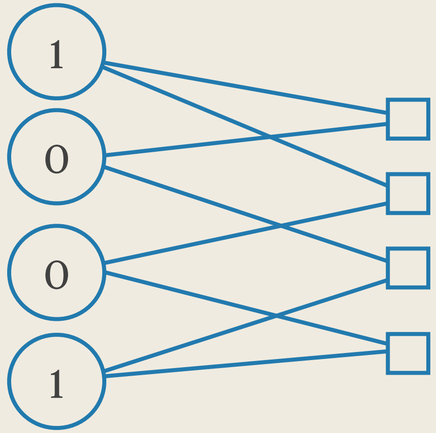
Reingold-Vadhan-Wigderson ’00

… and many more.

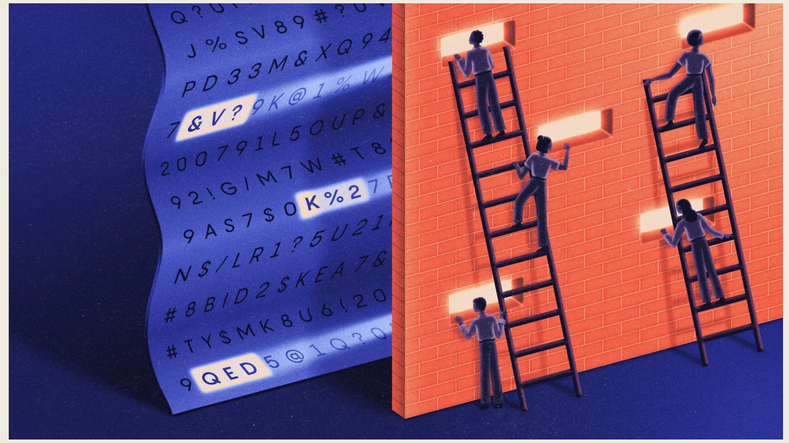
Applications of Expanders



Cyptographic Hash Functions



Expander Codes

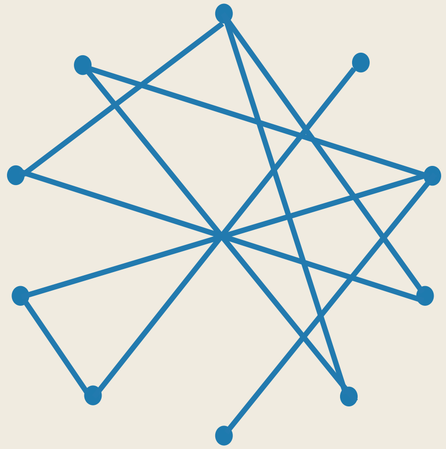


Probabilistically Checkable Proofs (PCP)

... and many more.

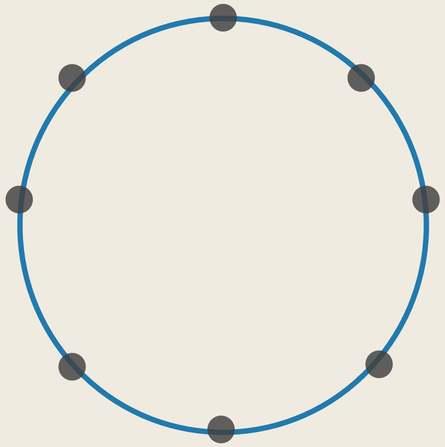
Image Credits – Kristina Armitage (Quanta)

Constructing Expanders with Symmetry



(Near)-Optimal Expanders

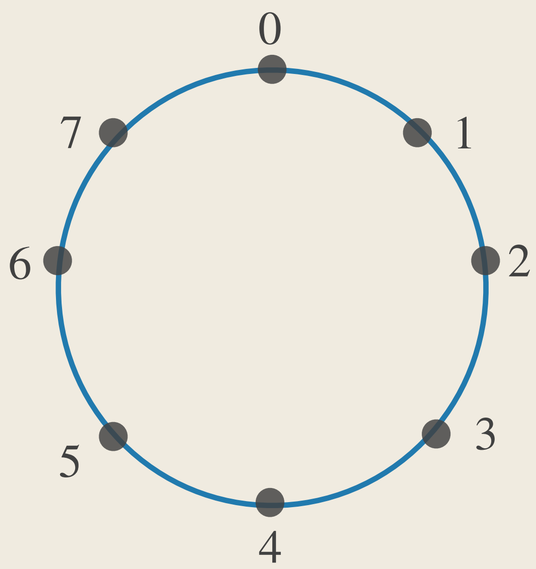
+



Symmetry

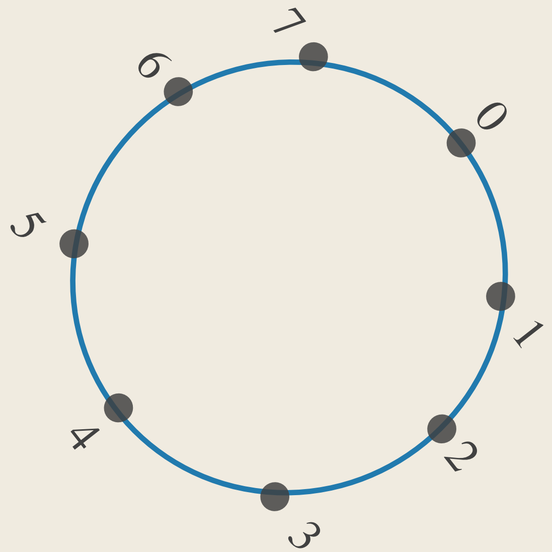
Symmetries of Graphs

Symmetries of Graphs



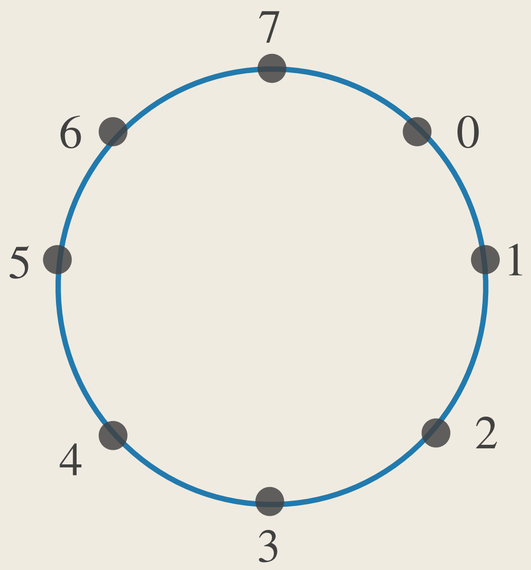
Cyclic Symmetry

Symmetries of Graphs



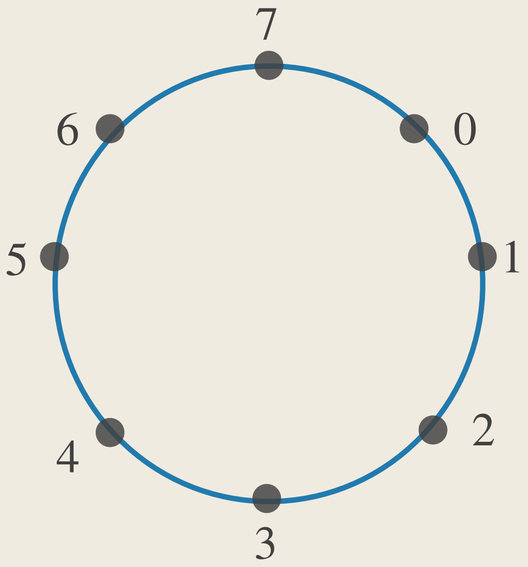
Cyclic Symmetry

Symmetries of Graphs

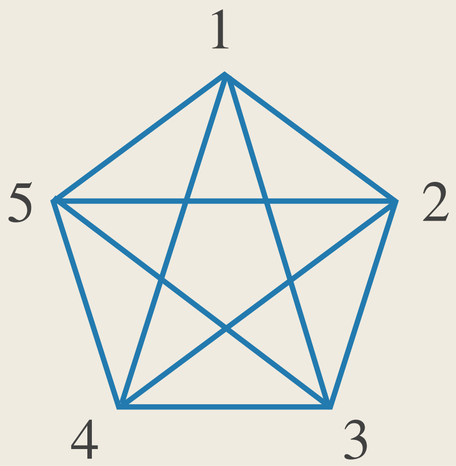


Cyclic Symmetry

Symmetries of Graphs

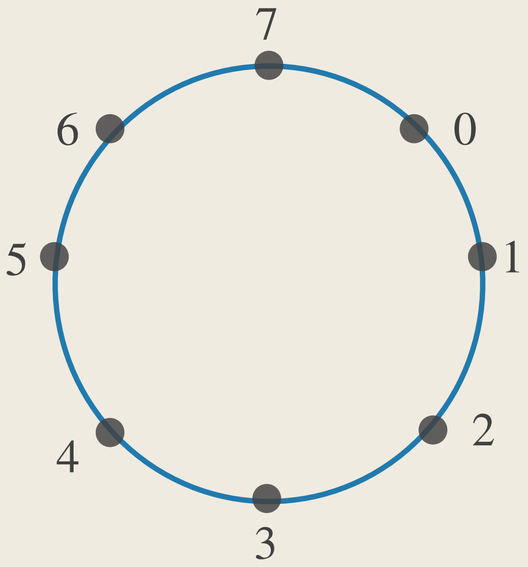


Cyclic Symmetry

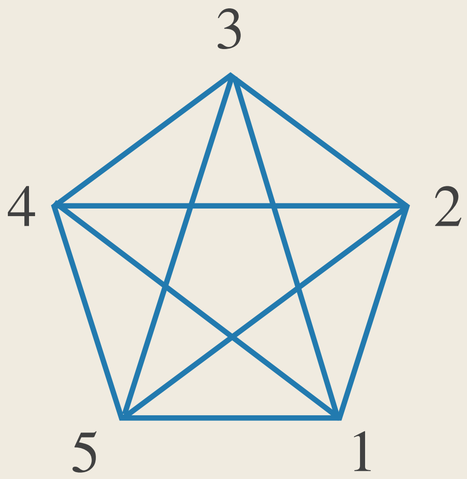


Permutation Symmetry

Symmetries of Graphs



Cyclic Symmetry

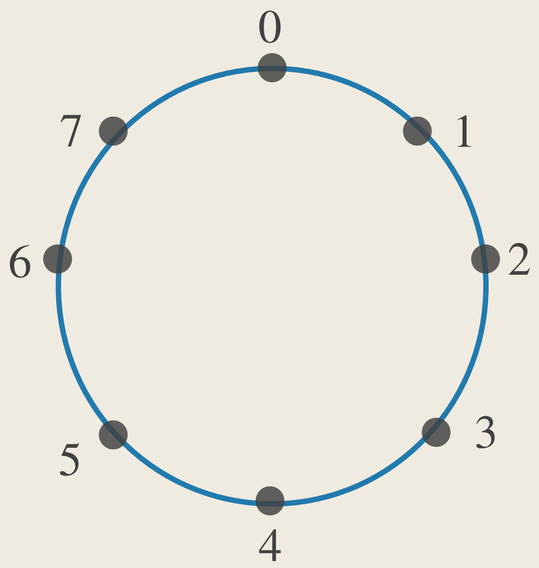


Permutation Symmetry

Symmetries aka Groups

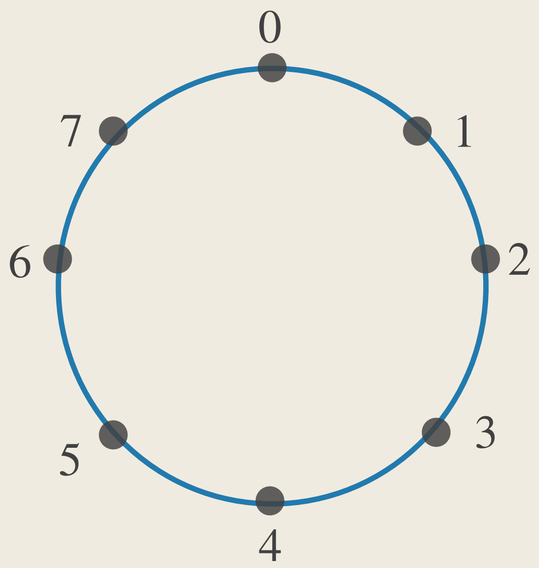
Group: Set & $*$ Operation

Symmetries aka Groups



Group: Set & $*$ Operation

Symmetries aka Groups



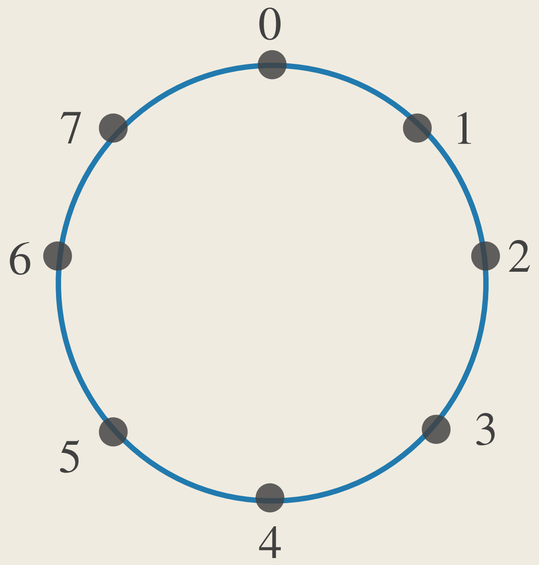
Cyclic Group ($\mathbb{Z}_n$)

Addition modulo n

$$1 * 7 = 0 \mod 8$$

Group: Set & $*$ Operation

Symmetries aka Groups

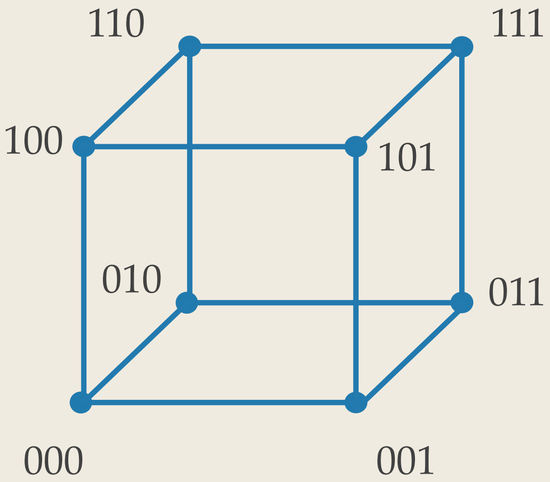


Cyclic Group ($\mathbb{Z}_n$)

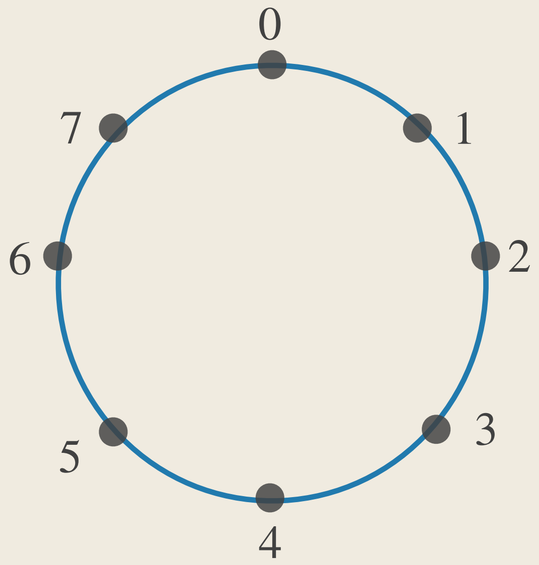
Addition modulo n

$$1 * 7 = 0 \mod 8$$

Group: Set & $*$ Operation



Symmetries aka Groups

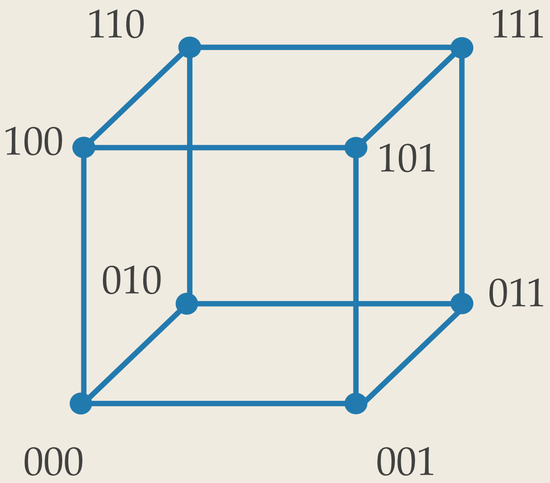


Cyclic Group ($\mathbb{Z}_n$)

Addition modulo n

$1 * 7 = 0 \mod 8$

Group: Set & $*$ Operation

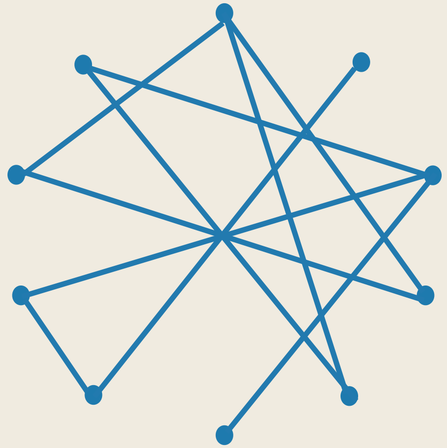


Hypercube ($\mathbb{Z}_2^n$)

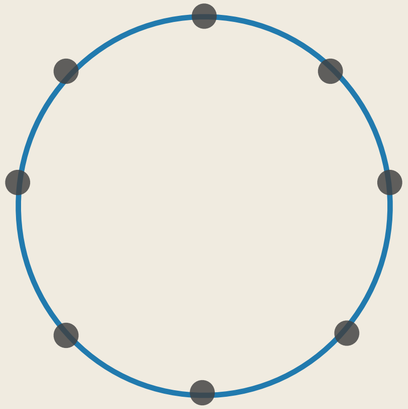
Bit-wise XOR.

$001 * 111 = 110.$

Constructing Expanders with Symmetry



+



(Near)-Optimal Expanders

Symmetry

Constructions?

Applications?

Symmetry and Expansion for:

1. Quantum Error Correction
2. Near-Optimal Pseudorandom Objects

Error Correcting Codes = Robust Redundancy

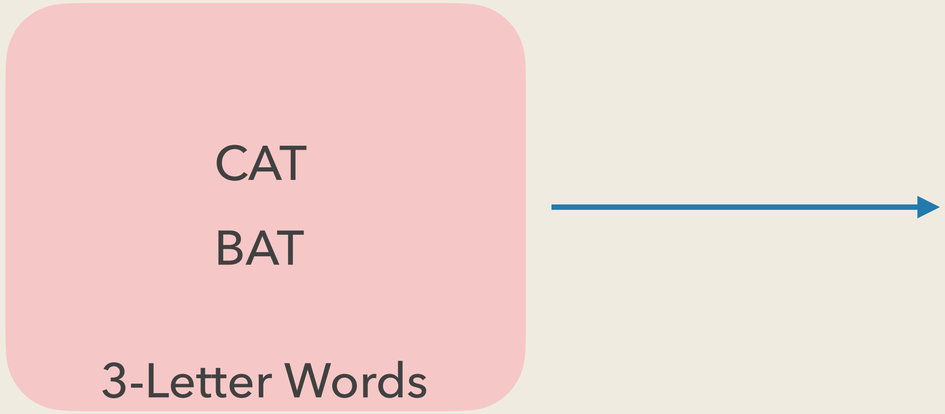
Error Correcting Codes = Robust Redundancy

CAT

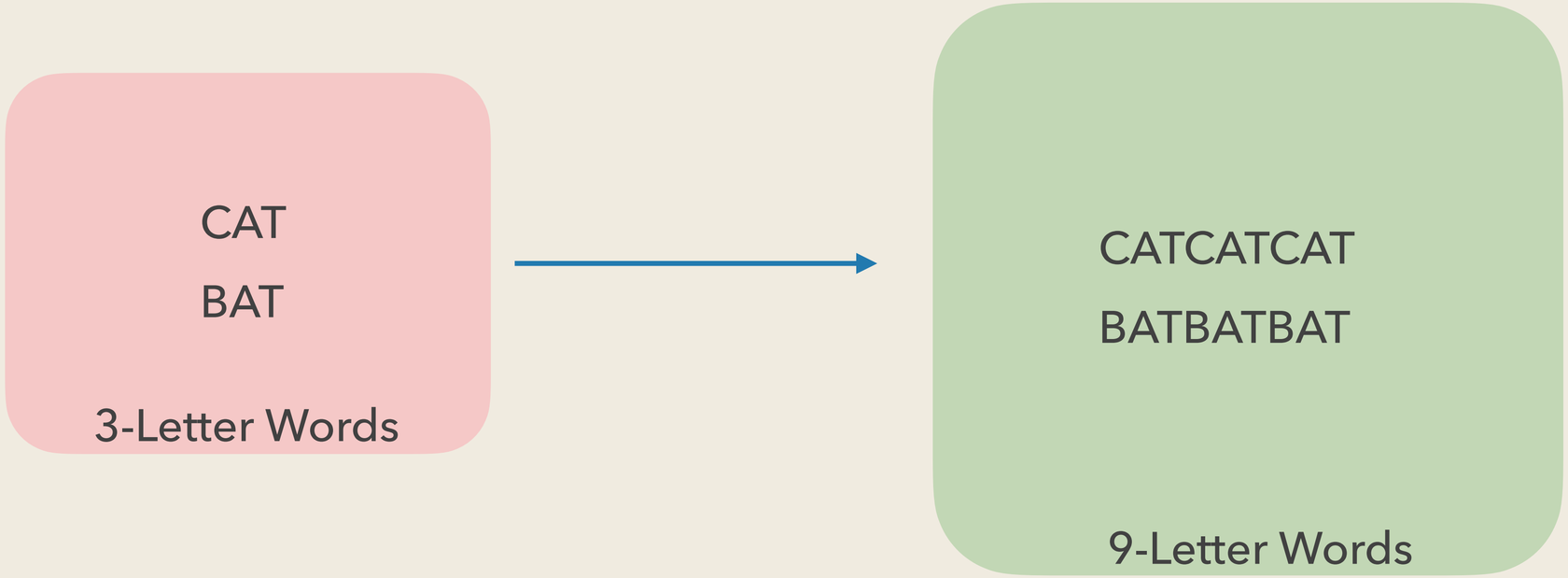
BAT

3-Letter Words

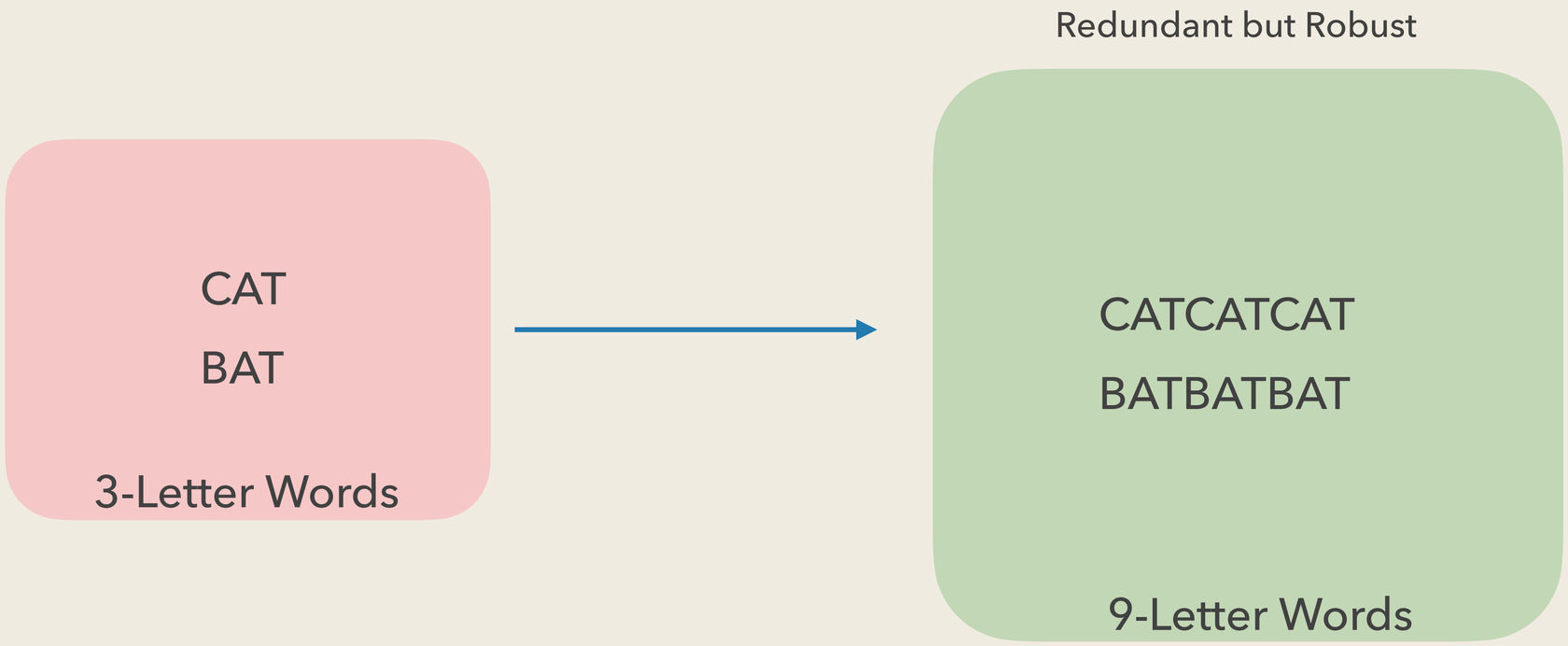
Error Correcting Codes = Robust Redundancy



Error Correcting Codes = Robust Redundancy

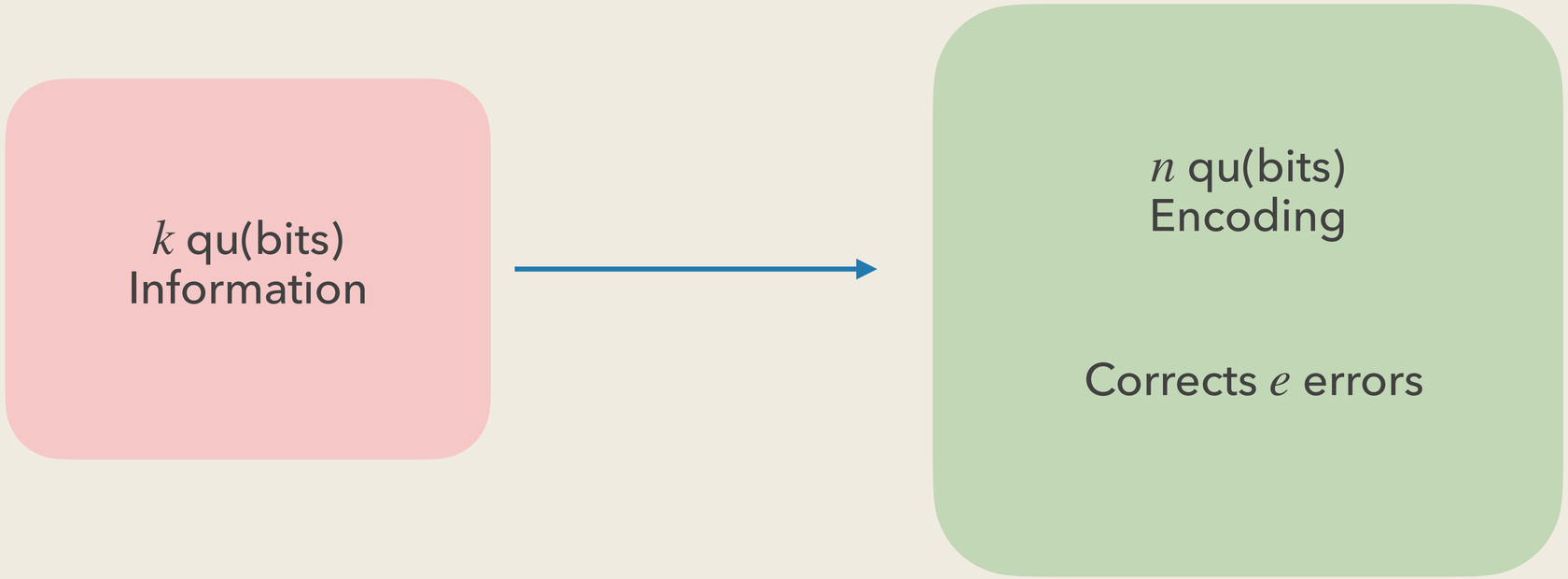


Error Correcting Codes = Robust Redundancy



Error Correcting Codes = Robust Redundancy

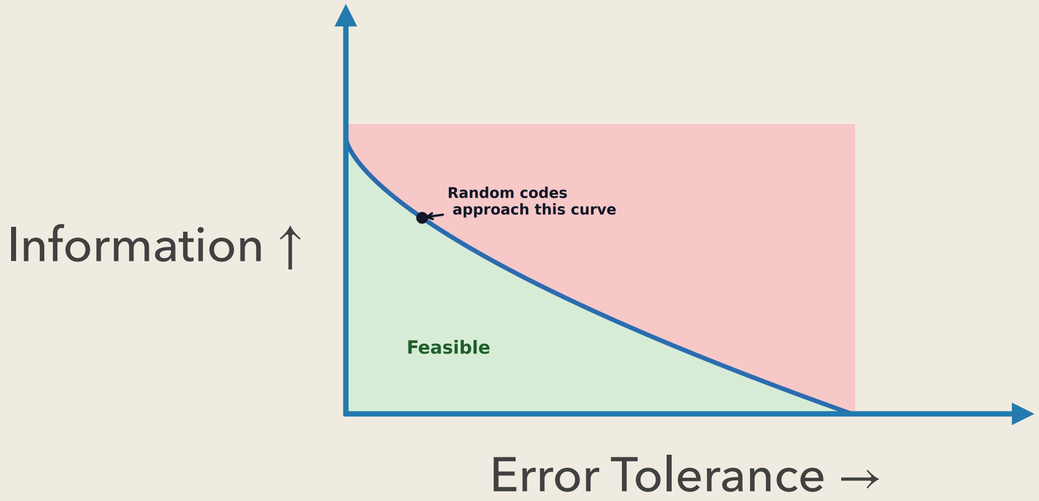
Error Correcting Codes = Robust Redundancy



Two Essential Properties

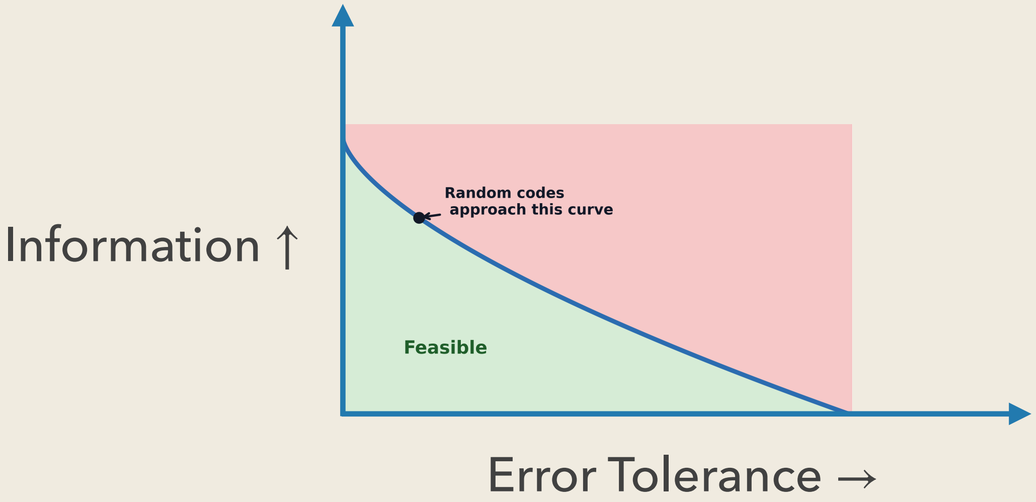
Two Essential Properties

Max Robustness for the budget

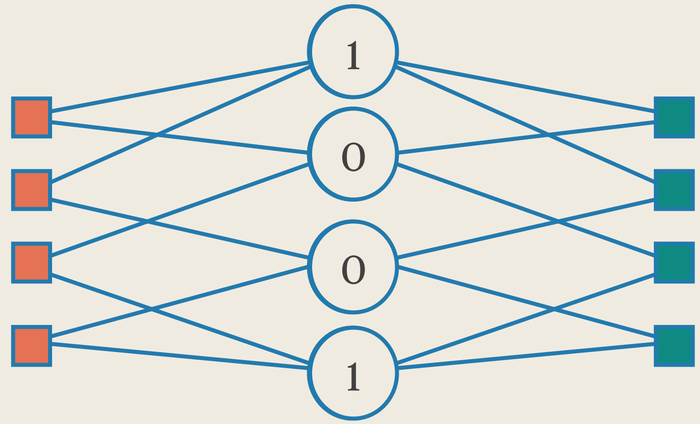


Two Essential Properties

Max Robustness for the budget

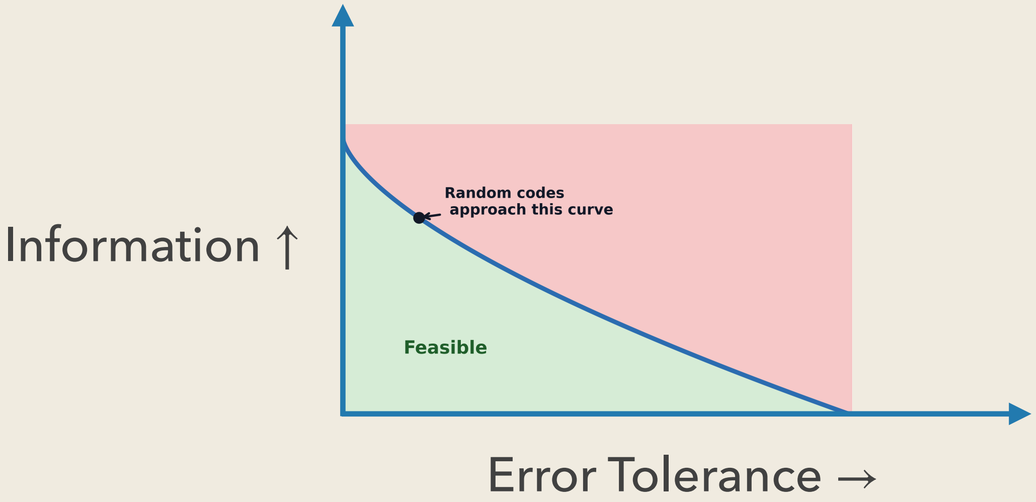


Fast Operations (Quantum LDPC)



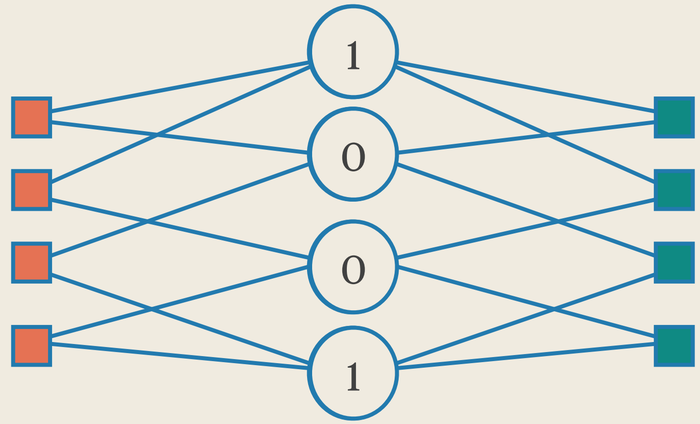
Two Essential Properties

Max Robustness for the budget



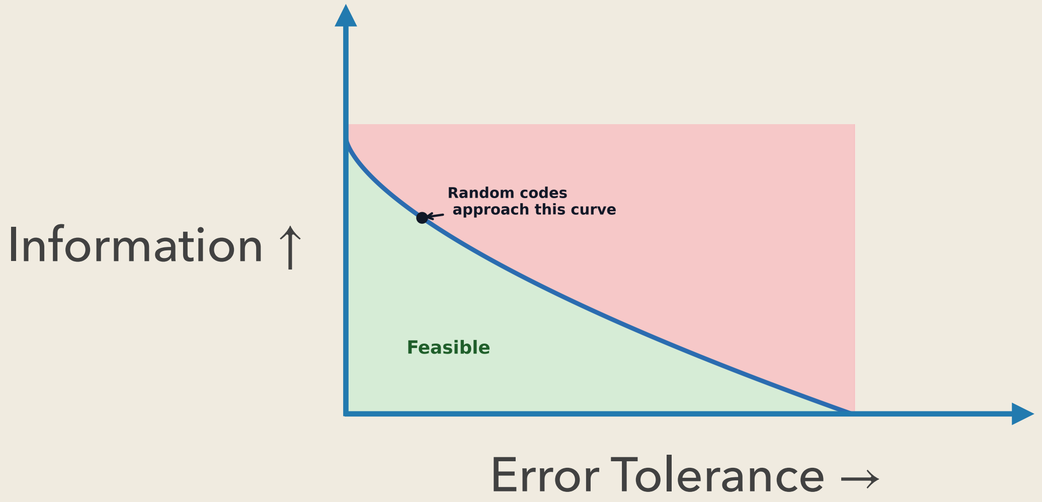
Fast Operations (Quantum LDPC)

" Very Sparse checks"



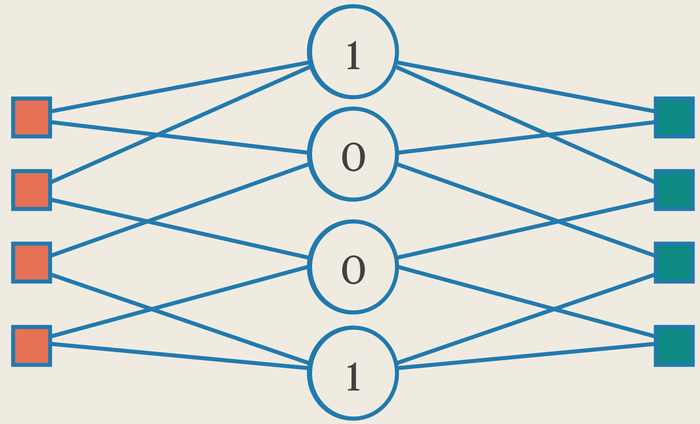
Two Essential Properties

Max Robustness for the budget



Fast Operations (Quantum LDPC)

" Very Sparse checks"



5G NR, WiFi 6 – Classical LDPC Codes!

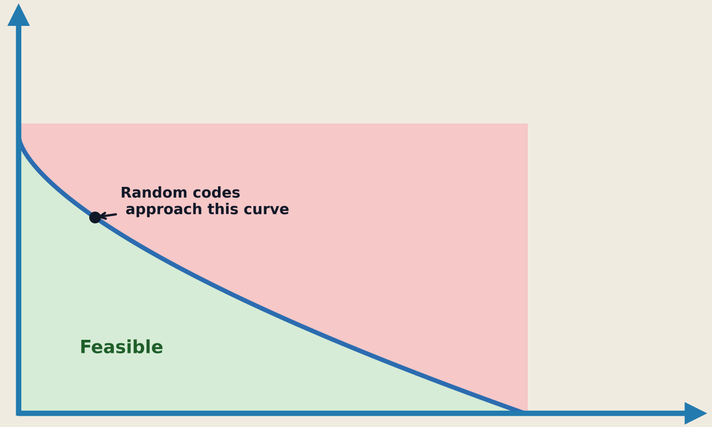
Getting both is very hard!

Goal 1: Max Robustness

Goal 2: Fast Operations

Getting both is very hard!

Max Robustness



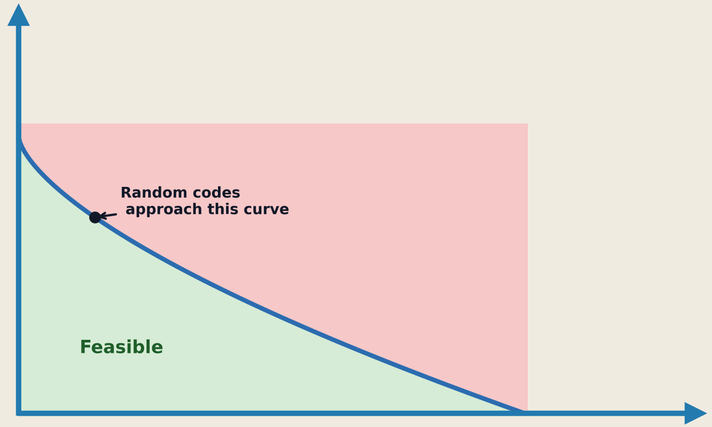
Goal 1: Max Robustness

Goal 2: Fast Operations

Getting both is very hard!

Max Robustness

Randomness



Goal 1: Max Robustness

Goal 2: Fast Operations

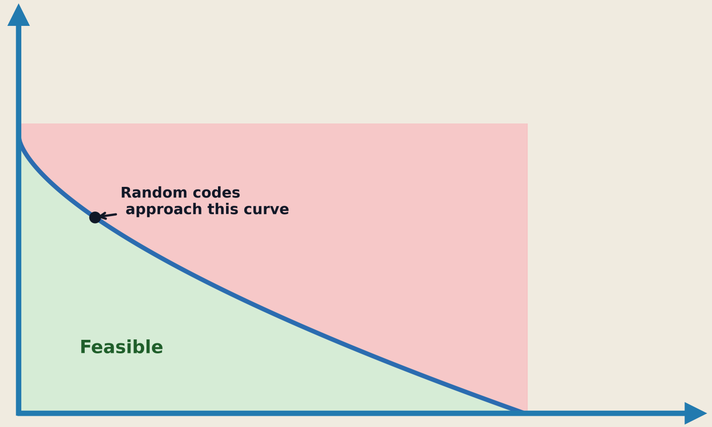
CAT → SXAKRDOPR

BAT → KRMFLERWP

Getting both is very hard!

Max Robustness

Randomness



✓ Goal 1: Max Robustness

✗ Goal 2: Fast Operations

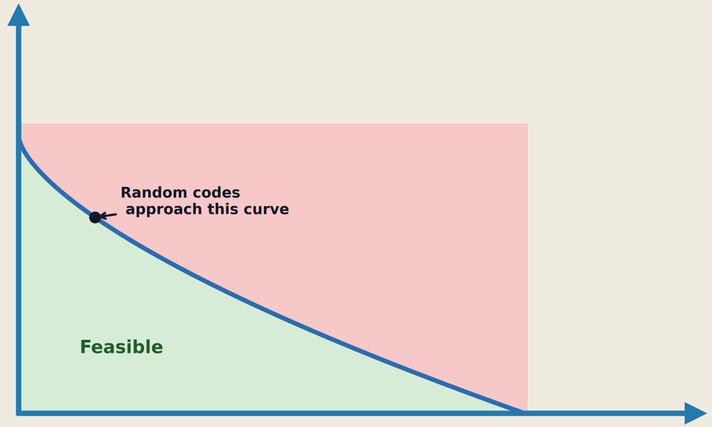
CAT → SXAKRDOPR

BAT → KRMFLERWP

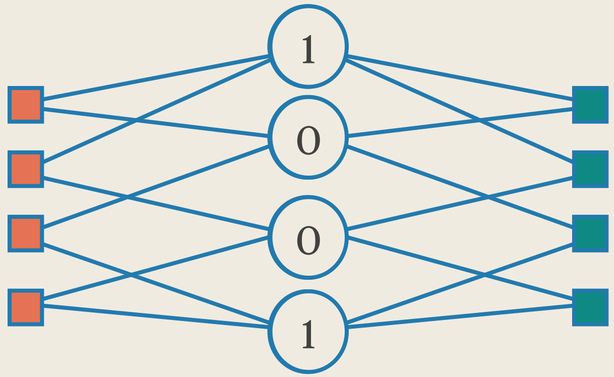
Getting both is very hard!

Max Robustness

Randomness



Fast Operations (QLDPC)



Goal 1: Max Robustness

Goal 2: Fast Operations

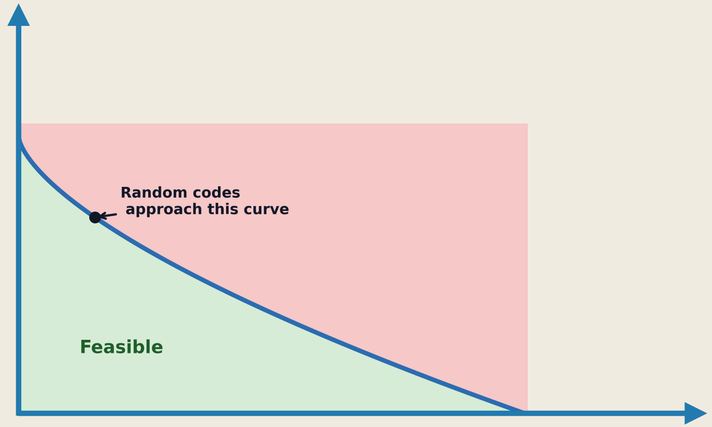
CAT → SXAKRDOPR

BAT → KRMFLERWP

Getting both is very hard!

Max Robustness

Randomness

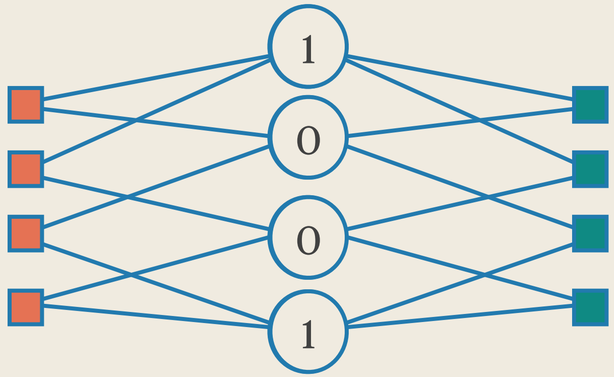


Goal 1: Max Robustness

Goal 2: Fast Operations

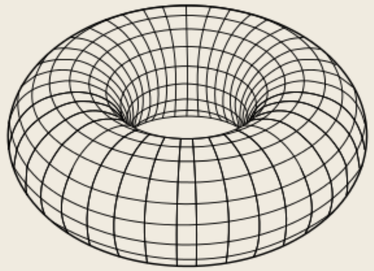
Fast Operations (QLDPC)

Symmetry



CAT → SXAKRDOPR

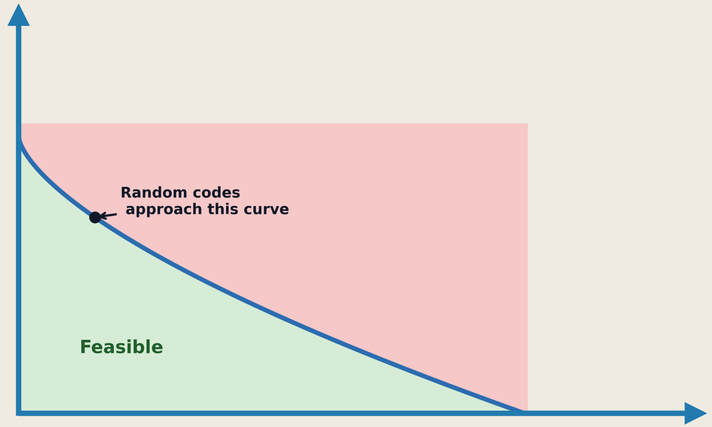
BAT → KRMFLERWP



Getting both is very hard!

Max Robustness

Randomness



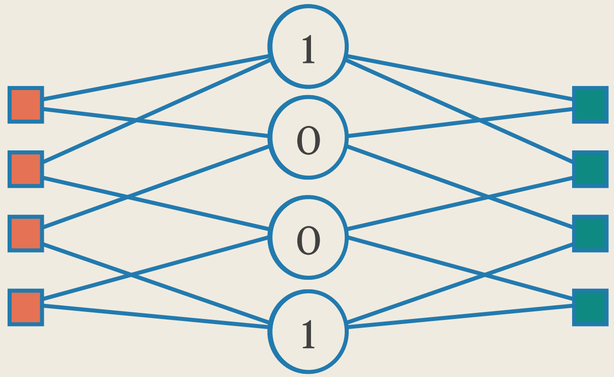
Random + Symmetric?

Goal 1: Max Robustness

Goal 2: Fast Operations

Fast Operations (QLDPC)

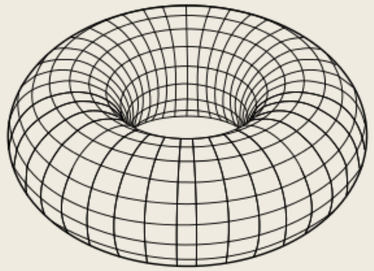
Symmetry



Robust Quantum LDPC Codes?

CAT → SXAKRDOPR

BAT → KRMFLERWP



Getting both is very hard!

Locally Testable Codes

Cyclic Codes

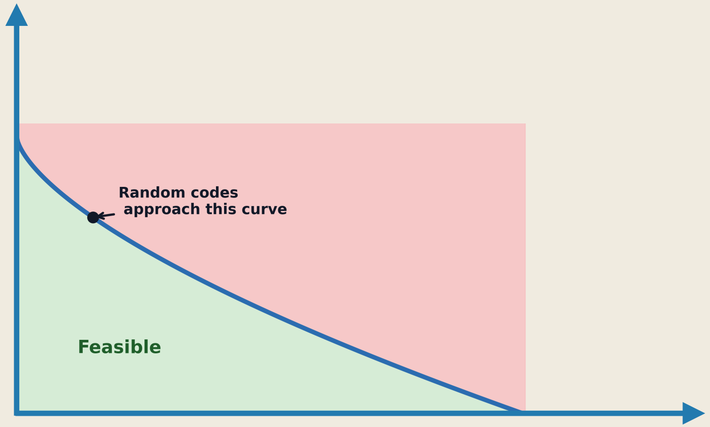
Max Robustness

Randomness

Fast Operations (QLDPC)

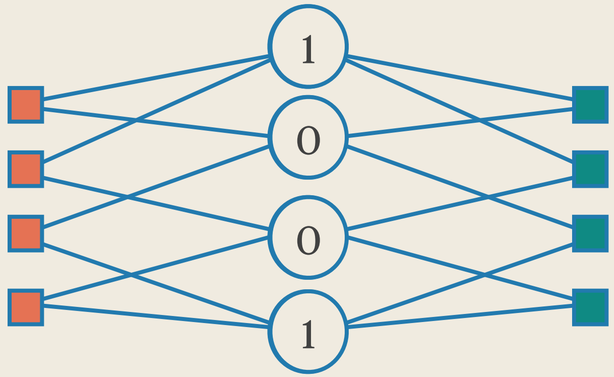
Symmetry

Random + Symmetric?



Goal 1: Max Robustness

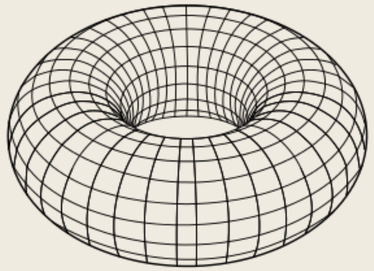
Goal 2: Fast Operations



Robust Quantum LDPC Codes?

CAT → SXAKRDOPR

BAT → KRMFLERWP



Expanders with Cyclic Symmetry are very useful!

Expanders with Cyclic Symmetry are very useful!

Expanders with Cyclic Symmetry?

Expanders with Cyclic Symmetry are very useful!

Expanders with Cyclic Symmetry?

[Pantaleev-Kalachev'21,22]

Quantum LDPC Codes

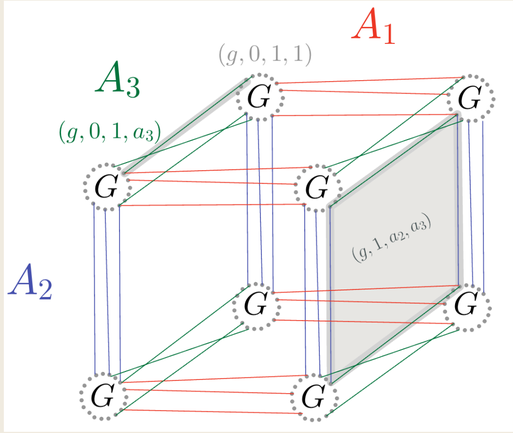
Quasi-cyclic LDPC codes

Expanders with Cyclic Symmetry are very useful!

Expanders with Cyclic Symmetry?

[Dinur-Lin-Vidick'24]

Cubical Complexes



[Pantaleev-Kalachev'21,22]

Quantum LDPC Codes

Quasi-cyclic LDPC codes

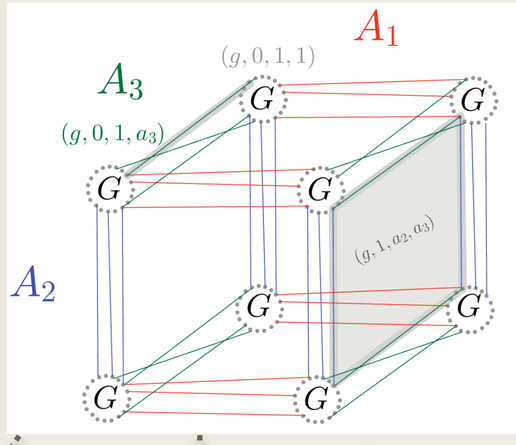
Image Credits – DLV'24

Expanders with Cyclic Symmetry are very useful!

Expanders with Cyclic Symmetry?

[Dinur-Lin-Vidick'24]

Cubical Complexes



[Pantaleev-Kalachev'21,22]

Quantum LDPC Codes

Quasi-cyclic LDPC codes

[Nguyen-Pattison'24]

Fault-tolerant Quantum Gadgets

[Dinur-Lin-Vidick'24]

Quantum Locally Testable Codes

[Hsieh-Lubotzky-Mohanty-Reiner-Zhang'25]

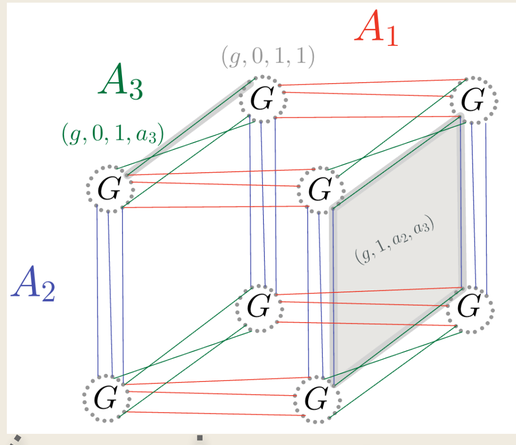
Lossless Expanders

Image Credits – DLV'24

We construct these graphs.

[Dinur-Lin-Vidick'24]

Cubical Complexes



[Pantaleev-Kalachev'21,22]

Quantum LDPC Codes

Quasi-cyclic LDPC codes

[Nguyen-Pattison'24]

Fault-tolerant Quantum Gadgets

[Dinur-Lin-Vidick'24]

Quantum Locally Testable Codes

[Hsieh-Lubotzky-Mohanty-Reiner-Zhang'25]

Lossless Expanders

Image Credits – DLV'24

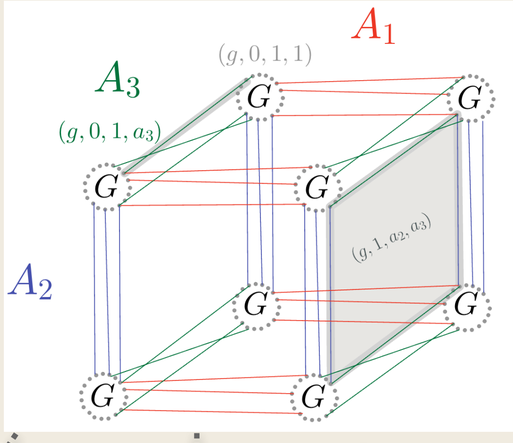
We construct these graphs.

Our Result [Jeronimo–**M**–O'Donnell–Paredes–Tulsiani ITCS'22]

Near-optimal Expanders with Cyclic symmetries

[Dinur-Lin-Vidick'24]

Cubical Complexes



[Pantaleev-Kalachev'21,22]

Quantum LDPC Codes

Quasi-cyclic LDPC codes

[Nguyen-Pattison'24]

Fault-tolerant Quantum Gadgets

[Dinur-Lin-Vidick'24]

Quantum Locally Testable Codes

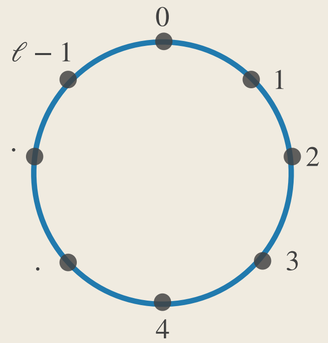
[Hsieh-Lubotzky-Mohanty-Reiner-Zhang'25]

Lossless Expanders

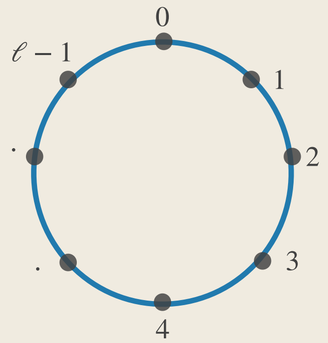
Image Credits – DLV'24

Cyclic Symmetry via Lifts

Cyclic Symmetry via Lifts

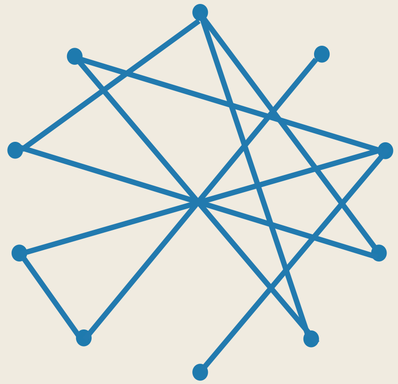


Cyclic Symmetry via Lifts

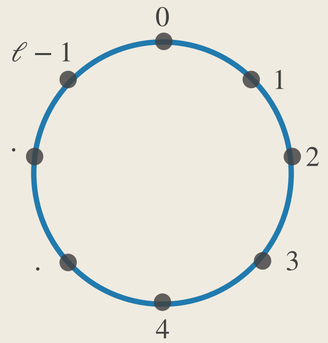


Graph with symmetries of ℓ-cycles

ℓ-lift

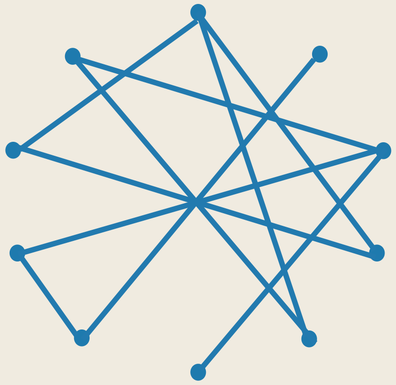


Cyclic Symmetry via Lifts



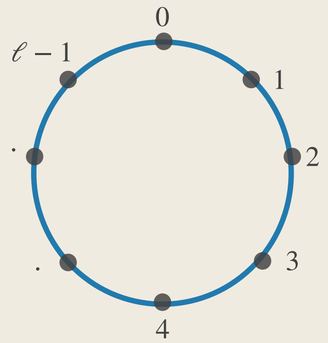
Graph with symmetries of ℓ-cycles

ℓ-lift



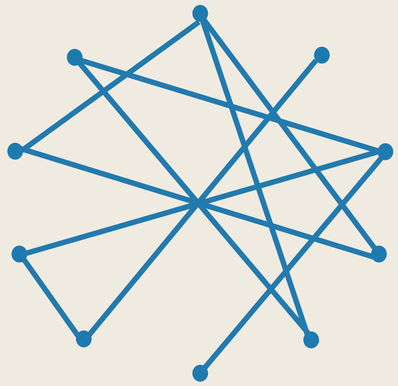
Large symmetries $\ell \rightarrow$ Large Error Tolerance

Cyclic Symmetry via Lifts



Graph with symmetries of ℓ-cycles

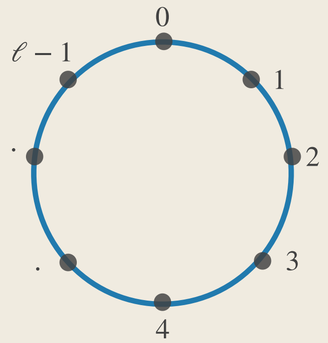
ℓ-lift



Large symmetries $\ell \to$ Large Error Tolerance

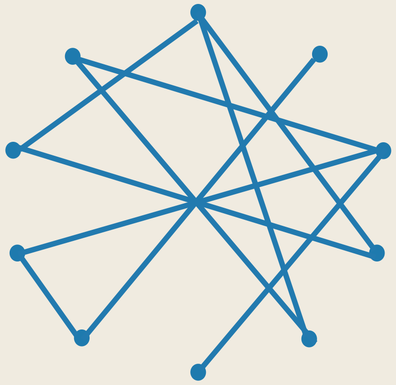
Previous explicit constructions only for $\ell = 2$.

Cyclic Symmetry via Lifts



Graph with symmetries of ℓ-cycles

ℓ-lift



Large symmetries $\ell \to$ Large Error Tolerance

Previous explicit constructions only for $\ell = 2$.

Max possible lift $\ell = 2^{O(n)}$.

Cyclic Symmetry via Lifts

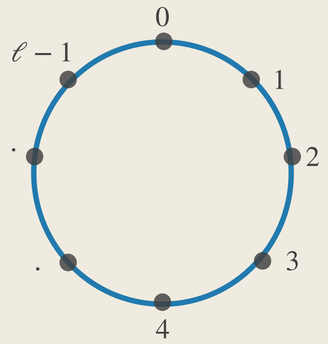
Our Result [Jeronimo–**M**–O'Donnell–Paredes–Tulsiani ITCS'22]

Near-optimal expanders with symmetries of ℓ -cycles for $\ell \geq 2^{\Omega(n)}$.

Large symmetries $\ell \to$ Large Error Tolerance

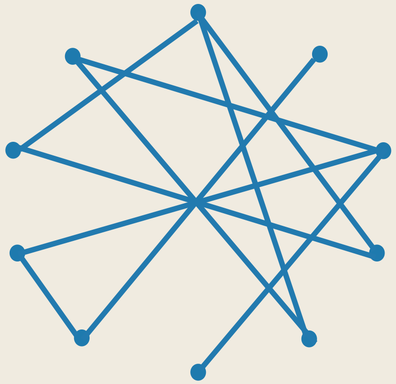
Previous explicit constructions only for $\ell = 2$.

Max possible lift $\ell = 2^{O(n)}$.



Graph with symmetries of ℓ-cycles

ℓ-lift



We construct these graphs.

Our Result [Jeronimo–**M**–O'Donnell–Paredes–Tulsiani ITCS'22]

Near-optimal Expanders with Cyclic symmetries

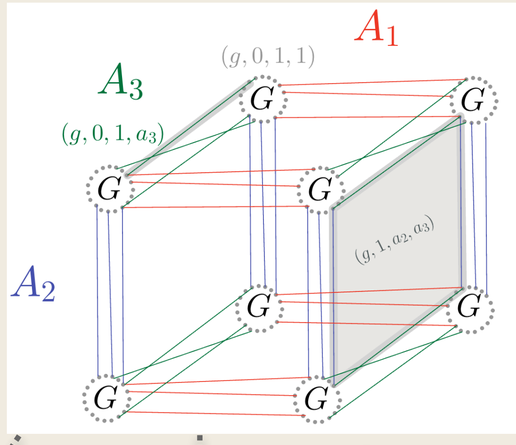
[Pantaleev-Kalachev'21,22]

Quantum LDPC Codes

Quasi-cyclic LDPC codes

[Dinur-Lin-Vidick'24]

Cubical Complexes



[Nguyen-Pattison'24]

Fault-tolerant Quantum Gadgets

[Dinur-Lin-Vidick'24]

Quantum Locally Testable Codes

[Hsieh-Lubotzky-Mohanty-Reiner-Zhang'25]

Lossless Expanders

Image Credits – DLV'24

Symmetry and Expansion for:

1. Quantum Error Correction
2. Near-Optimal Pseudorandom Objects

Expanders & Quantum Expanders

Expanders & Quantum Expanders

Expanders

Cryptography

Pseudorandomness

Complexity Theory

Expanders & Quantum Expanders

Expanders → Quantum Expanders

Cryptography

Pseudorandomness

Complexity Theory

Expanders & Quantum Expanders

Expanders → **Quantum Expanders**

Cryptography
Pseudorandomness
Complexity Theory

→

Quantum Cryptography
Quantum Pseudorandomness
Quantum Complexity Theory
Physics (Gapped 1-D local Hamiltonians)

Expanders to Quantum Expanders

Expanders

"Efficiently sample a uniform bitstring"

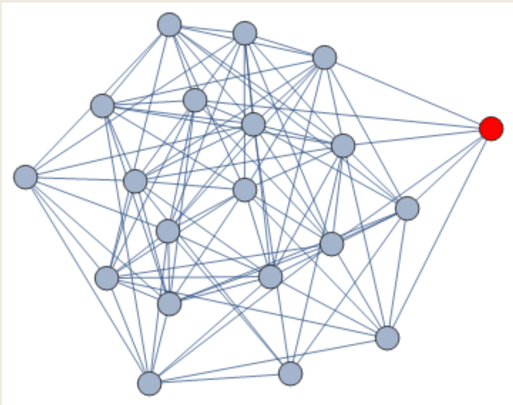


Image – 'rhermans' on Mathematica.SE

Expanders to Quantum Expanders

Expanders

"Efficiently sample a uniform bitstring"

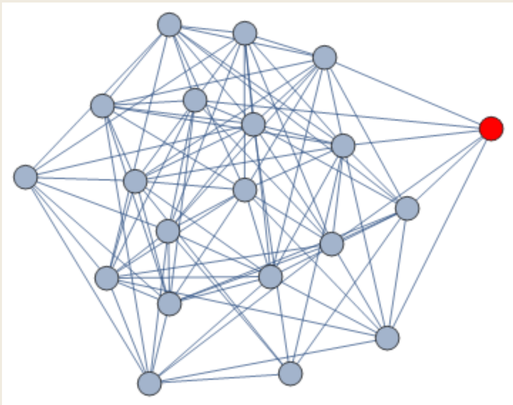


Image – 'rhermans' on Mathematica.SE

Expanders to Quantum Expanders

Expanders

"Efficiently sample a uniform bitstring"

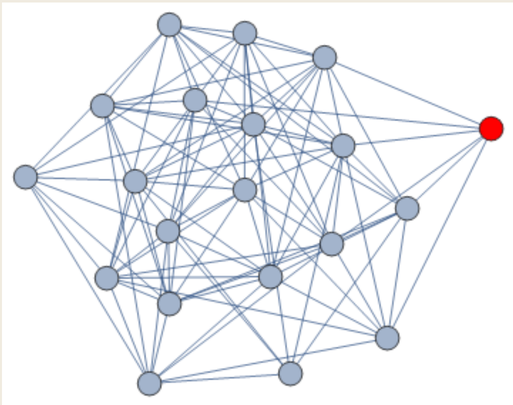
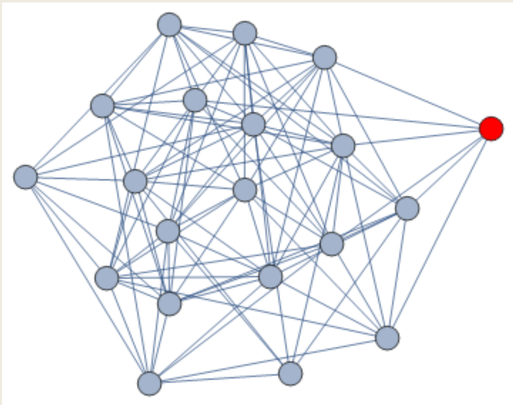


Image – 'rhermans' on Mathematica.SE

Expanders to Quantum Expanders

Expanders

"Efficiently sample a uniform bitstring"



Quantum Expanders

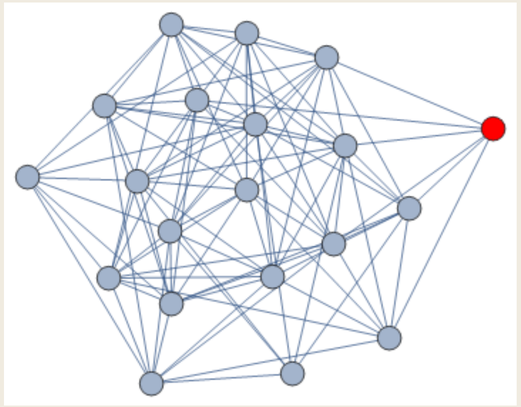
Efficiently prepare the "maximally mixed *state*"

Image – 'rhermans' on Mathematica.SE

Expanders to Quantum Expanders

Expanders

"Efficiently sample a uniform bitstring"



Quantum Expanders

Efficiently prepare the "maximally mixed *state*"

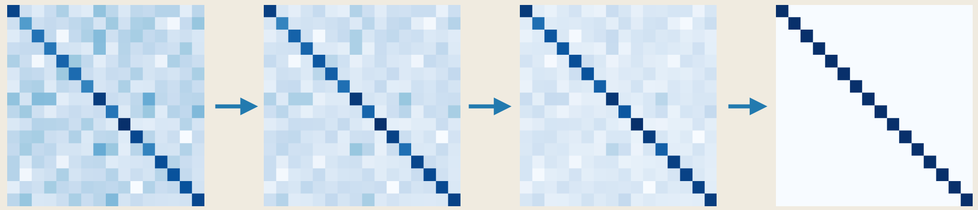
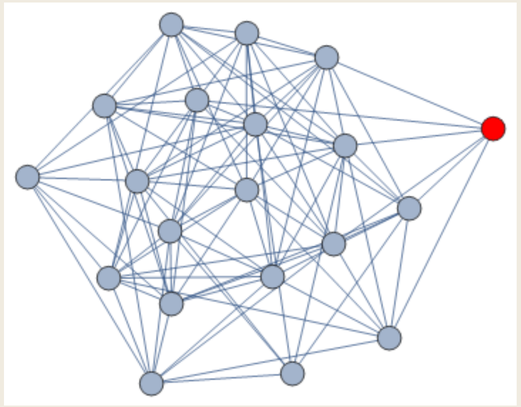


Image – 'rhermans' on Mathematica.SE

Expanders to Quantum Expanders

Expanders

"Efficiently sample a uniform bitstring"



Ramanujan (Optimal) Constructions

Quantum Expanders

Efficiently prepare the "maximally mixed *state*"

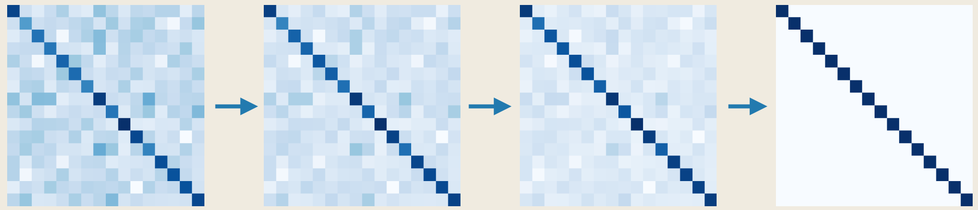
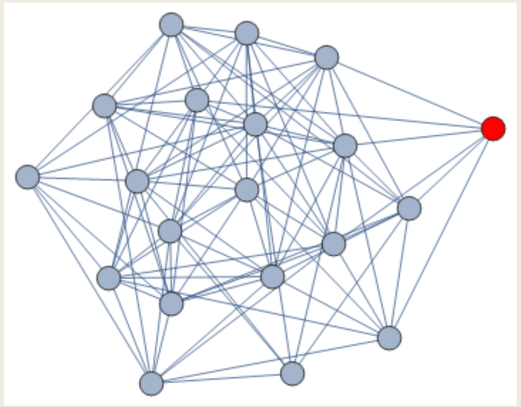


Image – 'rhermans' on Mathematica.SE

Expanders to Quantum Expanders

Expanders

"Efficiently sample a uniform bitstring"



Ramanujan (Optimal) Constructions

Quantum Expanders

Efficiently prepare the "maximally mixed *state*"

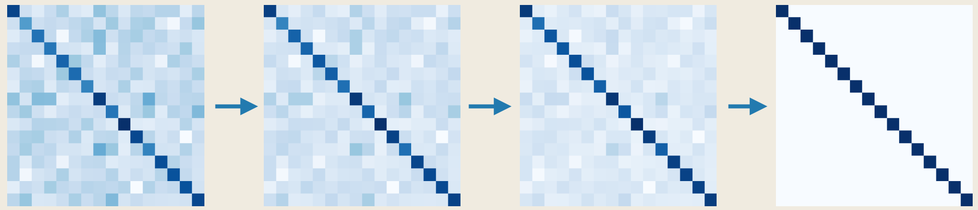
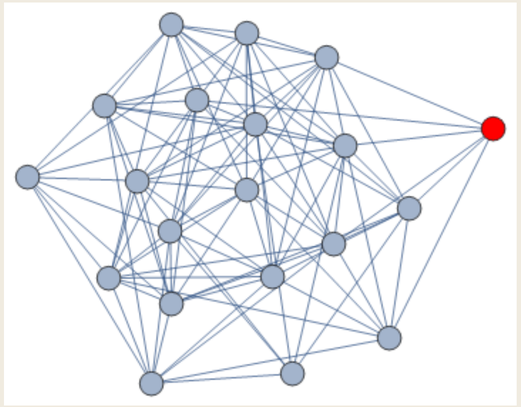


Image – 'rhermans' on Mathematica.SE

Expanders to Quantum Expanders

Expanders

"Efficiently sample a uniform bitstring"



Ramanujan (Optimal) Constructions

Quantum Expanders

Efficiently prepare the "maximally mixed *state*"

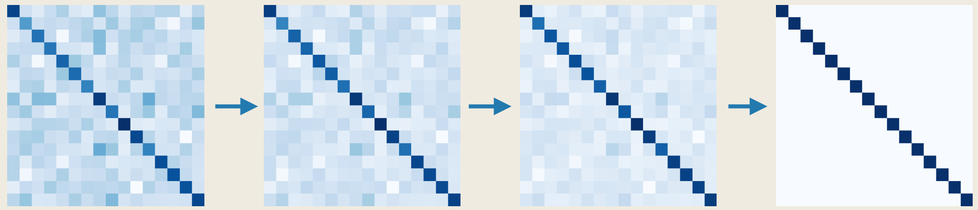
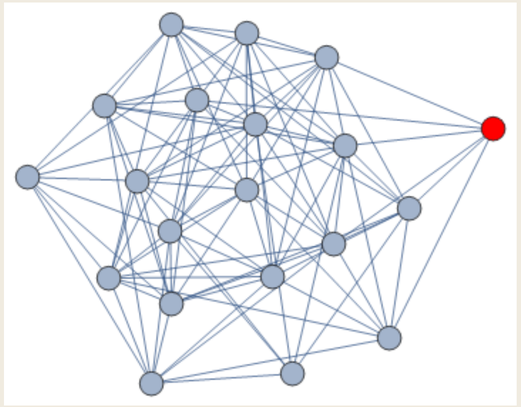


Image – 'rhermans' on Mathematica.SE

Expanders to Quantum Expanders

Expanders

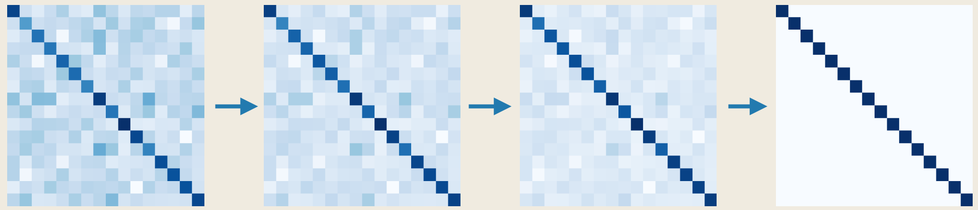
"Efficiently sample a uniform bitstring"



Ramanujan (Optimal) Constructions

Quantum Expanders

Efficiently prepare the "maximally mixed *state*"



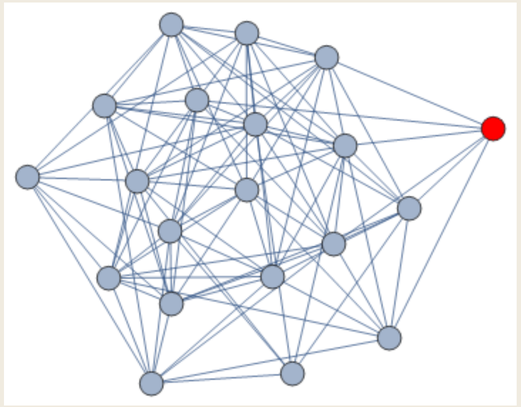
No Ramanujan or Near-Optimal Constructions

Image – 'rhermans' on Mathematica.SE

Expanders to Quantum Expanders

Expanders

"Efficiently sample a uniform bitstring"



Quantum Expanders

Efficiently prepare the "maximally mixed *state*"

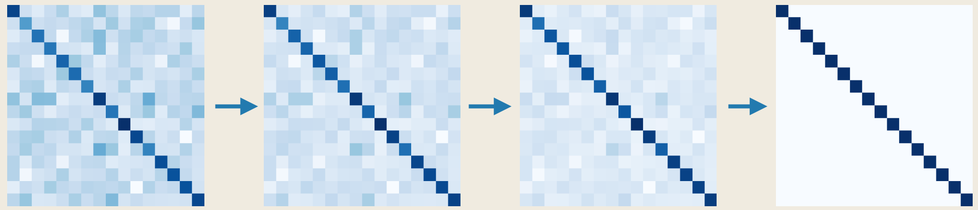
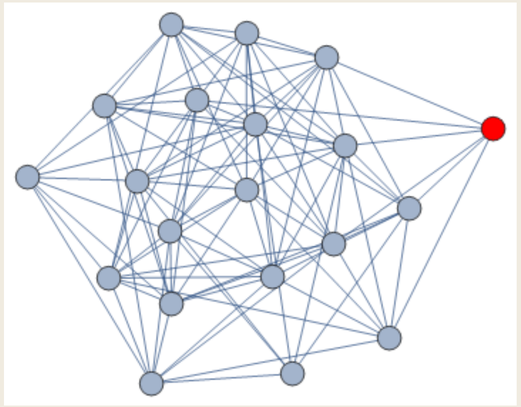


Image – 'rhermans' on Mathematica.SE

Expanders to Quantum Expanders

Expanders

"Efficiently sample a uniform bitstring"



Quantum Expanders

Efficiently prepare the "maximally mixed *state*"

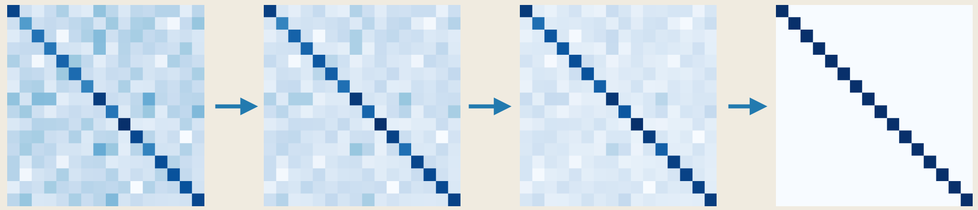
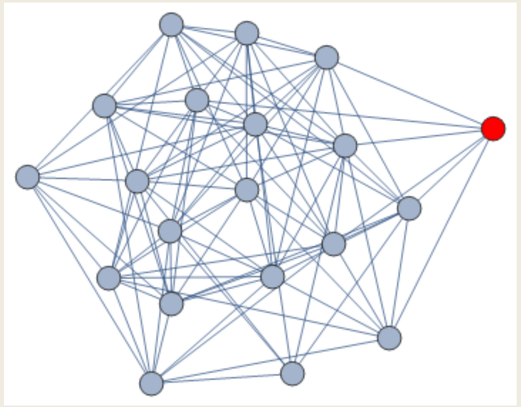


Image – 'rhermans' on Mathematica.SE

Expanders to Quantum Expanders

Expanders

"Efficiently sample a uniform bitstring"



[Harrow'07]

Expanders with
Permutation Symmetry

Quantum Expanders

Efficiently prepare the "maximally mixed *state*"

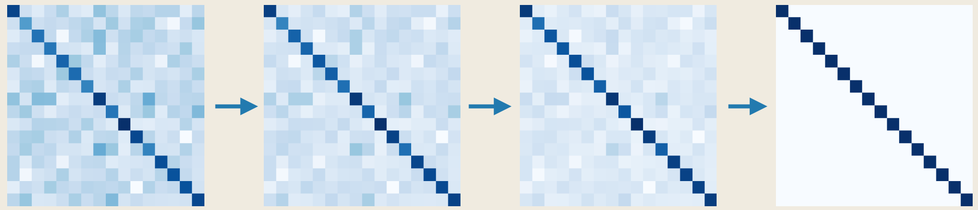
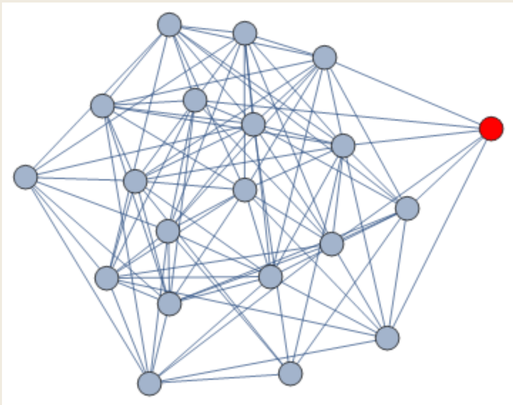


Image – 'rhermans' on Mathematica.SE

Expanders to Quantum Expanders

Expanders

"Efficiently sample a uniform bitstring"

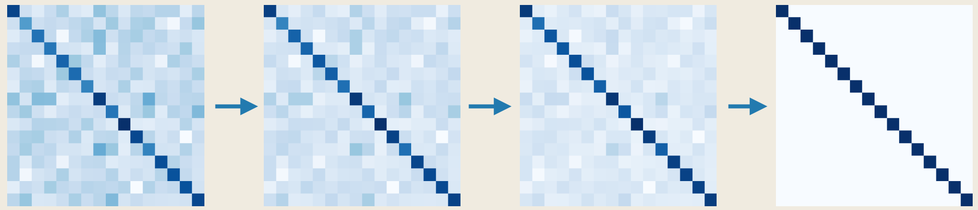


[Harrow'07]

Expanders with
Permutation Symmetry

Quantum Expanders

Efficiently prepare the "maximally mixed *state*"



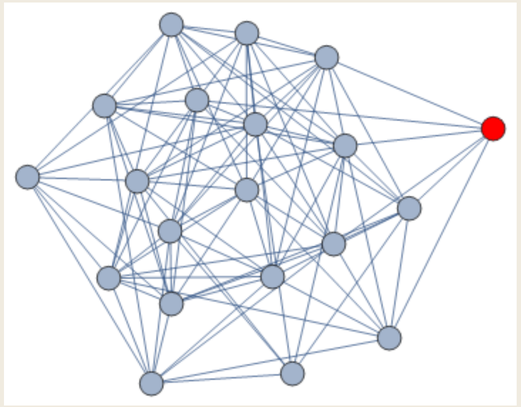
Near-optimal expanders with permutation symmetries?

Image – 'rhermans' on Mathematica.SE

Expanders to Quantum Expanders

Expanders

"Efficiently sample a uniform bitstring"

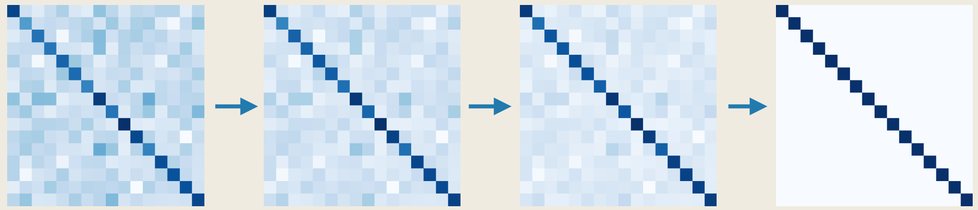


[Harrow'07]

Expanders with
Permutation Symmetry

Quantum Expanders

Efficiently prepare the "maximally mixed *state*"

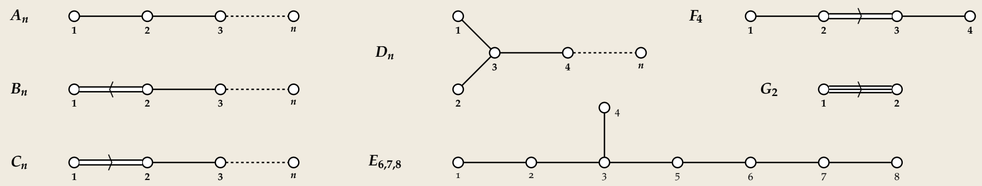


Near-optimal expanders with permutation symmetries?
Near-optimal expanders with *general* symmetries?

Image – 'rhermans' on Mathematica.SE

The Periodic Table Of Finite Simple Groups

Dynkin Diagrams of Simple Lie Algebras



$0, C_1, \mathbb{Z}_1$ **1** **1**																C_2 **2**
$A_1(4), A_1(5)$ A_5 **60**	$A_2(2)$ $A_1(7)$ **168**											$^2A_3(4)$ $B_2(3)$ **25 920**	$C_3(3)$ **4 585 351 680**	$D_4(2)$ **174 182 400**	$^2D_4(2^2)$ **197 406 720**	$G_2(2)'$ $^2A_2(9)$ **6 048**	C_3 **3**
$A_1(9), B_2(2)'$ A_6 **360**	$^2G_2(3)'$ $A_1(8)$ **504**											$B_2(4)$ **979 200**	$C_3(5)$ **228 501 000 000 000**	$D_4(3)$ **4 952 179 814 400**	$^2D_4(3^2)$ **10 151 968 619 520**	$^2A_2(16)$ **62 400**	C_5 **5**
A_7 **2 520**	$A_1(11)$ **660**	$E_6(2)$ **214 841 575 522 005 575 270 400**	$E_7(2)$ **7 997 476 042 075 799 759 100 487 262 680 802 918 400**	$E_8(2)$ **337 804 753 143 634 806 261 588 190 614 065 595 079 991 692 242 467 651 576 160 959 909 068 800 000**	$F_4(2)$ **3 311 126 603 366 400**	$G_2(3)$ **4 245 696**	$^3D_4(2^3)$ **211 341 312**	$^2E_6(2^2)$ **76 532 479 683 774 853 939 200**	$^2B_2(2^3)$ **29 120**	Tits* $^2F_4(2)'$ **17 971 200**	$^2G_2(3^3)$ **10 073 444 472**	$B_3(2)$ **1 451 520**	$C_4(3)$ **65 784 756 654 489 600**	$D_5(2)$ **23 499 295 948 800**	$^2D_5(2^2)$ **25 015 379 558 400**	$^2A_2(25)$ **126 000**	C_7 **7**
$A_3(2)$ A_8 **20 160**	$A_1(13)$ **1 092**	$E_6(3)$ **7 257 703 347 541 463 210 028 258 395 214 643 200**	$E_7(3)$ **1 271 375 236 818 136 742 240 479 753 139 021 644 554 379 203 770 766 254 617 395 200**	$E_8(3)$ **14 636 855 916 969 695 633 965 120 680 532 377 600**	$F_4(3)$ **5 734 420 792 816 671 844 761 600**	$G_2(4)$ **251 596 800**	$^3D_4(3^3)$ **20 560 831 566 912**	$^2E_6(3^2)$ **14 636 855 916 969 695 633 965 120 680 532 377 600**	$^2B_2(2^5)$ **32 537 600**	$^2F_4(2^3)$ **264 905 352 699 586 176 614 400**	$^2G_2(3^5)$ **49 825 657 439 340 552**	$B_2(5)$ **4 680 000**	$C_3(7)$ **273 457 218 604 953 600**	$D_4(5)$ **8 911 539 000 000 000 000**	$^2D_4(4^2)$ **67 536 471 195 648 000**	$^2A_3(9)$ **3 265 920**	C_{11} **11**
A_9 **181 440**	$A_1(17)$ **2 448**	$E_6(4)$ **85 528 710 781 342 640 103 833 619 055 142 765 466 746 880 000**	$E_7(4)$ **111 131 458 114 940 385 379 597 233 677 084 941 280 664 199 527 155 056 307 251 745 263 504 588 800 000 000**	$E_8(4)$ **191 797 261 142 671 737 734 639 757 897 312 906 421 357 307 454 216 557 313 358 441 448 166 298 469 477 195 648 512 000 441 348 266 298 469 477 195 648 512 000 492 217 656 441 474 908 160 000 000 000**	$F_4(4)$ **19 009 825 523 840 945 451 297 669 120 000**	$G_2(5)$ **5 859 000 000**	$^3D_4(4^3)$ **67 802 350 642 790 400**	$^2E_6(4^2)$ **85 696 576 147 617 709 485 896 772 387 584 983 695 360 000 000**	$^2B_2(2^7)$ **34 093 383 680**	$^2F_4(2^5)$ **1 318 633 155 799 591 447 702 161 609 782 722 560 000**	$^2G_2(3^7)$ **239 189 910 264 352 349 332 632**	$B_2(7)$ **138 297 600**	$C_3(9)$ **54 025 731 402 499 584 000**	$D_5(3)$ **1 289 512 799 941 305 139 200**	$^2D_4(5^2)$ **17 880 203 250 000 000 000**	$^2A_2(64)$ **5 515 776**	C_{13} **13**
A_n $\frac{n!}{2}$	$PSL_{n+1}(q), L_{n+1}(q)$ $A_n(q)$ $\frac{q^{n(n+1)/2}}{(n+1,q-1)}\prod_{i=1}^{n}(q^{i+1}-1)$	$E_6(q)$ $q^{36}(q^{12}-1)(q^9-1)(q^8-1)\frac{(q^6-1)(q^5-1)(q^2-1)}{(3,q-1)}$	$E_7(q)$ $\frac{q^{63}}{(2,q-1)}\prod_{i=2,6,8}^{8}(q^{2i}-1)$	$E_8(q)$ $q^{120}(q^{30}-1)(q^{24}-1)\frac{(q^{20}-1)(q^{18}-1)(q^{14}-1)}{(q^{12}-1)(q^8-1)(q^2-1)}$	$F_4(q)$ $q^{24}(q^{12}-1)(q^8-1)\frac{(q^6-1)}{(q^4-1)}$	$G_2(q)$ $q^6(q^6-1)(q^2-1)$	$^3D_4(q^3)$ $q^{12}(q^8+q^4+1)\frac{}{(q^6-1)(q^2-1)}$	$^2E_6(q^2)$ $q^{36}(q^{12}-1)(q^9+1)(q^8-1)\frac{(q^6+1)(q^5-1)(q^2-1)}{(3,q+1)}$	$^2B_2(2^{2n+1})$ $q^2(q^2+1)(q-1)$	$^2F_4(2^{2n+1})$ $q^{12}(q^6+1)(q^4-1)\frac{}{(q^4+1)(q-1)}$	$^2G_2(3^{2n+1})$ $q^3(q^3+1)(q-1)$	$O_{2n+1}(q), \Omega_{2n+1}(q)$ $B_n(q)$ $\frac{q^{n^2}}{(2,q-1)}\prod_{i=1}^{n}(q^{2i}-1)$	$PSp_{2n}(q)$ $C_n(q)$ $\frac{q^{n^2}}{(2,q-1)}\prod_{i=1}^{n}(q^{2i}-1)$	$O_{2n}^+(q)$ $D_n(q)$ $q^{n(n-1)}(q^n-1)\prod_{i=1}^{n-1}(q^{2i}-1)\frac{}{(4,q^n-1)}$	$O_{2n}^-(q)$ $^2D_n(q^2)$ $q^{n(n-1)}(q^n+1)\prod_{i=1}^{n-1}(q^{2i}-1)\frac{}{(4,q^n+1)}$	$PSU_{n+1}(q)$ $^2A_n(q^2)$ $\frac{q^{n(n+1)/2}}{(n+1,q+1)}\prod_{i=2}^{n+1}(q^i-(-1)^i)$	$\mathbb{Z}_p$ C_p p

- Alternating Groups
- Classical Chevalley Groups
- Chevalley Groups
- Classical Steinberg Groups
- Steinberg Groups
- Suzuki Groups
- Ree Groups and Tits Group*
- Sporadic Groups
- Cyclic Groups

*The Tits group $^2F_4(2)'$ is not a group of Lie type, but is the (index **2**) commutator subgroup of $^2F_4(2)$. It is usually given honorary Lie type status.

The groups starting on the second row are the classical groups. The sporadic suzuki group is unrelated to the families of Suzuki groups.

Copyright © 2012 Ivan Andrus.

Alternates†
Symbol
Order‡

†For sporadic groups and families, alternate names in the upper left are other names by which they may be known. For specific non-sporadic groups these are used to indicate isomorphims. All such isomorphisms appear on the table except the family $B_n(2^m) \cong C_n(2^m)$.

‡Finite simple groups are determined by their order with the following exceptions:
$B_n(q)$ and $C_n(q)$ for q odd, $n > 2$;
$A_8 \cong A_3(2)$ and $A_2(4)$ of order **20160**.

						$J(1), J(11)$ J_1 **175 560**	HJ J_2 **604 800**	HJM J_3 **50 232 960**	J_4 **86 775 571 046 077 562 880**	HS **44 352 000**	McL **898 128 000**	Fr, HHM, HTH He **4 030 387 200**	Ru **145 926 144 000**
M_{11} **7 920**	M_{12} **95 040**	M_{22} **443 520**	M_{23} **10 200 960**	M_{24} **244 823 040**									
Sz Suz **448 345 497 600**	$O'NS, O{-}S$ $O'N$ **460 815 505 920**	$\cdot 3$ Co_3 **495 766 656 000**	$\cdot 2$ Co_2 **42 305 421 312 000**	$\cdot 1$ Co_1 **4 157 776 806 543 360 000**	F_5, D HN **273 030 912 000 000**	LyS Ly **51 765 179 004 000 000**	F_3, E Th **90 745 943 887 872 000**	$M(22)$ Fi_{22} **64 561 751 654 400**	$M(23)$ Fi_{23} **4 089 470 473 293 004 800**	$F_{3+}, M(24)'$ Fi'_{24} **1 255 205 709 190 661 721 292 800**	F_2 B **4 154 781 481 226 426 191 177 580 544 000 000**	F_1, M_1 M **808 017 424 794 512 875 886 459 904 961 710 757 005 754 368 000 000 000**	

Image Credits:

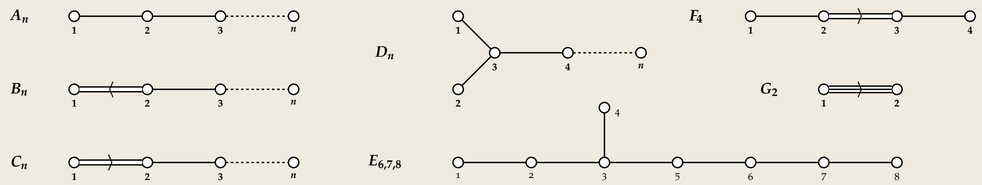
Ivan Andrus

The Periodic Table Of Finite Simple Groups

Cyclic

$0, C_1, \mathbb{Z}_1$
1
1

Dynkin Diagrams of Simple Lie Algebras



$A_1(4), A_1(5)$ **A_5** **60**	$A_2(2)$ **$A_1(7)$** **168**											$^2A_3(4)$ **$B_2(3)$** **25 920**	**$C_3(3)$** **4 585 351 680**	**$D_4(2)$** **174 182 400**	**$^2D_4(2^2)$** **197 406 720**	$G_2(2)'$ **$^2A_2(9)$** **6 048**
$A_1(9), B_2(2)'$ **A_6** **360**	$^2G_2(3)'$ **$A_1(8)$** **504**											**$B_2(4)$** **979 200**	**$C_3(5)$** **228 501 000 000 000**	**$D_4(3)$** **4 952 179 814 400**	**$^2D_4(3^2)$** **10 151 968 619 520**	**$^2A_2(16)$** **62 400**
A_7 **2 520**	**$A_1(11)$** **660**	**$E_6(2)$** **214 841 575 522 005 575 270 400**	**$E_7(2)$** **7 997 476 042 075 799 759 100 487 262 680 802 918 400**	**$E_8(2)$** **337 804 753 143 634 806 261 588 196 614 065 595 079 991 692 242 467 651 576 160 959 909 068 800 000**	**$F_4(2)$** **3 311 126 603 366 400**	**$G_2(3)$** **4 245 696**	**$^3D_4(2^3)$** **211 341 312**	**$^2E_6(2^2)$** **76 532 479 683 774 853 939 200**	**$^2B_2(2^3)$** **29 120**	Tits* **$^2F_4(2)'$** **17 971 200**	**$^2G_2(3^3)$** **10 073 444 472**	**$B_3(2)$** **1 451 520**	**$C_4(3)$** **65 784 756 654 489 600**	**$D_5(2)$** **23 499 295 948 800**	**$^2D_5(2^2)$** **25 015 379 558 400**	**$^2A_2(25)$** **126 000**
$A_3(2)$ **A_8** **20 160**	**$A_1(13)$** **1 092**	**$E_6(3)$** **7 257 703 347 541 463 210 028 258 395 214 643 200**	**$E_7(3)$** **1 271 375 236 818 136 742 240 479 753 139 021 644 554 379 203 770 766 254 617 395 200**	**$E_8(3)$** **14 650 052 912 703 932 331 899 032 409 971 660 352 140 886 784 948 153 101 922 649 391 826 616 580 150 109 976 711 383 949 982 164 694 680 626 420 940 800 000**	**$F_4(3)$** **5 734 420 792 816 671 844 761 600**	**$G_2(4)$** **251 596 800**	**$^3D_4(3^3)$** **20 560 831 566 912**	**$^2E_6(3^2)$** **14 636 855 916 969 695 633 965 120 680 532 377 600**	**$^2B_2(2^5)$** **32 537 600**	**$^2F_4(2^3)$** **264 905 352 699 586 176 614 400**	**$^2G_2(3^5)$** **49 825 657 439 340 552**	**$B_2(5)$** **4 680 000**	**$C_3(7)$** **273 457 218 604 953 600**	**$D_4(5)$** **8 911 539 000 000 000 000**	**$^2D_4(4^2)$** **67 536 471 195 648 000**	**$^2A_3(9)$** **3 265 920**
A_9 **181 440**	**$A_1(17)$** **2 448**	**$E_6(4)$** **85 528 710 781 342 640 103 833 619 055 142 765 466 746 880 000**	**$E_7(4)$** **111 131 458 114 940 383 379 597 233 677 884 941 280 664 199 527 155 056 307 251 745 263 504 588 800 000 000**	**$E_8(4)$** **191 797 292 142 671 737 754 639 757 897 512 936 421 357 307 454 216 557 313 358 497 191 826 616 580 150 109 976 711 383 161 548 268 298 469 677 195 648 512 000 492 217 656 441 474 908 160 000 000 000**	**$F_4(4)$** **19 009 825 523 840 945 451 297 669 120 000**	**$G_2(5)$** **5 859 000 000**	**$^3D_4(4^3)$** **67 802 350 642 790 400**	**$^2E_6(4^2)$** **85 696 576 147 617 709 485 896 772 387 584 983 695 360 000 000**	**$^2B_2(2^7)$** **34 093 383 680**	**$^2F_4(2^5)$** **1 318 633 155 799 591 447 702 161 609 782 722 560 000**	**$^2G_2(3^7)$** **239 189 910 264 352 349 332 632**	**$B_2(7)$** **138 297 600**	**$C_3(9)$** **54 025 731 402 499 584 000**	**$D_5(3)$** **1 289 512 799 941 305 139 200**	**$^2D_4(5^2)$** **17 880 203 250 000 000 000**	**$^2A_2(64)$** **5 515 776**
$PSL_{n+1}(q), L_{n+1}(q)$ **$A_n(q)$** **$\frac{n!}{2}$**		**$E_6(q)$**	**$E_7(q)$**	**$E_8(q)$**	**$F_4(q)$**	**$G_2(q)$**	**$^3D_4(q^3)$**	**$^2E_6(q^2)$**	**$^2B_2(2^{2n+1})$**	**$^2F_4(2^{2n+1})$**	**$^2G_2(3^{2n+1})$**	$O_{2n+1}(q), \Omega_{2n+1}(q)$ **$B_n(q)$**	$PSp_{2n}(q)$ **$C_n(q)$**	$O_{2n}^+(q)$ **$D_n(q)$**	$O_{2n}^-(q)$ **$^2D_n(q^2)$**	$PSU_{n+1}(q)$ **$^2A_n(q^2)$**
																$\mathbb{Z}_p$ **C_p** **p**

Alternating Groups
Classical Chevalley Groups
Chevalley Groups
Classical Steinberg Groups
Steinberg Groups
Suzuki Groups
Ree Groups and Tits Group*
Sporadic Groups
Cyclic Groups

*The Tits group $^2F_4(2)'$ is not a group of Lie type, but is the (index **2**) commutator subgroup of $^2F_4(2)$. It is usually given honorary Lie type status.

The groups starting on the second row are the classical groups. The sporadic suzuki group is unrelated to the families of Suzuki groups.

Copyright © 2012 Ivan Andrus.

Alternates†
Symbol
Order‡

†For sporadic groups and families, alternate names in the upper left are other names by which they may be known. For specific non-sporadic groups these are used to indicate isomorphims. All such isomorphisms appear on the table except the family $B_n(2^m) \cong C_n(2^m)$.

‡Finite simple groups are determined by their order with the following exceptions:
$B_n(q)$ and $C_n(q)$ for q odd, $n > 2$;
$A_8 \cong A_3(2)$ and $A_2(4)$ of order **20160**.

M_{11} **7 920**	**M_{12}** **95 040**	**M_{22}** **443 520**	**M_{23}** **10 200 960**	**M_{24}** **244 823 040**	$J(1), J(11)$ **J_1** **175 560**	HJ **J_2** **604 800**	HJM **J_3** **50 232 960**	**J_4** **86 775 571 046 077 562 880**	**HS** **44 352 000**	**McL** **898 128 000**	Fr, HHM, HTH **He** **4 030 387 200**	**Ru** **145 926 144 000**

Sz **Suz** **448 345 497 600**	$O'NS, O–S$ **$O'N$** **460 815 505 920**	·3 **Co_3** **495 766 656 000**	·2 **Co_2** **42 305 421 312 000**	·1 **Co_1** **4 157 776 806 543 360 000**	F_5, D **HN** **273 030 912 000 000**	LyS **Ly** **51 765 179 004 000 000**	F_3, E **Th** **90 745 943 887 872 000**	$M(22)$ **Fi_{22}** **64 561 751 654 400**	$M(23)$ **Fi_{23}** **4 089 470 473 293 004 800**	$F_{3+}, M(24)'$ **Fi'_{24}** **1 255 205 709 190 661 721 292 800**	F_2 **B** **4 154 781 481 226 426 191 177 580 544 000 000**	F_1, M_1 **M** **808 017 424 794 512 875 886 459 904 961 710 757 005 754 368 000 000 000**

Image Credits:

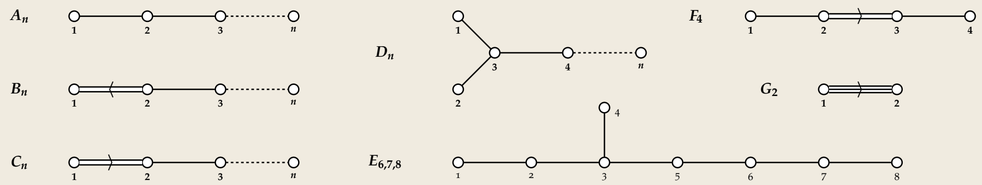
Ivan Andrus

The Periodic Table Of Finite Simple Groups

Permutation

Cyclic

Dynkin Diagrams of Simple Lie Algebras



$0, C_1, \mathbb{Z}_1$ **1** 1																
$A_1(4), A_1(5)$ A_5 60	$A_2(2)$ $A_1(7)$ 168											$^2A_3(4)$ $B_2(3)$ 25 920	$C_3(3)$ 4 585 351 680	$D_4(2)$ 174 182 400	$^2D_4(2^2)$ 197 406 720	$G_2(2)'$ $^2A_2(9)$ 6 048	C_2 2
$A_1(9), B_2(2)'$ A_6 360	$^2G_2(3)'$ $A_1(8)$ 504											$B_2(4)$ 979 200	$C_3(5)$ 228 501 000 000 000	$D_4(3)$ 4 952 179 814 400	$^2D_4(3^2)$ 10 151 968 619 520	$^2A_2(16)$ 62 400	C_5 5
A_7 2 520	$A_1(11)$ 660	$E_6(2)$ 214 841 575 522 005 575 270 400	$E_7(2)$ 7 997 476 042 075 799 759 100 487 262 680 802 918 400	$E_8(2)$ 337 804 753 143 634 806 261 588 190 614 065 595 079 991 692 242 467 651 576 160 959 909 068 800 000	$F_4(2)$ 3 311 126 603 366 400	$G_2(3)$ 4 245 696	$^3D_4(2^3)$ 211 341 312	$^2E_6(2^2)$ 76 532 479 683 774 853 939 200	$^2B_2(2^3)$ 29 120	Tits* $^2F_4(2)'$ 17 971 200	$^2G_2(3^3)$ 10 073 444 472	$B_3(2)$ 1 451 520	$C_4(3)$ 65 784 756 654 489 600	$D_5(2)$ 23 499 295 948 800	$^2D_5(2^2)$ 25 015 379 558 400	$^2A_2(25)$ 126 000	C_7 7
$A_3(2)$ A_8 20 160	$A_1(13)$ 1 092	$E_6(3)$ 7 257 703 347 541 463 210 028 258 395 214 643 200	$E_7(3)$ 1 271 375 236 818 136 742 240 479 753 139 021 644 554 379 203 770 766 254 617 395 200	$E_8(3)$ 14 636 855 916 969 695 633 965 120 680 532 377 600	$F_4(3)$ 5 734 420 792 816 671 844 761 600	$G_2(4)$ 251 596 800	$^3D_4(3^3)$ 20 560 831 566 912	$^2E_6(3^2)$ 14 636 855 916 969 695 633 965 120 680 532 377 600	$^2B_2(2^5)$ 32 537 600	$^2F_4(2^3)$ 264 905 352 699 586 176 614 400	$^2G_2(3^5)$ 49 825 657 439 340 552	$B_2(5)$ 4 680 000	$C_3(7)$ 273 457 218 604 953 600	$D_4(5)$ 8 911 539 000 000 000 000	$^2D_4(4^2)$ 67 536 471 195 648 000	$^2A_3(9)$ 3 265 920	C_{11} 11
A_9 181 440	$A_1(17)$ 2 448	$E_6(4)$ 85 528 710 781 342 640 103 833 619 055 142 765 466 746 880 000	$E_7(4)$ 111 131 458 114 940 385 379 597 233 677 884 941 280 664 199 527 155 056 307 251 745 263 504 588 800 000 000	$E_8(4)$ 191 797 261 142 671 737 754 639 757 897 312 906 421 357 307 404 216 557 313 358 341 348 246 298 469 677 195 648 512 000 492 217 656 441 474 908 160 000 000 000	$F_4(4)$ 19 009 825 523 840 945 451 297 669 120 000	$G_2(5)$ 5 859 000 000	$^3D_4(4^3)$ 67 802 350 642 790 400	$^2E_6(4^2)$ 85 696 576 147 617 709 485 896 772 387 584 983 695 360 000 000	$^2B_2(2^7)$ 34 093 383 680	$^2F_4(2^5)$ 1 318 633 155 799 591 447 702 161 609 782 722 560 000	$^2G_2(3^7)$ 239 189 910 264 352 349 332 632	$B_2(7)$ 138 297 600	$C_3(9)$ 54 025 731 402 499 584 000	$D_5(3)$ 1 289 512 799 941 305 139 200	$^2D_4(5^2)$ 17 880 203 250 000 000 000	$^2A_3(64)$ 5 515 776	C_{13} 13
A_n $\frac{n!}{2}$	$PSL_{n+1}(q), L_{n+1}(q)$ $A_n(q)$ $\frac{q^{n(n+1)/2}}{(n+1,q-1)}\prod_{i=1}^{n}(q^{i+1}-1)$	$E_6(q)$ $\frac{q^{36}-1)(q^9-1)(q^8-1)}{(q^6-1)(q^5-1)(q^2-1)}$ $(3,q-1)$	$E_7(q)$ $\frac{q^{63}}{(2,q-1)}\prod_{i=2,6,8}^{18}(q^{2i}-1)$	$E_8(q)$ $\frac{q^{120}(q^{30}-1)(q^{24}-1)}{(q^{20}-1)(q^{18}-1)(q^{14}-1)}$ $(q^{12}-1)(q^8-1)(q^2-1)$	$F_4(q)$ $q^{24}(q^{12}-1)(q^8-1)$ $(q^6-1)(q^2-1)$	$G_2(q)$ $q^6(q^6-1)(q^2-1)$	$^3D_4(q^3)$ $q^{12}(q^8+q^4+1)$ $(q^4-1)(q^2-1)$	$^2E_6(q^2)$ $\frac{q^{36}(q^{12}-1)(q^9+1)(q^8-1)}{(q^6-1)(q^5+1)(q^2-1)}$ $(3,q+1)$	$^2B_2(2^{2n+1})$ $q^2(q^2+1)(q-1)$	$^2F_4(2^{2n+1})$ $q^{12}(q^6+1)(q^4-1)$ $(q^4+1)(q-1)$	$^2G_2(3^{2n+1})$ $q^3(q^3+1)(q-1)$	$O_{2n+1}(q), \Omega_{2n+1}(q)$ $B_n(q)$ $\frac{q^{n^2}}{(2,q-1)}\prod_{i=1}^{n}(q^{2i}-1)$	$PSp_{2n}(q)$ $C_n(q)$ $\frac{q^{n^2}}{(2,q-1)}\prod_{i=1}^{n}(q^{2i}-1)$	$O_{2n}^+(q)$ $D_n(q)$ $\frac{q^{n(n-1)}(q^n-1)}{(4,q^n-1)}\prod_{i=1}^{n-1}(q^{2i}-1)$	$O_{2n}^-(q)$ $^2D_n(q^2)$ $\frac{q^{n(n-1)}(q^n+1)}{(4,q^n+1)}\prod_{i=1}^{n-1}(q^{2i}-1)$	$PSU_{n+1}(q)$ $^2A_n(q^2)$ $\frac{q^{n(n+1)/2}}{(n+1,q+1)}\prod_{i=2}^{n+1}(q^i-(-1)^i)$	$\mathbb{Z}_p$ C_p p

- Alternating Groups
- Classical Chevalley Groups
- Chevalley Groups
- Classical Steinberg Groups
- Steinberg Groups
- Suzuki Groups
- Ree Groups and Tits Group*
- Sporadic Groups
- Cyclic Groups

*The Tits group $^2F_4(2)'$ is not a group of Lie type, but is the (index **2**) commutator subgroup of $^2F_4(2)$. It is usually given honorary Lie type status.

The groups starting on the second row are the classical groups. The sporadic suzuki group is unrelated to the families of Suzuki groups.

Copyright © 2012 Ivan Andrus.

Alternates†
Symbol
Order‡

†For sporadic groups and families, alternate names in the upper left are other names by which they may be known. For specific non-sporadic groups these are used to indicate isomorphims. All such isomorphisms appear on the table except the family $B_n(2^m) \cong C_n(2^m)$.

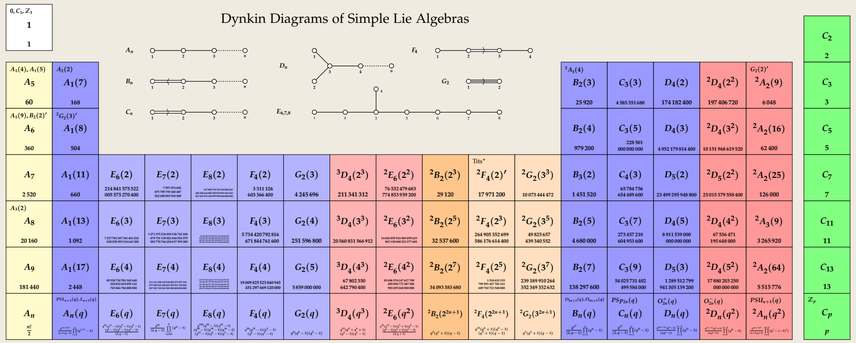
‡Finite simple groups are determined by their order with the following exceptions:
$B_n(q)$ and $C_n(q)$ for q odd, $n > 2$;
$A_8 \cong A_3(2)$ and $A_2(4)$ of order **20160**.

						$J(1), J(11)$ J_1 175 560	HJ J_2 604 800	HJM J_3 50 232 960	J_4 86 775 571 046 077 562 880	HS 44 352 000	McL 898 128 000	Fr, HHM, HTH He 4 030 387 200	Ru 145 926 144 000
M_{11} 7 920	M_{12} 95 040	M_{22} 443 520	M_{23} 10 200 960	M_{24} 244 823 040									
Sz Suz 448 345 497 600	$O'NS, O–S$ $O'N$ 460 815 505 920	$\cdot 3$ Co_3 495 766 656 000	$\cdot 2$ Co_2 42 305 421 312 000	$\cdot 1$ Co_1 4 157 776 806 543 360 000	F_5, D HN 273 030 912 000 000	LyS Ly 51 765 179 004 000 000	F_3, E Th 90 745 943 887 872 000	$M(22)$ Fi_{22} 64 561 751 654 400	$M(23)$ Fi_{23} 4 089 470 473 293 004 800	$F_{3+}, M(24)'$ Fi'_{24} 1 255 205 709 190 661 721 292 800	F_2 B 4 154 781 481 226 426 191 177 580 544 000 000	F_1, M_1 M 808 017 424 794 512 875 886 459 904 961 710 757 005 754 368 000 000 000	

Image Credits:

Ivan Andrus

The Periodic Table Of Finite Simple Groups



$\mathbb{Z}_1$, C_1, Z_1 **1** 1															C_2 **2**
$A_1(4)$, $A_1(5)$ A_5 **60**	$A_1(2)$ $A_1(7)$ **168**										${}^2A_3(4)$ $B_2(3)$ **25 920**	$C_3(3)$ **4 585 351 680**	$D_4(2)$ **174 182 400**	${}^2D_4(2^2)$ **197 406 720**	$G_2(2)'$ ${}^2A_2(9)$ **6 048**	C_3 **3**
$A_1(9)$, $B_2(2)'$ A_6 **360**	${}^2G_2(3)'$ $A_1(8)$ **504**										$B_2(4)$ **979 200**	$C_3(5)$ **228 501 000 000 000**	$D_4(3)$ **4 952 379 014 400**	${}^2D_4(3^2)$ **10 151 960 619 520**	${}^2A_2(16)$ **62 400**	C_5 **5**
A_7 **2 520**	$A_1(11)$ **660**	$E_6(2)$ **214 641 375 522 005 375 270 400**	$E_7(2)$ **[illegible]**	$E_8(2)$ **[illegible]**	$F_4(2)$ **3 311 126 603 366 400**	$G_2(3)$ **4 245 696**	${}^3D_4(2^3)$ **211 341 312**	${}^2E_6(2^2)$ **76 532 479 683 774 853 939 200**	${}^2B_2(2^3)$ **29 120**	Tits* ${}^2F_4(2)'$ **17 971 200**	${}^2G_2(3^3)$ **10 073 444 472**	$B_3(2)$ **1 451 520**	$C_4(3)$ **65 784 756 654 609 600**	$D_5(2)$ **23 499 295 948 800**	${}^2D_5(2^2)$ **254 013 379 554 400**	${}^2A_2(25)$ **126 000**	C_7 **7**
$A_1(2)$ A_8 **20 160**	$A_1(13)$ **1 092**	$E_6(3)$ **[illegible]**	$E_7(3)$ **[illegible]**	$E_8(3)$ **[illegible]**	$F_4(3)$ **5 734 420 792 816 671 844 761 600**	$G_2(4)$ **251 596 800**	${}^3D_4(3^3)$ **20 560 031 364 912**	${}^2E_6(3^2)$ **[illegible]**	${}^2B_2(2^5)$ **32 537 600**	${}^2F_4(2^3)$ **264 905 352 699 586 176 614 400**	${}^2G_2(3^5)$ **48 825 657 639 360 532**	$B_2(5)$ **4 680 000**	$C_3(7)$ **273 457 218 604 953 600**	$D_5(3)$ **8 911 539 000 000 000 000**	${}^2D_4(4^2)$ **67 336 471 195 648 000**	${}^2A_3(9)$ **3 265 920**	C_{11} **11**
A_9 **181 440**	$A_1(17)$ **2 448**	$E_6(4)$ **[illegible]**	$E_7(4)$ **[illegible]**	$E_8(4)$ **[illegible]**	$F_4(4)$ **19 009 625 323 840 943 455 297 669 120 000**	$G_2(5)$ **5 859 000 000**	${}^3D_4(4^3)$ **67 802 350 642 790 400**	${}^2E_6(4^2)$ **[illegible]**	${}^2B_2(2^7)$ **34 093 383 680**	${}^2F_4(2^5)$ **[illegible]**	${}^2G_2(3^7)$ **239 189 910 264 352 349 332 632**	$B_2(7)$ **138 297 600**	$C_3(9)$ **54 025 733 402 499 584 000**	$D_5(3)$ **1 209 312 799 961 305 159 200**	${}^2D_4(5^2)$ **17 860 203 250 000 000 000**	${}^2A_2(64)$ **5 515 776**	C_{13} **13**
A_n $\frac{n!}{2}$	$PSL_{n+1}(q)$, $L_{n+1}(q)$ $A_n(q)$ $\frac{1}{\gcd(n+1,q-1)}\prod_{i=2}^{n+1}(q^i-1)$	$E_6(q)$ $\frac{(q^{12}-1)(q^9-1)(q^8-1)(q^5-1)}{\gcd(3,q-1)}$	$E_7(q)$ $\frac{(q^{18}-1)(q^{14}-1)(q^{12}-1)(q^8-1)(q^6-1)(q^2-1)}{\gcd(2,q-1)}$	$E_8(q)$ $(q^{30}-1)(q^{24}-1)(q^{20}-1)(q^{18}-1)(q^{14}-1)(q^{12}-1)(q^8-1)(q^2-1)$	$F_4(q)$ $(q^{12}-1)(q^8-1)(q^6-1)(q^2-1)$	$G_2(q)$ $(q^6-1)(q^2-1)$	${}^3D_4(q^3)$ $q^{12}(q^8+q^4+1)$ $(q^8+q^4+1)(q^6-1)(q^2-1)$	${}^2E_6(q^2)$ $\frac{(q^{12}-1)(q^9+1)(q^8-1)(q^5+1)}{\gcd(3,q+1)}$	${}^2B_2(2^{2n+1})$ $q^2(q^2+1)(q-1)$	${}^2F_4(2^{2n+1})$ $q^{24}(q^6+1)(q^4-1)(q^3+1)(q-1)$	${}^2G_2(3^{2n+1})$ $q^6(q^2+1)(q-1)$	$O_{2m+1}(q)$, $O_{2m+1}(q)$ $B_n(q)$ $\frac{1}{\gcd(2,q-1)}\prod_{i=1}^{n}(q^{2i}-1)$	$PSp_{2m}(q)$ $C_n(q)$ $\frac{1}{\gcd(2,q-1)}\prod_{i=1}^{n}(q^{2i}-1)$	$O_{2n}^+(q)$ $D_n(q)$ $\frac{1}{\gcd(4,q^n-1)}\prod_{i=1}^{n-1}(q^{2i}-1)$	$O_{2n}^-(q)$ ${}^2D_n(q^2)$ $\frac{1}{\gcd(4,q^n+1)}\prod_{i=1}^{n-1}(q^{2i}-1)$	$PSU_{n+1}(q)$ ${}^2A_n(q^2)$ $\frac{1}{\gcd(n+1,q+1)}\prod_{i=2}^{n+1}(q^i-(-1)^i)$	$\mathbb{Z}_p$ C_p p

						$J(1)$, $J(11)$ J_1 **175 560**	HJ J_2 **604 800**	HJM J_3 **50 232 960**	J_4 **86 775 571 046 077 562 880**	HS **44 352 000**	McL **898 128 000**	s, HHM, HTH He **4 030 387 200**	Ru **145 926 144 000**
M_{11} **7 920**	M_{12} **95 040**	M_{22} **443 520**	M_{23} **10 200 960**	M_{24} **244 823 040**									
Sz **448 345 497 600**	$O'N$, O-S **460 815 505 920**	-3 Co_3 **495 766 656 000**	-2 Co_2 **42 305 421 312 000**	-1 Co_1 **4 157 776 806 543 360 000**	F_5, D HN **273 030 912 000 000**	Ly **51 765 179 004 000 000**	Th **90 745 943 887 872 000**	$M(22)$ Fi_{22} **64 561 751 654 400**	$M(23)$ Fi_{23} **4 089 470 473 293 004 800**	F_{3+}, $M(24)'$ Fi'_{24} **1 255 205 709 190 661 721 292 800**	F_2 B **4 154 781 481 226 426 191 177 580 544 000 000**	F_1, M_1 M **808 017 424 794 512 875 886 459 904 961 710 757 005 754 368 000 000**	

- Alternating Groups
- Classical Chevalley Groups
- Chevalley Groups
- Classical Steinberg Groups
- Steinberg Groups
- Suzuki Groups
- Ree Groups and Tits Group*
- Sporadic Groups
- Cyclic Groups

*The Tits group ${}^2F_4(2)'$ is not a group of Lie type, but is the (index 2) commutator subgroup of ${}^2F_4(2)$. It is usually given honorary Lie type status.

The groups starting on the second row are the classical groups. The sporadic swedi group is unrelated to the families of Suzuki groups.

Copyright © 2012 Ivan Andrus.

Alternates†
Symbol
Order‡

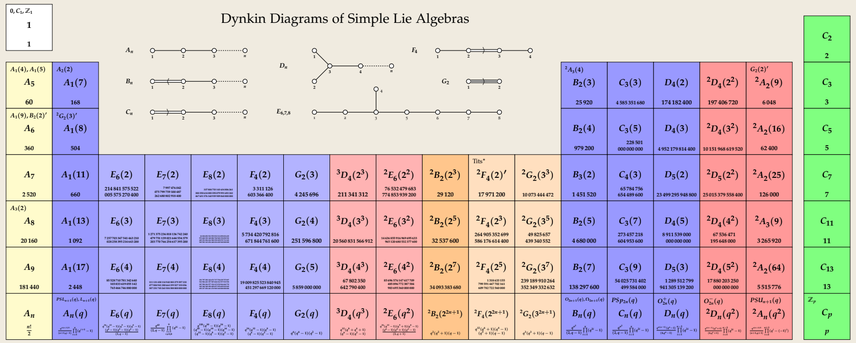
†For sporadic groups and families, alternate names in the upper left are other names by which they may be known. For specific non-sporadic groups, the name listed is the most frequently used. All non-isomorphic groups on the table except the families $B_n(2^m)$ is ${}^2C_n(2^m)$.

‡Finite simple groups are determined by their order with the following exceptions:
$B_n(q)$ and $C_n(q)$ for q odd, $n > 2$;
A_7, $A_1(2)$ and $A_2(4)$ of order 20160.

Definitions ▶ Quantum Error Correction ▶ Other Pseudorandom Objects ▶ Future

Image Credits: Ivan Andrus

The Periodic Table Of Finite Simple Groups



$\mathbb{Z}_1$, C_1, Z_1 **1** 1															C_2 **2**
$A_1(4)$, $A_1(5)$ A_5 **60**	$A_1(2)$ $A_1(7)$ **168**										${}^2A_3(4)$ $B_2(3)$ **25 920**	$C_3(3)$ **4 585 351 680**	$D_4(2)$ **174 182 400**	${}^2D_4(2^2)$ **197 406 720**	$G_2(2)'$ ${}^2A_2(9)$ **6 048**	C_3 **3**
$A_1(9)$, $B_2(2)'$ A_6 **360**	${}^2G_2(3)'$ $A_1(8)$ **504**										$B_2(4)$ **979 200**	$C_3(5)$ **228 501 000 000 000**	$D_4(3)$ **4 952 379 014 400**	${}^2D_4(3^2)$ **10 151 960 619 520**	${}^2A_2(16)$ **62 400**	C_5 **5**
A_7 **2 520**	$A_1(11)$ **660**	$E_6(2)$ **214 641 375 522 005 375 270 400**	$E_7(2)$ **[illegible]**	$E_8(2)$ **[illegible]**	$F_4(2)$ **3 311 126 603 366 400**	$G_2(3)$ **4 245 696**	${}^3D_4(2^3)$ **211 341 312**	${}^2E_6(2^2)$ **76 532 479 683 774 853 939 200**	${}^2B_2(2^3)$ **29 120**	Tits* ${}^2F_4(2)'$ **17 971 200**	${}^2G_2(3^3)$ **10 073 444 472**	$B_3(2)$ **1 451 520**	$C_4(3)$ **65 784 756 654 609 600**	$D_5(2)$ **23 499 295 948 800**	${}^2D_5(2^2)$ **254 013 379 554 400**	${}^2A_2(25)$ **126 000**	C_7 **7**
$A_1(2)$ A_8 **20 160**	$A_1(13)$ **1 092**	$E_6(3)$ **[illegible]**	$E_7(3)$ **[illegible]**	$E_8(3)$ **[illegible]**	$F_4(3)$ **5 734 420 792 816 671 844 761 600**	$G_2(4)$ **251 596 800**	${}^3D_4(3^3)$ **20 560 031 364 912**	${}^2E_6(3^2)$ **[illegible]**	${}^2B_2(2^5)$ **32 537 600**	${}^2F_4(2^3)$ **264 905 352 699 586 176 614 400**	${}^2G_2(3^5)$ **48 825 657 639 360 532**	$B_2(5)$ **4 680 000**	$C_3(7)$ **273 457 218 604 953 600**	$D_5(3)$ **8 911 539 000 000 000 000**	${}^2D_4(4^2)$ **67 336 471 195 648 000**	${}^2A_3(9)$ **3 265 920**	C_{11} **11**
A_9 **181 440**	$A_1(17)$ **2 448**	$E_6(4)$ **[illegible]**	$E_7(4)$ **[illegible]**	$E_8(4)$ **[illegible]**	$F_4(4)$ **19 009 625 323 840 943 455 297 669 120 000**	$G_2(5)$ **5 859 000 000**	${}^3D_4(4^3)$ **67 802 350 642 790 400**	${}^2E_6(4^2)$ **[illegible]**	${}^2B_2(2^7)$ **34 093 383 680**	${}^2F_4(2^5)$ **[illegible]**	${}^2G_2(3^7)$ **239 189 910 264 352 349 332 632**	$B_2(7)$ **138 297 600**	$C_3(9)$ **54 025 733 402 499 584 000**	$D_5(3)$ **1 209 312 799 961 305 159 200**	${}^2D_4(5^2)$ **17 860 203 250 000 000 000**	${}^2A_2(64)$ **5 515 776**	C_{13} **13**
A_n $\frac{n!}{2}$	$PSL_{n+1}(q)$, $L_{n+1}(q)$ $A_n(q)$ $\frac{1}{\gcd(n+1,q-1)}\prod_{i=2}^{n+1}(q^i-1)$	$E_6(q)$ $\frac{(q^{12}-1)(q^9-1)(q^8-1)(q^5-1)}{\gcd(3,q-1)}$	$E_7(q)$ $\frac{(q^{18}-1)(q^{14}-1)(q^{12}-1)(q^8-1)(q^6-1)(q^2-1)}{\gcd(2,q-1)}$	$E_8(q)$ $(q^{30}-1)(q^{24}-1)(q^{20}-1)(q^{18}-1)(q^{14}-1)(q^{12}-1)(q^8-1)(q^2-1)$	$F_4(q)$ $(q^{12}-1)(q^8-1)(q^6-1)(q^2-1)$	$G_2(q)$ $(q^6-1)(q^2-1)$	${}^3D_4(q^3)$ $q^{12}(q^8+q^4+1)$ $(q^8+q^4+1)(q^6-1)(q^2-1)$	${}^2E_6(q^2)$ $\frac{(q^{12}-1)(q^9+1)(q^8-1)(q^5+1)}{\gcd(3,q+1)}$	${}^2B_2(2^{2n+1})$ $q^2(q^2+1)(q-1)$	${}^2F_4(2^{2n+1})$ $q^{24}(q^6+1)(q^4-1)(q^3+1)(q-1)$	${}^2G_2(3^{2n+1})$ $q^6(q^2+1)(q-1)$	$O_{2m+1}(q)$, $O_{2m+1}(q)$ $B_n(q)$ $\frac{1}{\gcd(2,q-1)}\prod_{i=1}^{n}(q^{2i}-1)$	$PSp_{2m}(q)$ $C_n(q)$ $\frac{1}{\gcd(2,q-1)}\prod_{i=1}^{n}(q^{2i}-1)$	$O_{2n}^+(q)$ $D_n(q)$ $\frac{1}{\gcd(4,q^n-1)}\prod_{i=1}^{n-1}(q^{2i}-1)$	$O_{2n}^-(q)$ ${}^2D_n(q^2)$ $\frac{1}{\gcd(4,q^n+1)}\prod_{i=1}^{n-1}(q^{2i}-1)$	$PSU_{n+1}(q)$ ${}^2A_n(q^2)$ $\frac{1}{\gcd(n+1,q+1)}\prod_{i=2}^{n+1}(q^i-(-1)^i)$	$\mathbb{Z}_p$ C_p p

						$J(1)$, $J(11)$ J_1 **175 560**	HJ J_2 **604 800**	HJM J_3 **50 232 960**	J_4 **86 775 571 046 077 562 880**	HS **44 352 000**	McL **898 128 000**	s, HHM, HTH He **4 030 387 200**	Ru **145 926 144 000**
M_{11} **7 920**	M_{12} **95 040**	M_{22} **443 520**	M_{23} **10 200 960**	M_{24} **244 823 040**									
Sz **448 345 497 600**	$O'N$, O-S **460 815 505 920**	-3 Co_3 **495 766 656 000**	-2 Co_2 **42 305 421 312 000**	-1 Co_1 **4 157 776 806 543 360 000**	F_5, D HN **273 030 912 000 000**	Ly **51 765 179 004 000 000**	Th **90 745 943 887 872 000**	$M(22)$ Fi_{22} **64 561 751 654 400**	$M(23)$ Fi_{23} **4 089 470 473 293 004 800**	F_{3+}, $M(24)'$ Fi'_{24} **1 255 205 709 190 661 721 292 800**	F_2 B **4 154 781 481 226 426 191 177 580 544 000 000**	F_1, M_1 M **808 017 424 794 512 875 886 459 904 961 710 757 005 754 368 000 000**	

- Alternating Groups
- Classical Chevalley Groups
- Chevalley Groups
- Classical Steinberg Groups
- Steinberg Groups
- Suzuki Groups
- Ree Groups and Tits Group*
- Sporadic Groups
- Cyclic Groups

*The Tits group ${}^2F_4(2)'$ is not a group of Lie type, but is the (index 2) commutator subgroup of ${}^2F_4(2)$. It is usually given honorary Lie type status.

The groups starting on the second row are the classical groups. The sporadic swedi group is unrelated to the families of Suzuki groups.

Copyright © 2012 Ivan Andrus.

Alternates†
Symbol
Order‡

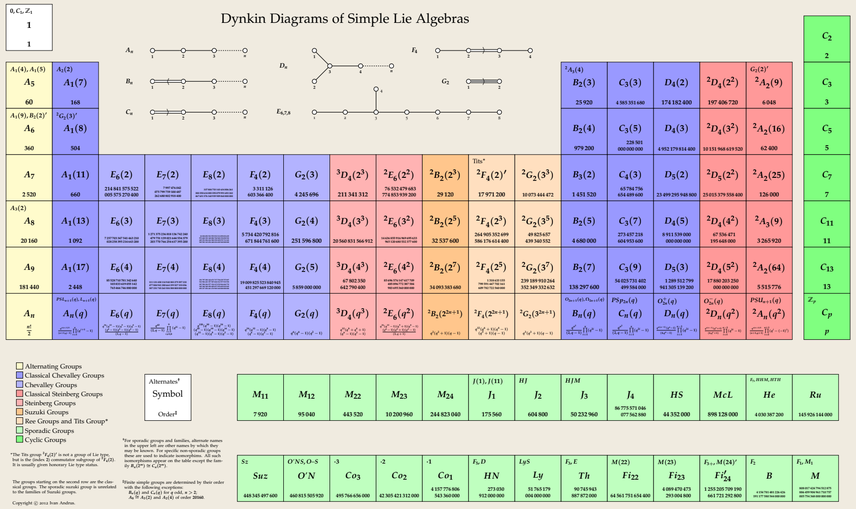
†For sporadic groups and families, alternate names in the upper left are other names by which they may be known. For specific non-sporadic groups, the name listed is the most frequently used. All non-isomorphic groups on the table except the families $B_n(2^m)$ is ${}^2C_n(2^m)$.

‡Finite simple groups are determined by their order with the following exceptions:
$B_n(q)$ and $C_n(q)$ for q odd, $n > 2$;
A_7, $A_1(2)$ and $A_2(4)$ of order 20160.

Definitions ▶ Quantum Error Correction ▶ Other Pseudorandom Objects ▶ Future

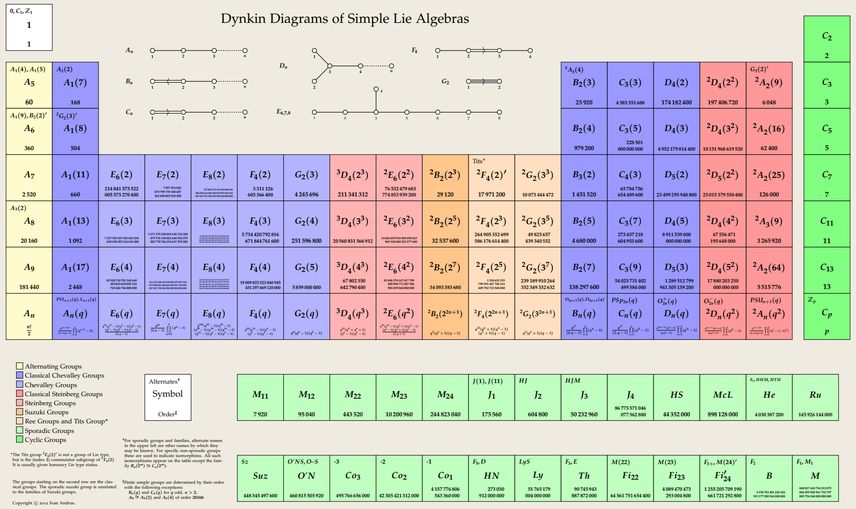
Image Credits: Ivan Andrus

The Periodic Table Of Finite Simple Groups



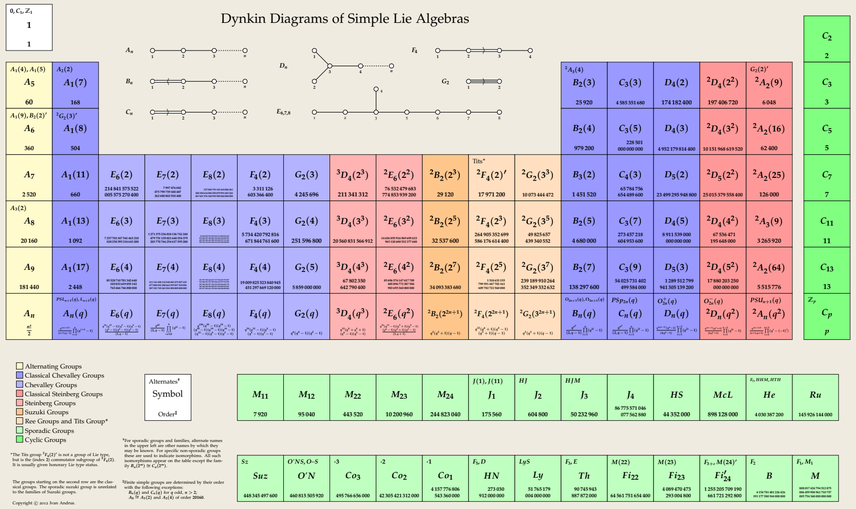
θ, C_e, Z_1															C_2
1															2
$A_1(4), A_1(8)$	$A_2(2)$										${}^2A_3(4)$				C_3
A_5	$A_1(7)$										$B_2(3)$	$C_3(3)$	$D_4(2)$	${}^2D_4(2^2)$	${}^2A_2(9)$	3
60	168										25 920	4 585 351 680	174 182 400	197 406 720	6 048	
$A_1(9), B_2(2)'$	${}^2G_2(3)'$														C_5
A_6	$A_1(8)$										$B_2(4)$	$C_3(5)$	$D_4(3)$	${}^2D_4(3^2)$	${}^2A_2(16)$	5
360	504										979 200	228 501 000 000 000	4 952 379 014 400	10 151 960 619 520	62 400	
A_7	$A_1(11)$	$E_6(2)$	$E_7(2)$	$E_8(2)$	$F_4(2)$	$G_2(3)$	${}^3D_4(2^3)$	${}^2E_6(2^2)$	${}^2B_2(2^3)$	Tits* ${}^2F_4(2)'$	${}^2G_2(3^3)$	$B_3(2)$	$C_4(3)$	$D_5(2)$	${}^2D_5(2^2)$	${}^2A_2(25)$	C_7
2 520	660	214 641 375 522 005 375 270 400			3 311 126 603 366 400	4 245 696	211 343 312	76 532 479 683 774 853 939 200	29 120	17 971 200	10 073 444 472	1 451 520	65 784 756 654 699 600	23 499 295 948 800	25 015 379 556 400	126 000	7
$A_7(2)$															C_{11}		
A_8	$A_1(13)$	$E_6(3)$	$E_7(3)$	$E_8(3)$	$F_4(3)$	$G_2(4)$	${}^3D_4(3^3)$	${}^2E_6(3^2)$	${}^2B_2(2^5)$	${}^2F_4(2^3)$	${}^2G_2(3^5)$	$B_2(5)$	$C_3(7)$	$D_6(5)$	${}^2D_4(4^2)$	${}^2A_3(9)$	11
20 160	1 092				5 734 420 792 816 671 844 761 600	251 596 800	20 560 031 364 912		32 537 600	264 905 352 699 586 176 614 400	49 825 657 639 360 532	4 680 000	273 457 218 604 953 600	8 911 339 000 000 000 000	67 336 471 195 648 000	3 265 920	
A_9	$A_1(17)$	$E_6(4)$	$E_7(4)$	$E_8(4)$	$F_4(4)$	$G_2(5)$	${}^3D_4(4^3)$	${}^2E_6(4^2)$	${}^2B_2(2^7)$	${}^2F_4(2^5)$	${}^2G_2(3^7)$	$B_2(7)$	$C_3(9)$	$D_5(3)$	${}^2D_4(5^2)$	${}^2A_2(64)$	C_{13}
181 440	2 448				19 089 825 323 840 943 451 297 469 120 000	5 859 000 000	67 802 350 642 790 400		34 093 383 680		239 189 910 264 352 349 332 632	138 297 600	54 025 733 402 499 584 000	1 599 312 799 961 305 159 200	17 860 203 250 000 000 000	5 515 776	13
A_n	$PSL_{n+1}(q), \Lambda_{n+1}(q)$											$O_{2n+1}(q), O_{2n+1}(q)$	$PSp_{2n}(q)$	$O_{2n}^+(q)$	$O_{2n}^-(q)$	$PSU_{n+1}(q)$	$\mathbb{Z}_p$
$\frac{n!}{2}$	$A_n(q)$	$E_6(q)$	$E_7(q)$	$E_8(q)$	$F_4(q)$	$G_2(q)$	${}^3D_4(q^3)$	${}^2E_6(q^2)$	${}^2B_2(2^{2n+1})$	${}^2F_4(2^{2n+1})$	${}^2G_2(3^{2n+1})$	$B_n(q)$	$C_n(q)$	$D_n(q)$	${}^2D_n(q^2)$	${}^2A_n(q^2)$	C_p
																	p

The Periodic Table Of Finite Simple Groups



θ, C_e, Z_1															C_2
1															2
$A_1(4), A_1(8)$	$A_2(2)$										${}^2A_3(4)$				C_3
A_5	$A_1(7)$										$B_2(3)$	$C_3(3)$	$D_4(2)$	${}^2D_4(2^2)$	${}^2A_2(9)$	3
60	168										25 920	4 585 351 680	174 182 400	197 406 720	6 048	
$A_1(9), B_2(2)'$	${}^2G_2(3)'$														C_5
A_6	$A_1(8)$										$B_2(4)$	$C_3(5)$	$D_4(3)$	${}^2D_4(3^2)$	${}^2A_2(16)$	5
360	504										979 200	228 501 000 000 000	4 952 379 014 400	10 151 960 619 520	62 400	
A_7	$A_1(11)$	$E_6(2)$	$E_7(2)$	$E_8(2)$	$F_4(2)$	$G_2(3)$	${}^3D_4(2^3)$	${}^2E_6(2^2)$	${}^2B_2(2^3)$	Tits* ${}^2F_4(2)'$	${}^2G_2(3^3)$	$B_3(2)$	$C_4(3)$	$D_5(2)$	${}^2D_5(2^2)$	${}^2A_2(25)$	C_7
2 520	660	214 641 375 522 005 375 270 400			3 311 126 603 366 400	4 245 696	211 343 312	76 532 479 683 774 853 939 200	29 120	17 971 200	10 073 444 472	1 451 520	65 784 756 654 699 600	23 499 295 948 800	25 015 379 556 400	126 000	7
$A_7(2)$															C_{11}		
A_8	$A_1(13)$	$E_6(3)$	$E_7(3)$	$E_8(3)$	$F_4(3)$	$G_2(4)$	${}^3D_4(3^3)$	${}^2E_6(3^2)$	${}^2B_2(2^5)$	${}^2F_4(2^3)$	${}^2G_2(3^5)$	$B_2(5)$	$C_3(7)$	$D_6(5)$	${}^2D_4(4^2)$	${}^2A_3(9)$	11
20 160	1 092				5 734 420 792 816 671 844 761 600	251 596 800	20 560 031 364 912		32 537 600	264 905 352 699 586 176 614 400	49 825 657 639 360 532	4 680 000	273 457 218 604 953 600	8 911 339 000 000 000 000	67 336 471 195 648 000	3 265 920	
A_9	$A_1(17)$	$E_6(4)$	$E_7(4)$	$E_8(4)$	$F_4(4)$	$G_2(5)$	${}^3D_4(4^3)$	${}^2E_6(4^2)$	${}^2B_2(2^7)$	${}^2F_4(2^5)$	${}^2G_2(3^7)$	$B_2(7)$	$C_3(9)$	$D_5(3)$	${}^2D_4(5^2)$	${}^2A_2(64)$	C_{13}
181 440	2 448				19 089 825 323 840 943 451 297 469 120 000	5 859 000 000	67 802 350 642 790 400		34 093 383 680		239 189 910 264 352 349 332 632	138 297 600	54 025 733 402 499 584 000	1 599 312 799 961 305 159 200	17 860 203 250 000 000 000	5 515 776	13
A_n	$PSL_{n+1}(q), \Lambda_{n+1}(q)$											$O_{2n+1}(q), O_{2n+1}(q)$	$PSp_{2n}(q)$	$O_{2n}^+(q)$	$O_{2n}^-(q)$	$PSU_{n+1}(q)$	$\mathbb{Z}_p$
$\frac{n!}{2}$	$A_n(q)$	$E_6(q)$	$E_7(q)$	$E_8(q)$	$F_4(q)$	$G_2(q)$	${}^3D_4(q^3)$	${}^2E_6(q^2)$	${}^2B_2(2^{2n+1})$	${}^2F_4(2^{2n+1})$	${}^2G_2(3^{2n+1})$	$B_n(q)$	$C_n(q)$	$D_n(q)$	${}^2D_n(q^2)$	${}^2A_n(q^2)$	C_p
																	p

The Periodic Table Of Finite Simple Groups



Lubotzky, Phillips, Sarnak, Kassabov, Nikolov, Breuillard–Green–Tao:

There are expanders with symmetries of each of these finite simple groups.

Breuillard and Lubotzky ‘18:

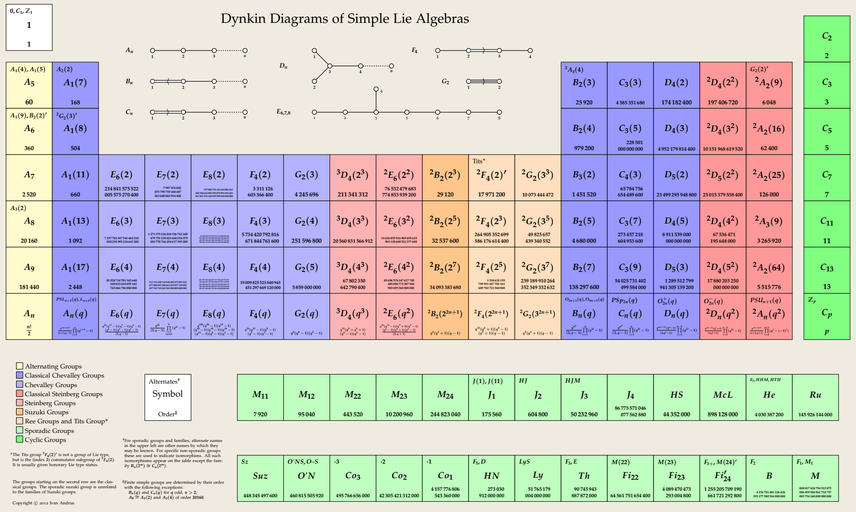
But are there optimal ones?

Optimal — Not Expander

$2\sqrt{\deg(X)-1}$ — $\deg(X)$

Image Credits: Ivan Andrus

The Periodic Table Of Finite Simple Groups



Lubotzky, Phillips, Sarnak, Kassabov, Nikolov, Breuillard–Green–Tao:

There are expanders with symmetries of each of these finite simple groups.

Breuillard and Lubotzky '18:

But are there optimal ones?

Our Work: Near-optimal ones, yes!

Optimal — Not Expander

$2\sqrt{\deg(X)-1}$ — $\deg(X)$

Image Credits: Ivan Andrus

Amplifying Expansion while maintaining symmetry

Amplifying Expansion while maintaining symmetry

Theorem [Jeronimo–**M**–Roy-Wigderson FOCS'22, SICOMP'25]:

For any group G,

Amplifying Expansion while maintaining symmetry

Theorem [Jeronimo–**M**–Roy-Wigderson FOCS'22, SICOMP'25]:

For any group G,

Any Weak Expander

with G -symmetry*

$\left(\lambda(X) = 0.9\deg(X)\right)$

* Cayley graph on G

Amplifying Expansion while maintaining symmetry

Theorem [Jeronimo–**M**–Roy-Wigderson FOCS'22, SICOMP'25]:

For any group G,

Any Weak Expander with G -symmetry*

$\left(\lambda(X) = 0.9\deg(X)\right)$

Near-Optimal Expander with G-symmetry*

$\left(\lambda(Y) \approx \deg(Y)^{\frac{1}{2}+o(1)}\right)$

* Cayley graph on G

Amplifying Expansion while maintaining symmetry

Theorem [Jeronimo–**M**–Roy-Wigderson FOCS'22, SICOMP'25]:

For any group G,

Any Weak Expander with G -symmetry*

$\left(\lambda(X) = 0.9\deg(X)\right)$

Efficient, randomness-free →

Near-Optimal Expander with G-symmetry*

$\left(\lambda(Y) \approx \deg(Y)^{\frac{1}{2}+o(1)}\right)$

* Cayley graph on G

Amplifying Expansion while maintaining symmetry

Theorem [Jeronimo–**M**–Roy-Wigderson FOCS'22, SICOMP'25]:

For any group G,

Any Weak Expander with G -symmetry*

$\left(\lambda(X) = 0.9\deg(X)\right)$

Efficient, randomness-free →

Near-Optimal Expander with G-symmetry*

$\left(\lambda(Y) \approx \deg(Y)^{\frac{1}{2}+o(1)}\right)$

* Cayley graph on G

Amplifying Expansion while maintaining symmetry

Theorem [Jeronimo–**M**–Roy-Wigderson FOCS'22, SICOMP'25]:

For any group G,

Any Weak Expander with G -symmetry*

$\left(\lambda(X) = 0.9\deg(X)\right)$

Efficient, randomness-free

Near-Optimal Expander with G-symmetry*

$\left(\lambda(Y) \approx \deg(Y)^{\frac{1}{2}+o(1)}\right)$

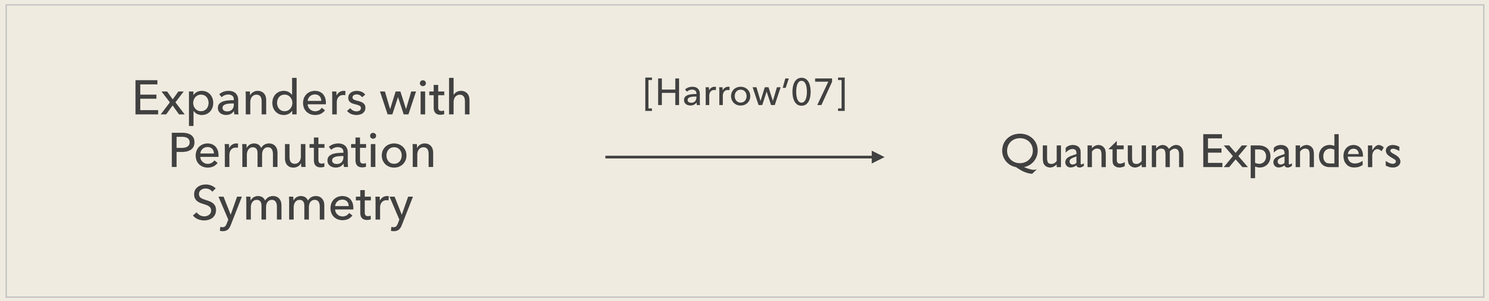
* Cayley graph on G

Optimal — Not Expander

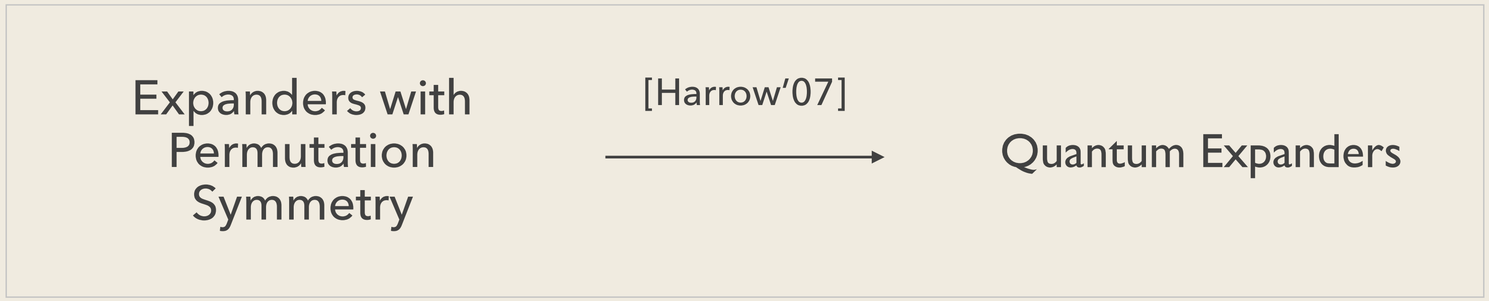
$2\sqrt{\deg(X)-1}$ — $\deg(X)$

Symmetry helps with extra structure ...

Symmetry helps with extra structure ...



Symmetry helps with extra structure ...



Corollary

Near-Optimal Expanders with Permutation Symmetry → Near-Optimal Quantum Expanders

Symmetry helps with extra structure ...

State-of-the-art explicit construction of these objects!

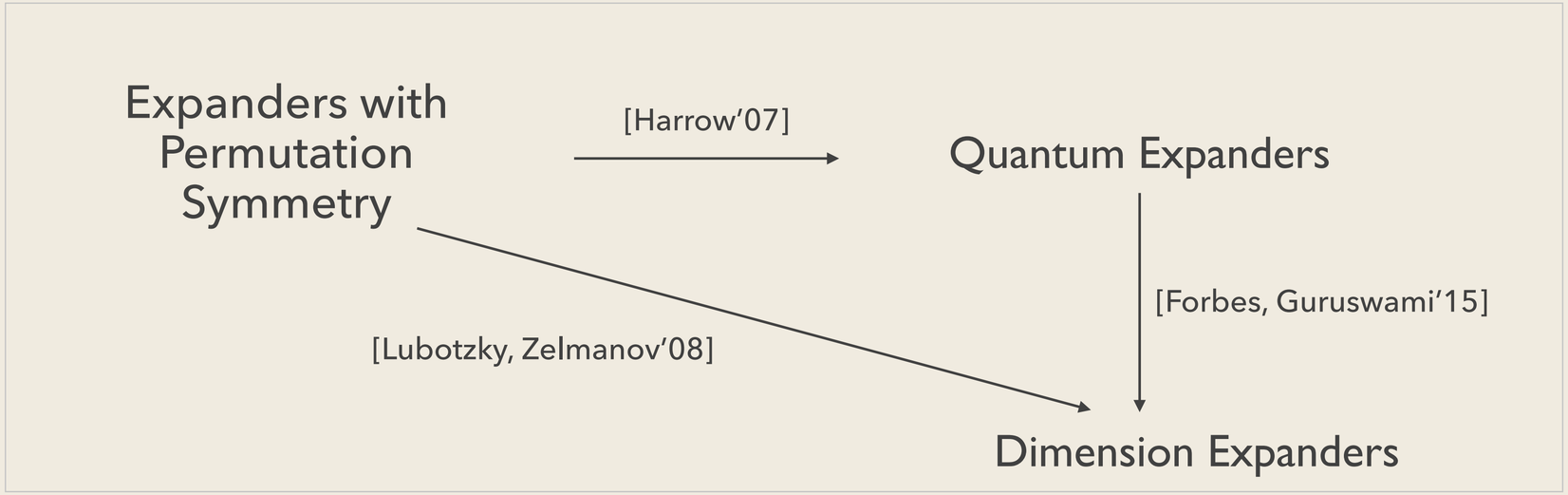
Expanders with Permutation Symmetry —[Harrow'07]→ Quantum Expanders

Corollary

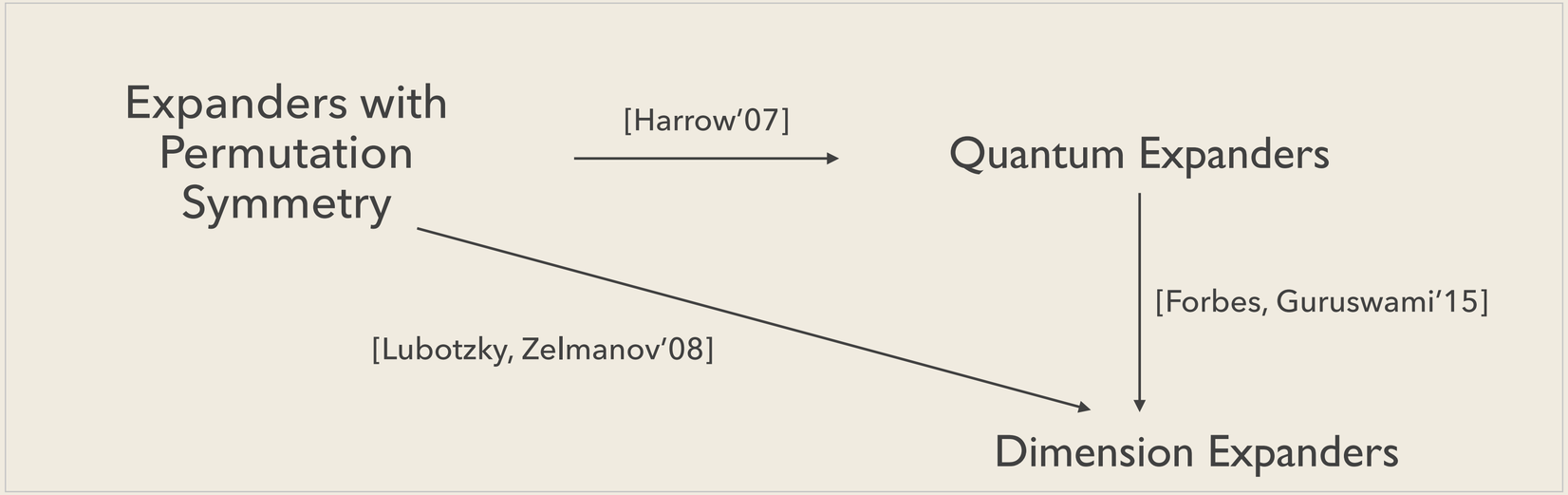
Near-Optimal Expanders with Permutation Symmetry → Near-Optimal Quantum Expanders

Symmetry helps with extra structure ...

Symmetry helps with extra structure ...



Symmetry helps with extra structure …

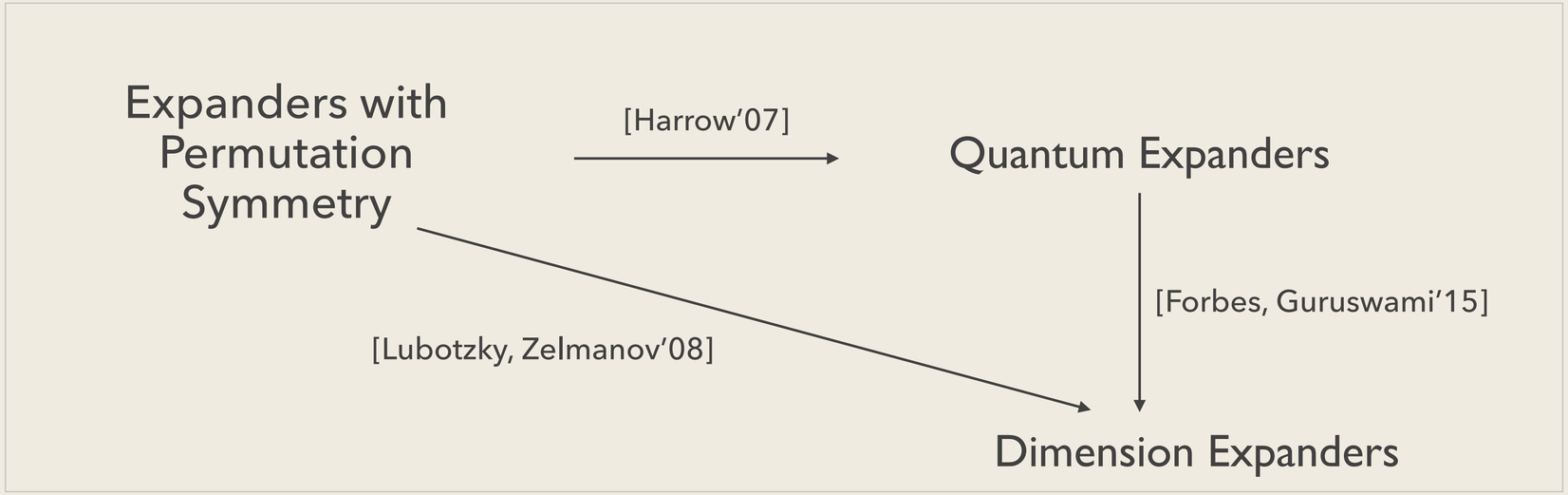


Corollary

Near-Optimal Expanders with Permutation Symmetry → Near-Optimal Dimension Expanders

Symmetry helps with extra structure …

State-of-the-art explicit construction of these objects!



Corollary

Near-Optimal Expanders with Permutation Symmetry → Near-Optimal Dimension Expanders

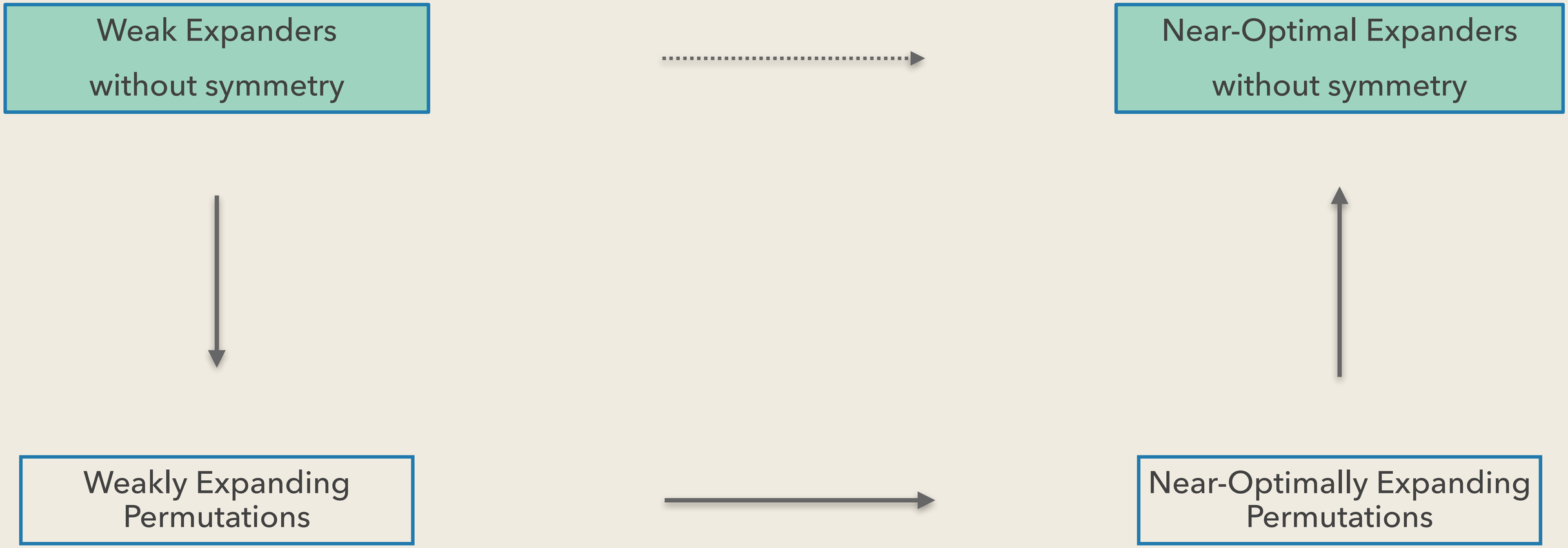
Using symmetry when there is no symmetry

Using symmetry when there is no symmetry

Weak Expanders without symmetry

Near-Optimal Expanders without symmetry

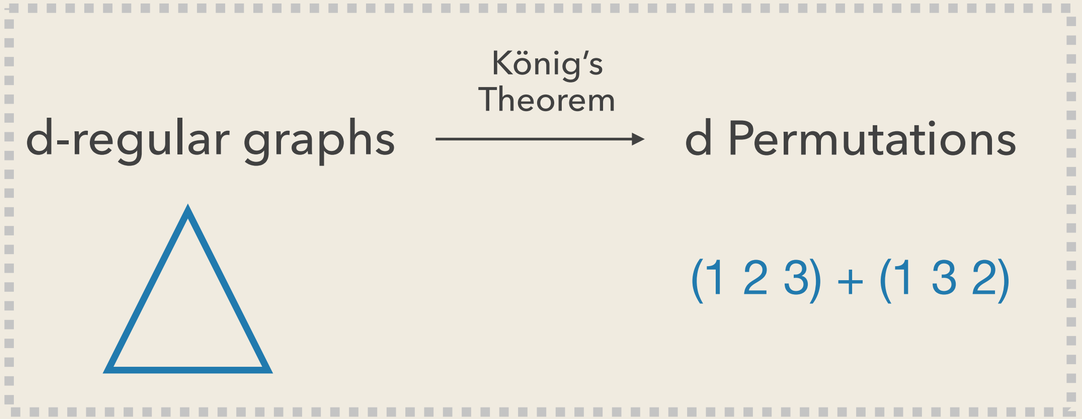
Using symmetry when there is no symmetry



Using symmetry when there is no symmetry

Weak Expanders without symmetry

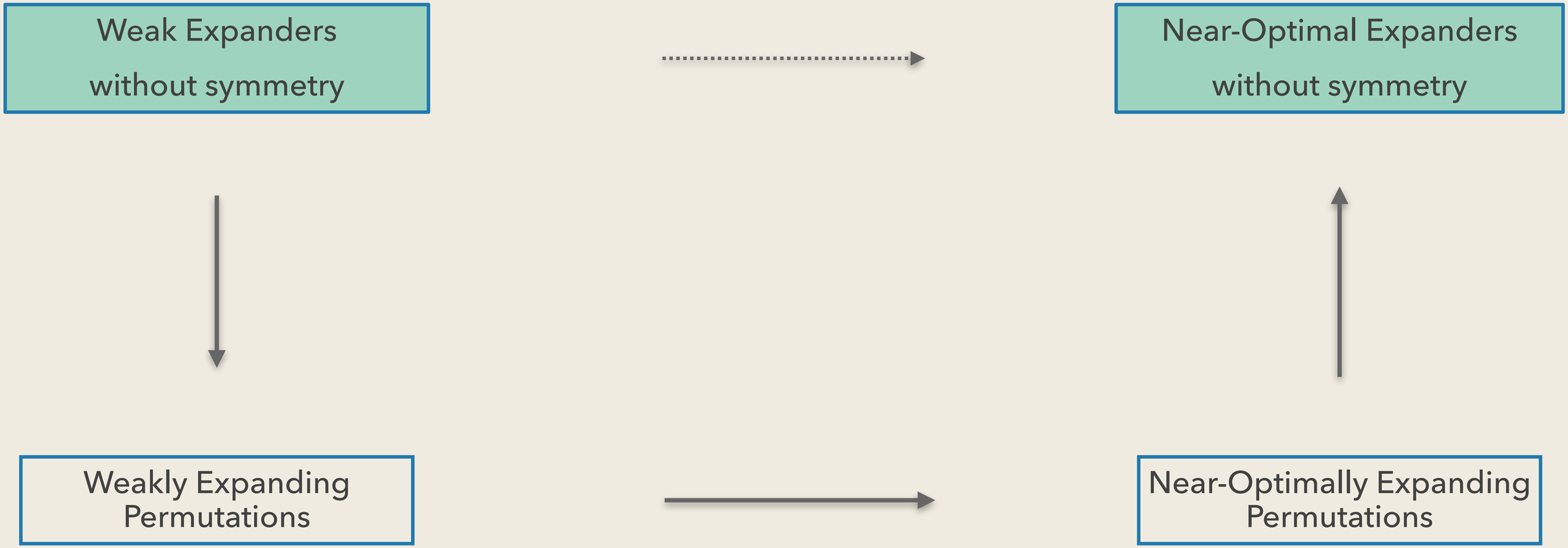
Near-Optimal Expanders without symmetry



Weakly Expanding Permutations

Near-Optimally Expanding Permutations

Using symmetry when there is no symmetry



Uses of Permutation Symmetric Expanders

Uses of Permutation Symmetric Expanders

Our Result [J**M**RW'22]:

Symmetry Preserving Expansion Amplification

Uses of Permutation Symmetric Expanders

Our Result [J**M**RW'22]:

Symmetry Preserving Expansion Amplification

Near-Optimal Expanders for all Finite Simple Groups

General Expander Amplification

"Planted" Quantum Codes [GK'23]

Uses of Permutation Symmetric Expanders

<u>Our Result [J**M**RW'22]:</u>

Symmetry Preserving Expansion Amplification

<u>Corollary:</u>

Near-optimal Expanders with Permutation symmetries

Near-Optimal Expanders for all Finite Simple Groups

General Expander Amplification

"Planted" Quantum Codes [GK'23]

Uses of Permutation Symmetric Expanders

Our Result [J**M**RW'22]:

Symmetry Preserving Expansion Amplification

Corollary:

Near-optimal Expanders with Permutation symmetries

Near-Optimal Expanders for all Finite Simple Groups

General Expander Amplification

"Planted" Quantum Codes [GK'23]

Near-Optimal Quantum Expanders

Near-Optimal Dimension Expanders

Near-Optimal Monotone Expanders

Bridging the two domains

PSEUDORANDOMNESS

SYMMETRY

Bridging the two domains

PSEUDORANDOMNESS

Analytic, Combinatorial

SYMMETRY

Bridging the two domains

PSEUDORANDOMNESS

Analytic, Combinatorial

SYMMETRY

Algebraic

Bridging the two domains

PSEUDORANDOMNESS

Analytic, Combinatorial

SYMMETRY

Algebraic

REPRESENTATION THEORY

GENERAL FOURIER ANALYSIS

Bridging the two domains

PSEUDORANDOMNESS

Analytic, Combinatorial

SYMMETRY

Algebraic

Groups

REPRESENTATION THEORY

GENERAL FOURIER ANALYSIS

Bridging the two domains

PSEUDORANDOMNESS

Analytic, Combinatorial

Matrices (Operators)

SYMMETRY

Algebraic

Groups

REPRESENTATION THEORY

GENERAL FOURIER ANALYSIS

Bridging the two domains

PSEUDORANDOMNESS

Analytic, Combinatorial

Matrices (Operators)

SYMMETRY

Algebraic

Groups

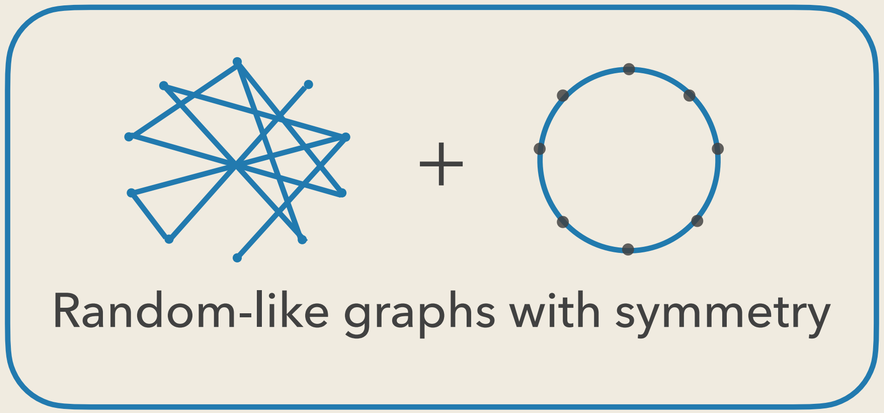
REPRESENTATION THEORY

GENERAL FOURIER ANALYSIS

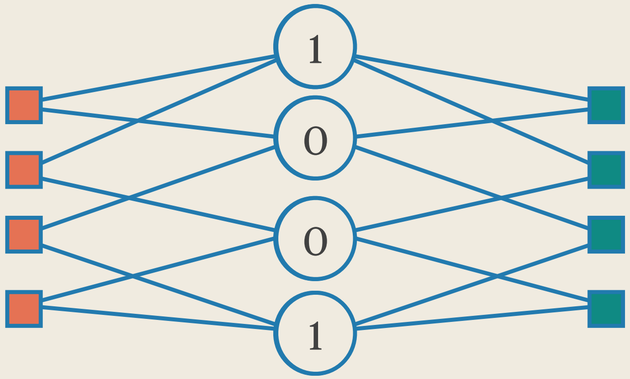
What I do – Study *representations* quantitatively

Summary

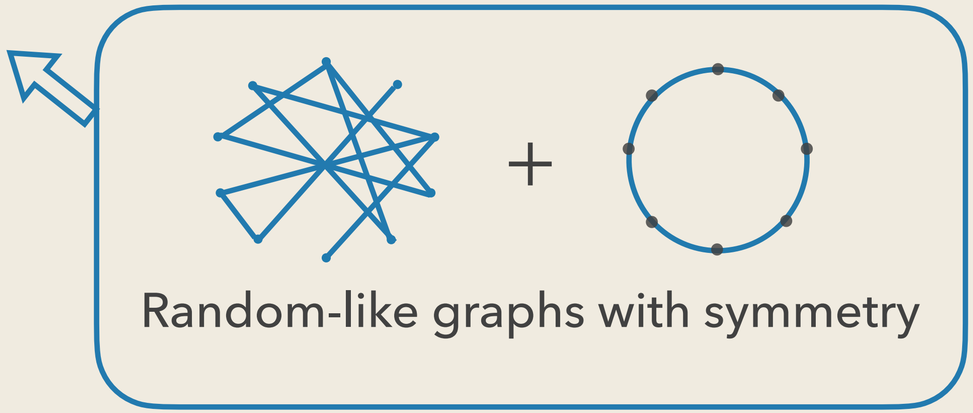
What we saw today



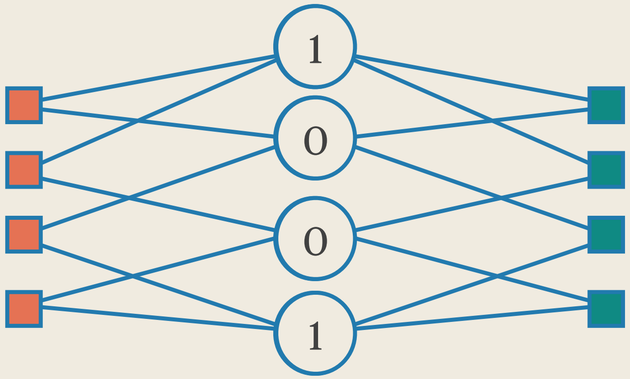
What we saw today



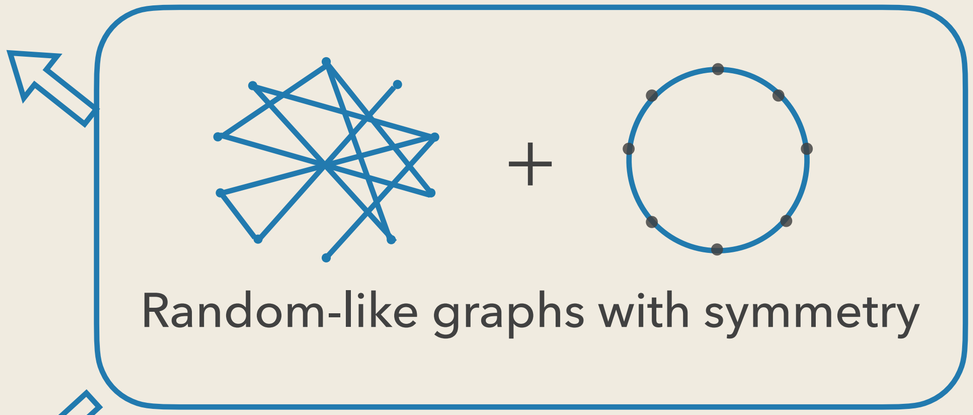
Quantum LDPC Codes

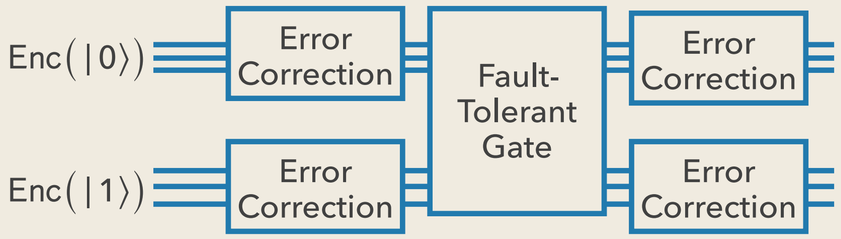


What we saw today



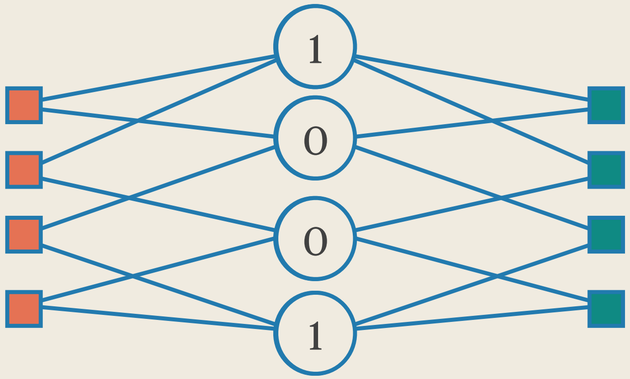
Quantum LDPC Codes



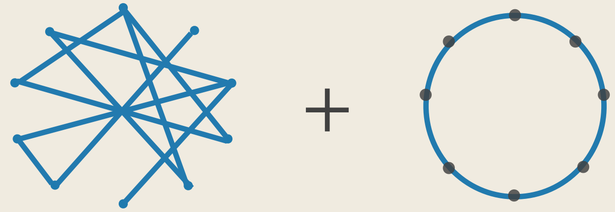


Fault-tolerant Quantum Gadgets

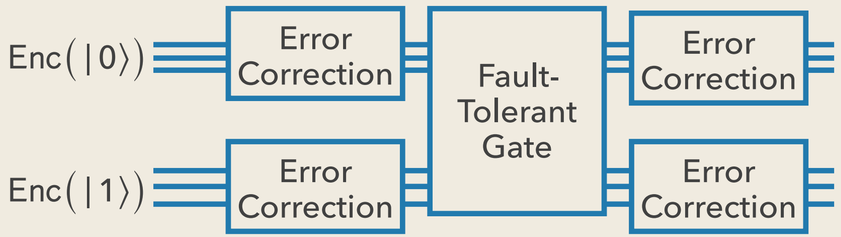
What we saw today



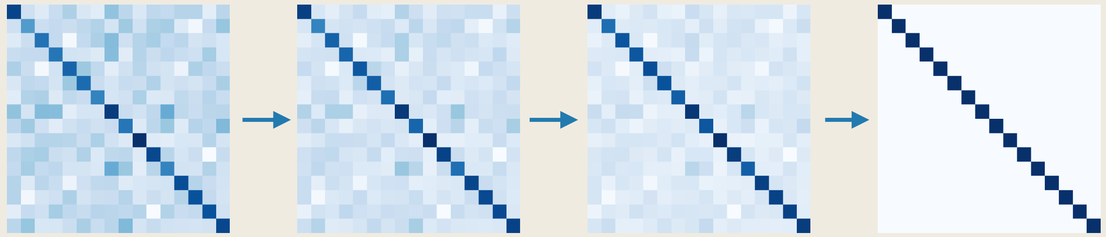
Quantum LDPC Codes



Random-like graphs with symmetry

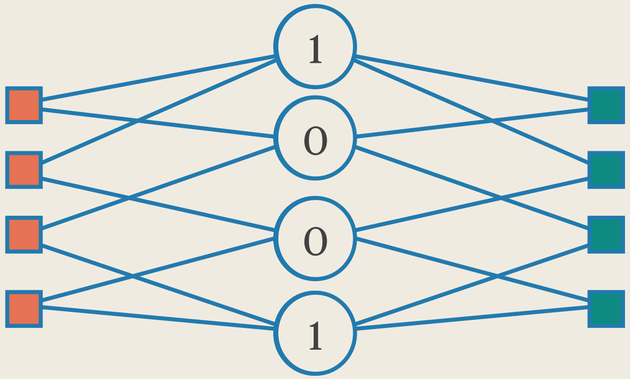


Fault-tolerant Quantum Gadgets



Near-Optimal Quantum Expanders (Fast Mixing Channel)

What we saw today

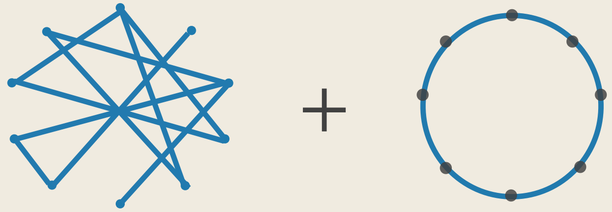


Quantum LDPC Codes

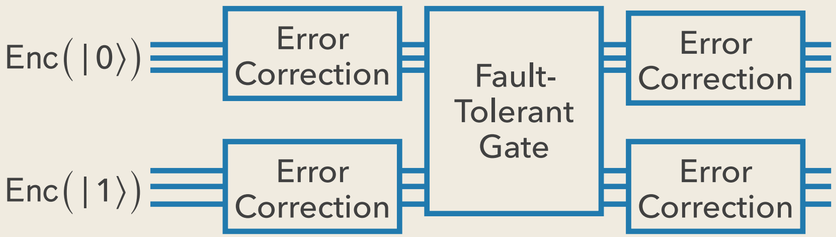
Weak Expanders

Generic Amplification Machinery

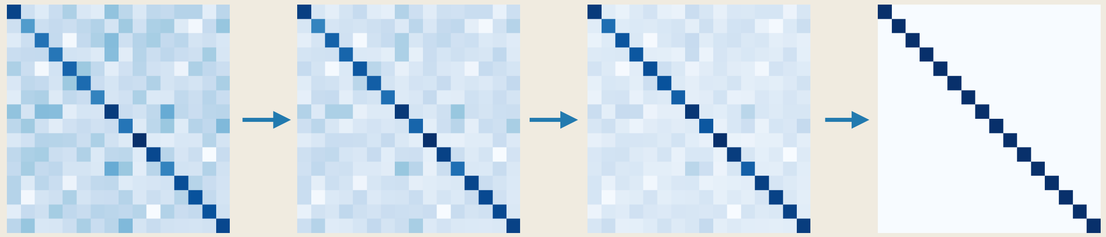
Near-Optimal Expanders



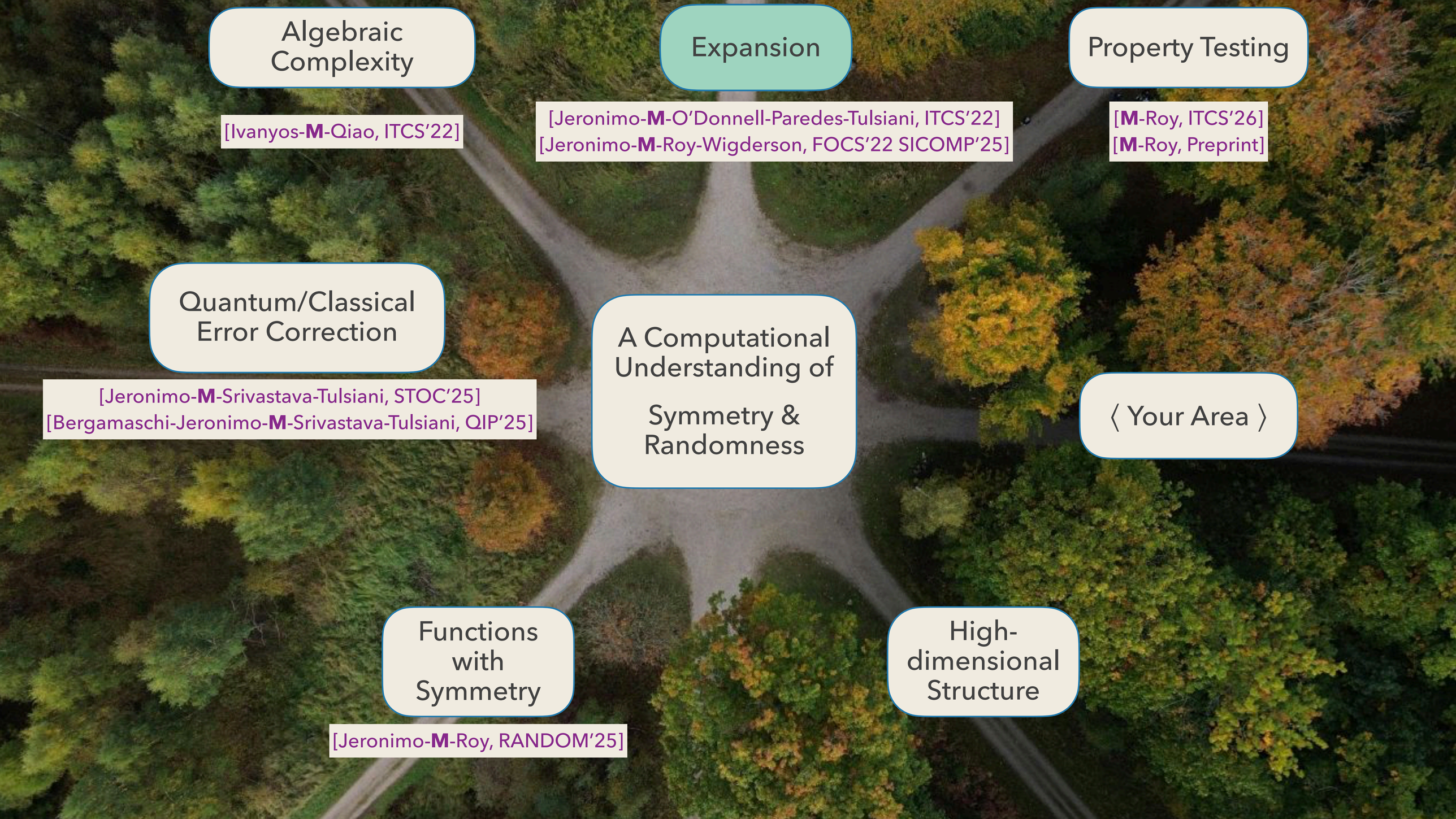
Random-like graphs with symmetry

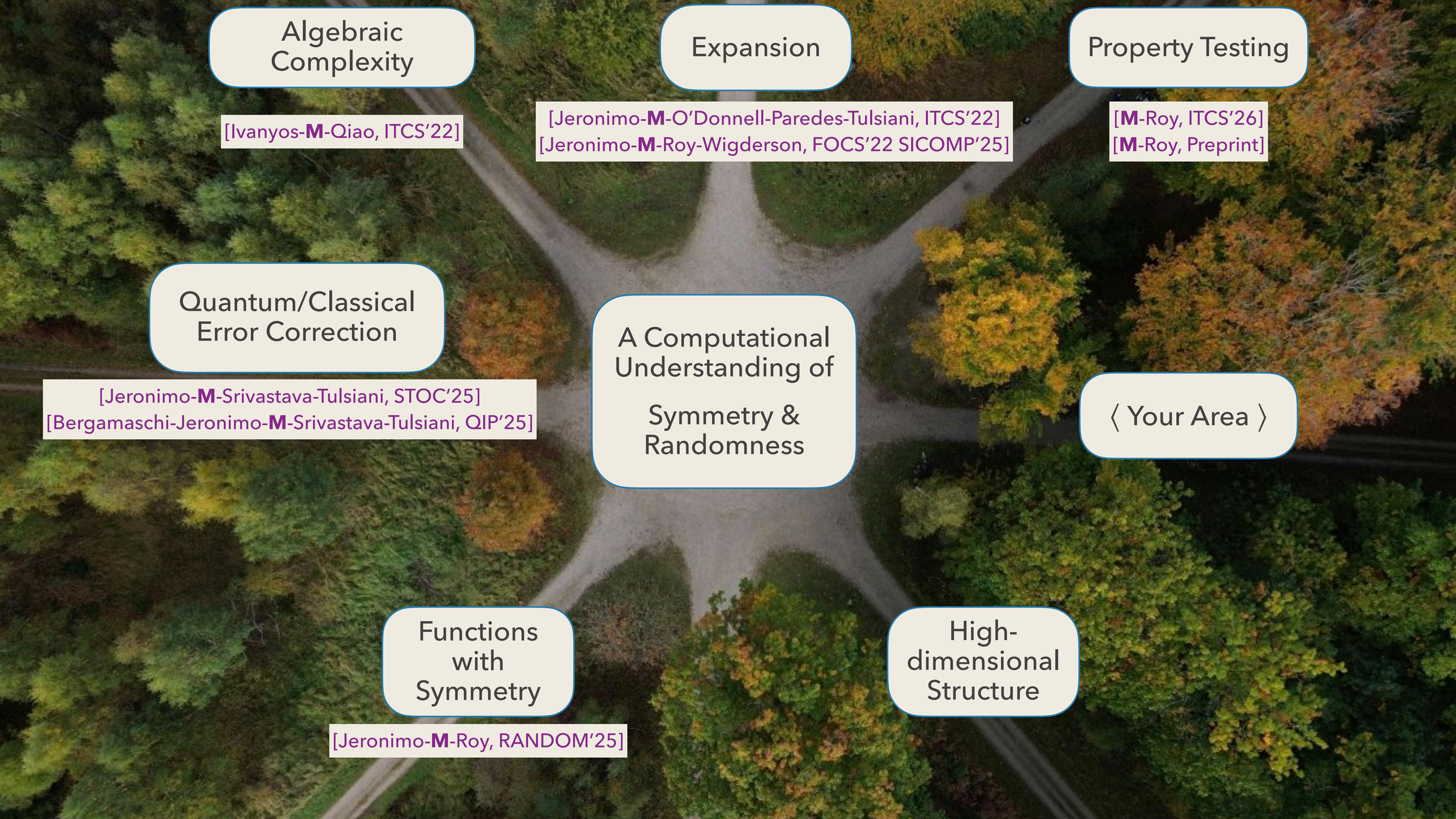


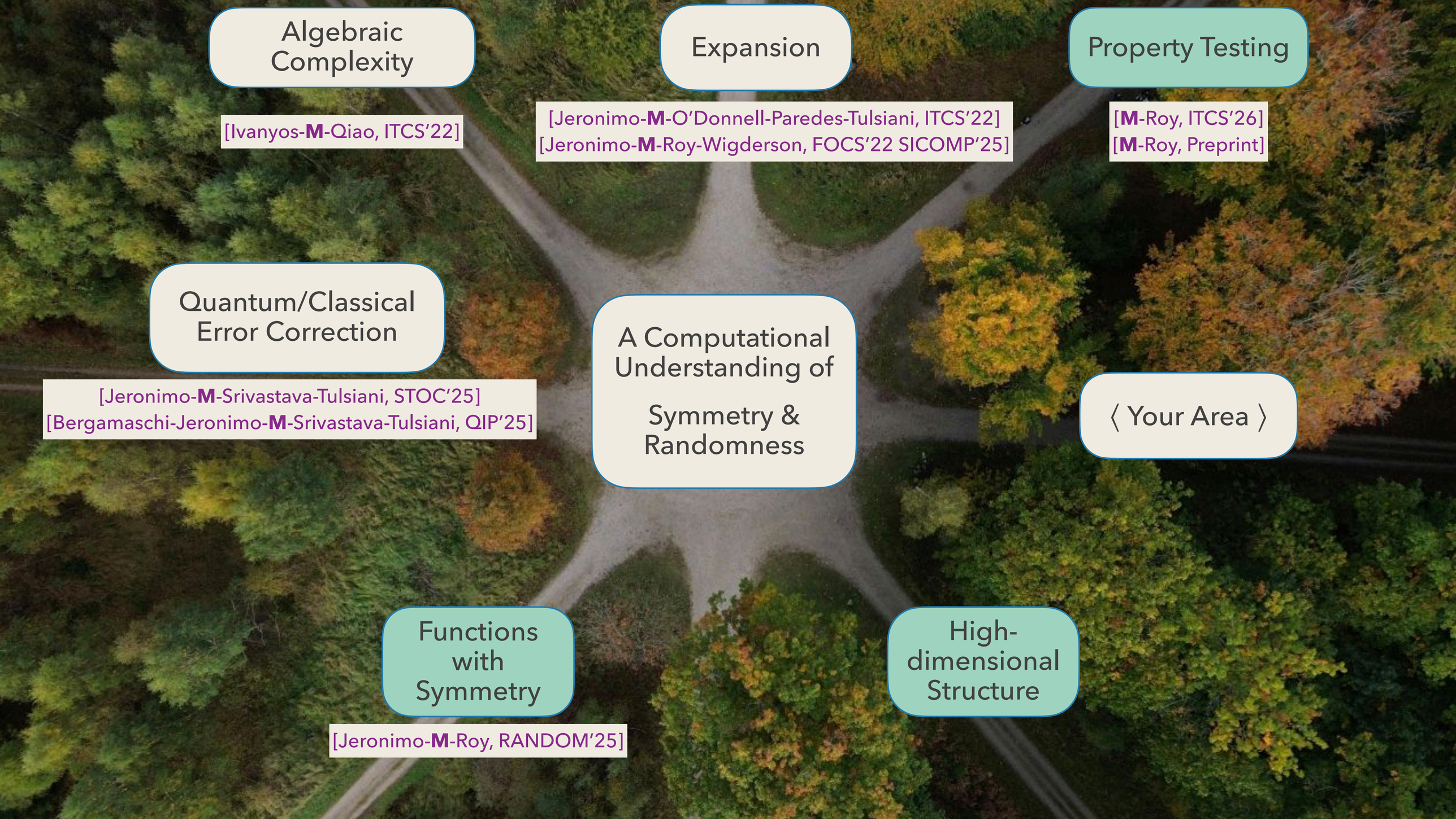
Fault-tolerant Quantum Gadgets



Near-Optimal Quantum Expanders (Fast Mixing Channel)


Algebraic Complexity
[Ivanyos-**M**-Qiao, ITCS'22]
Expansion
[Jeronimo-**M**-O'Donnell-Paredes-Tulsiani, ITCS'22]
[Jeronimo-**M**-Roy-Wigderson, FOCS'22 SICOMP'25]
Property Testing
[**M**-Roy, ITCS'26]
[**M**-Roy, Preprint]
Quantum/Classical Error Correction
[Jeronimo-**M**-Srivastava-Tulsiani, STOC'25]
[Bergamaschi-Jeronimo-**M**-Srivastava-Tulsiani, QIP'25]
A Computational Understanding of
Symmetry & Randomness
⟨ Your Area ⟩
Functions with Symmetry
[Jeronimo-**M**-Roy, RANDOM'25]
High-dimensional Structure


Algebraic Complexity
[Ivanyos-**M**-Qiao, ITCS'22]
Expansion
[Jeronimo-**M**-O'Donnell-Paredes-Tulsiani, ITCS'22]
[Jeronimo-**M**-Roy-Wigderson, FOCS'22 SICOMP'25]
Property Testing
[**M**-Roy, ITCS'26]
[**M**-Roy, Preprint]
Quantum/Classical Error Correction
[Jeronimo-**M**-Srivastava-Tulsiani, STOC'25]
[Bergamaschi-Jeronimo-**M**-Srivastava-Tulsiani, QIP'25]
A Computational Understanding of
Symmetry & Randomness
⟨ Your Area ⟩
Functions with Symmetry
[Jeronimo-**M**-Roy, RANDOM'25]
High-dimensional Structure

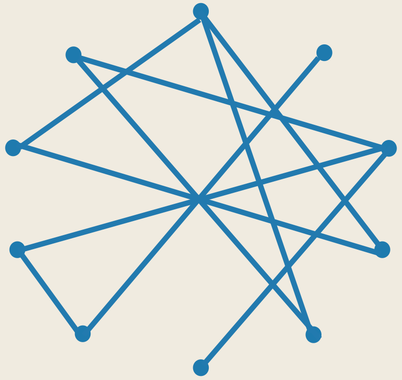

Algebraic Complexity
[Ivanyos-**M**-Qiao, ITCS'22]
Expansion
[Jeronimo-**M**-O'Donnell-Paredes-Tulsiani, ITCS'22]
[Jeronimo-**M**-Roy-Wigderson, FOCS'22 SICOMP'25]
Property Testing
[**M**-Roy, ITCS'26]
[**M**-Roy, Preprint]
Quantum/Classical Error Correction
[Jeronimo-**M**-Srivastava-Tulsiani, STOC'25]
[Bergamaschi-Jeronimo-**M**-Srivastava-Tulsiani, QIP'25]
A Computational Understanding of
Symmetry & Randomness
⟨ Your Area ⟩
Functions with Symmetry
[Jeronimo-**M**-Roy, RANDOM'25]
High-dimensional Structure

Future Work

High-
Dimensional
Structure
Functions
with
Symmetry

High-
Dimensional
Structure
Functions
with
Symmetry

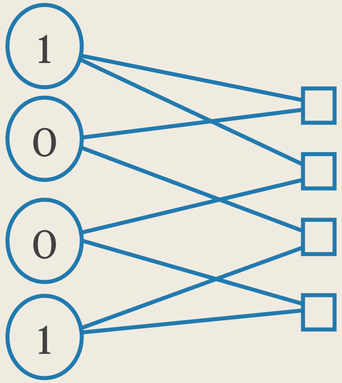
A duality between graphs and codes



1-D Expander

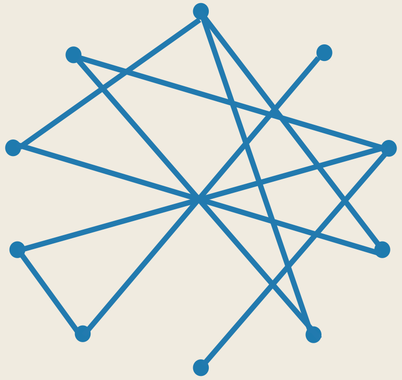
[Gallagher'63

Sipser-Spielman'91]



Classical 1-D Code

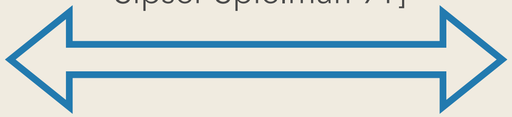
A duality between graphs and codes

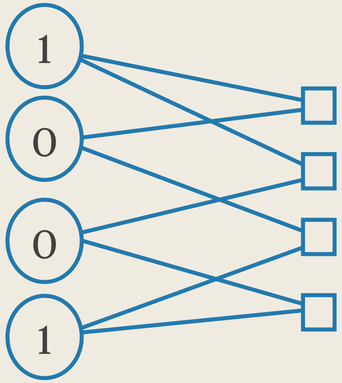


1-D Expander

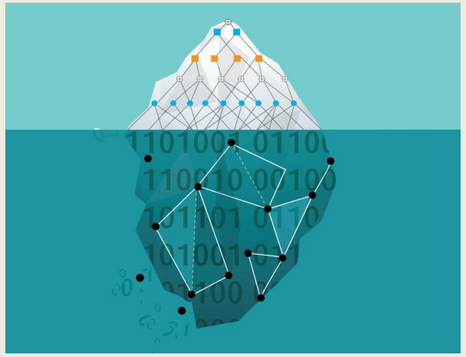
[Gallagher'63

Sipser-Spielman'91]

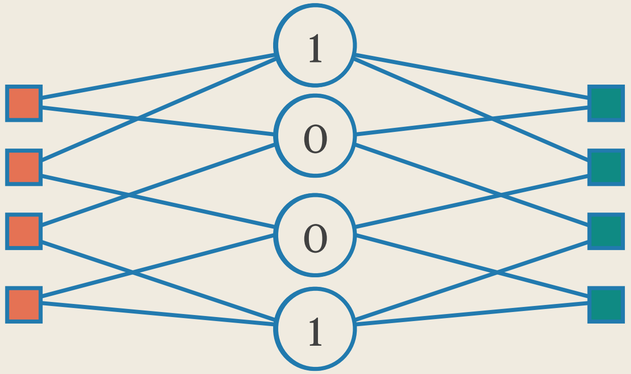




Classical 1-D Code

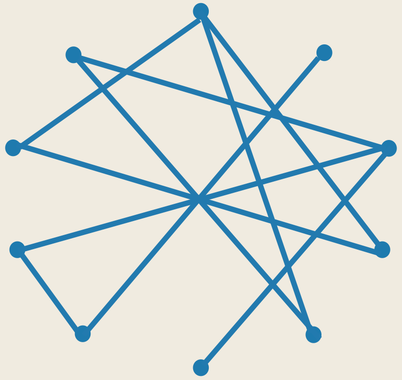


High-Dimensional Expander



Quantum"2-D" Codes

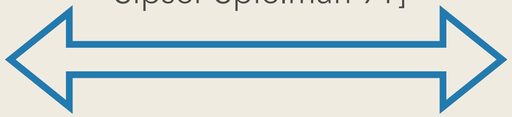
A duality between graphs and codes

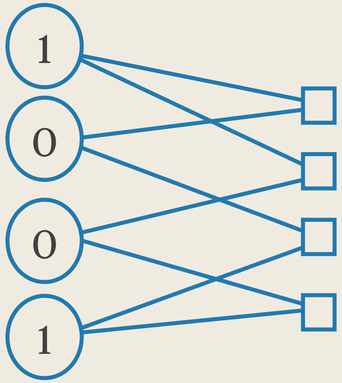


1-D Expander

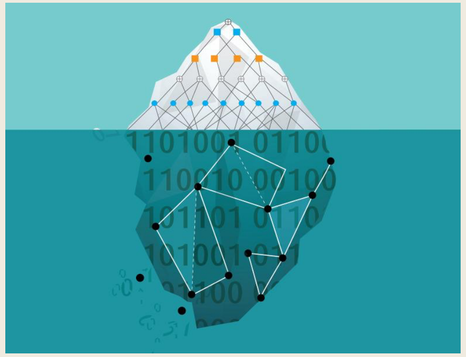
[Gallagher'63

Sipser-Spielman'91]

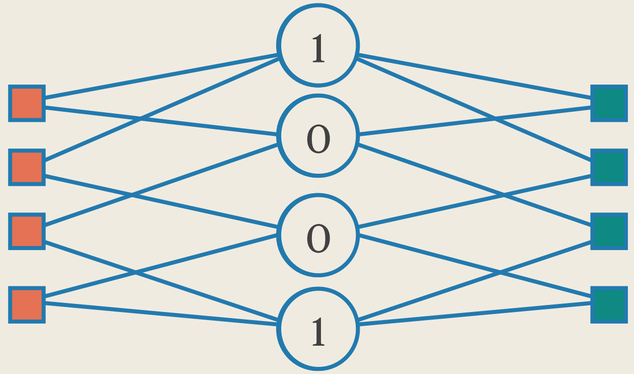




Classical 1-D Code

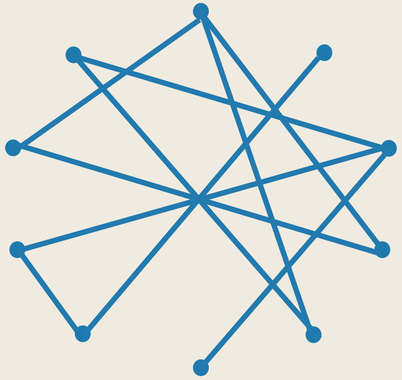


High-Dimensional Expander



Quantum"2-D" Codes

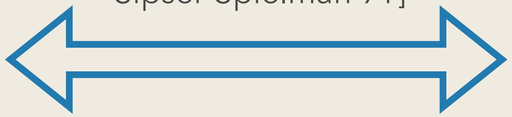
A duality between graphs and codes



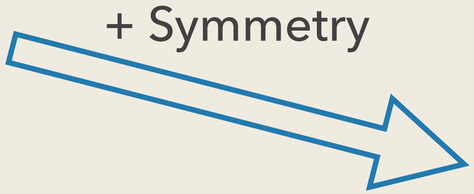
1-D Expander

[Gallagher'63

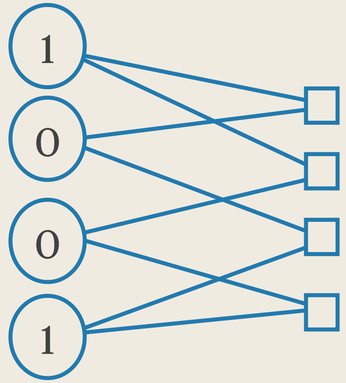
Sipser-Spielman'91]



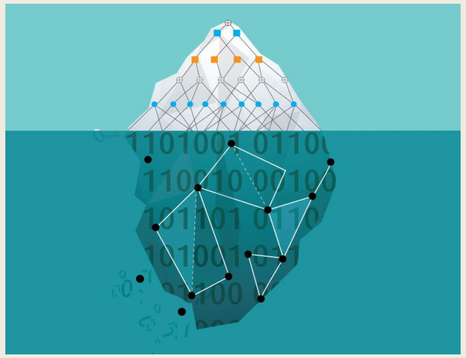
+ Symmetry



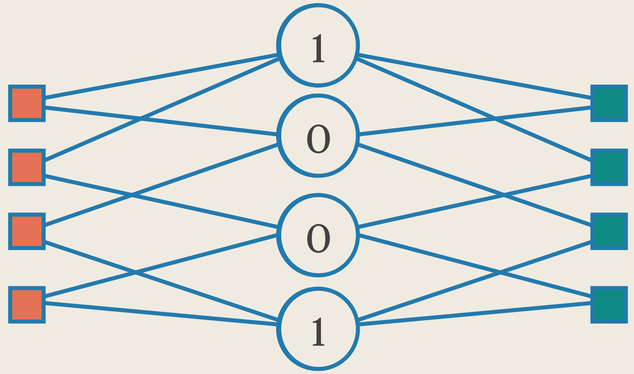
?



Classical 1-D Code

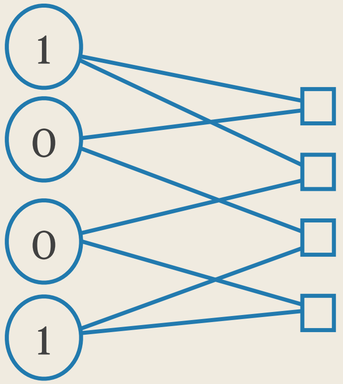


High-Dimensional Expander

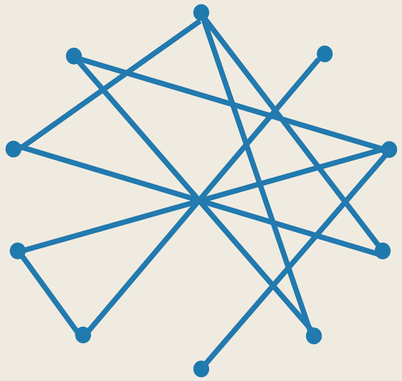


Quantum"2-D" Codes

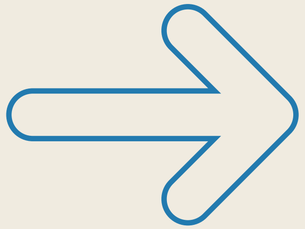
High-dimensional structure is very useful

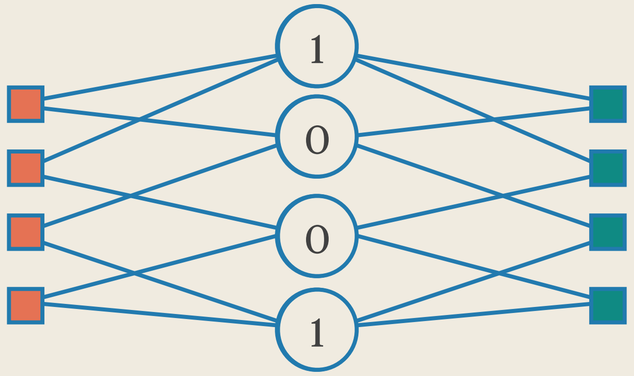


Classical 1-D Code

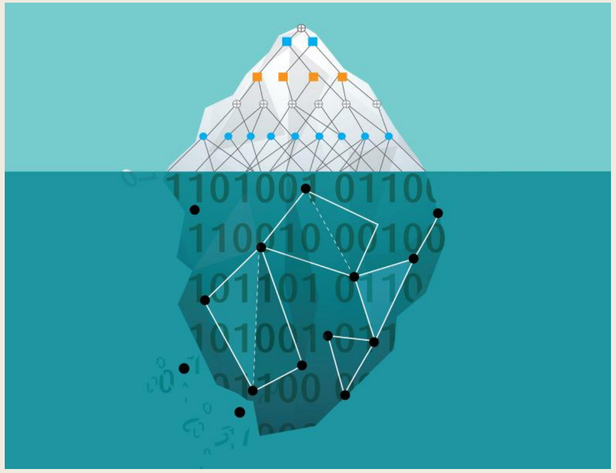


1-D Expander



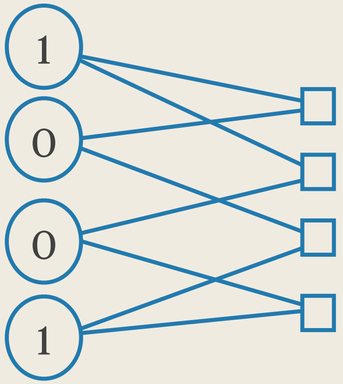


Quantum"2-D" Codes

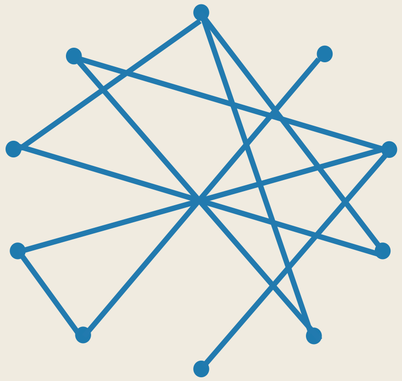


High-Dimensional Expander

High-dimensional structure is very useful

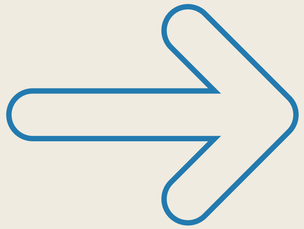


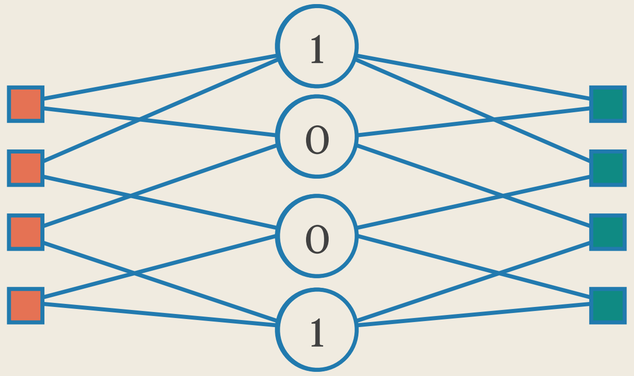
Classical 1-D Code



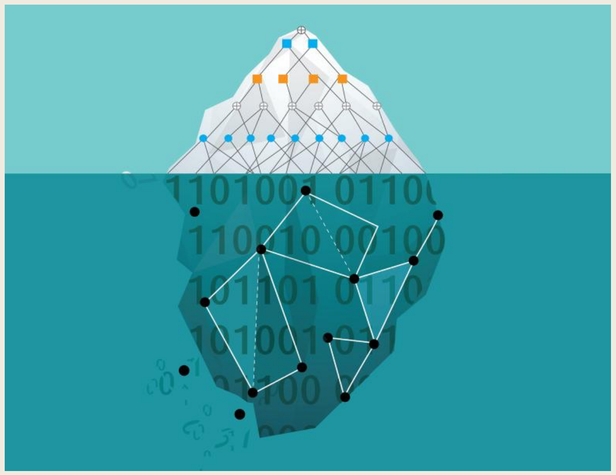
1-D Expander

QUANTUM COMPUTATION



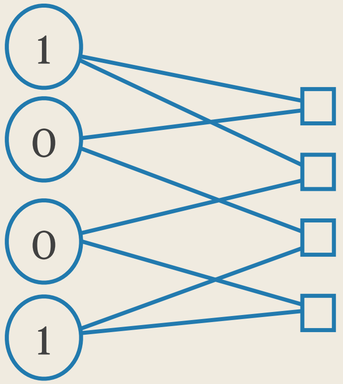


Quantum"2-D" Codes



High-Dimensional Expander

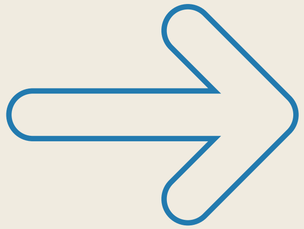
High-dimensional structure is very useful

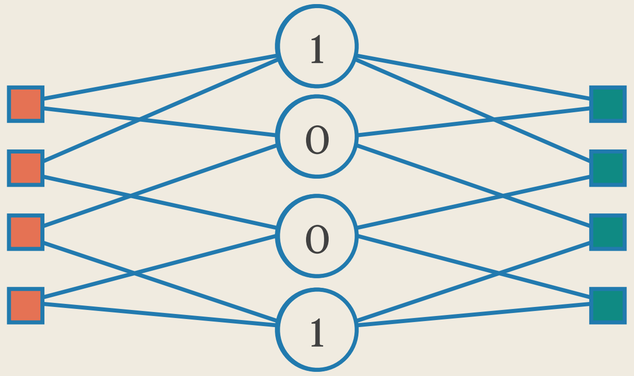


Classical 1-D Code

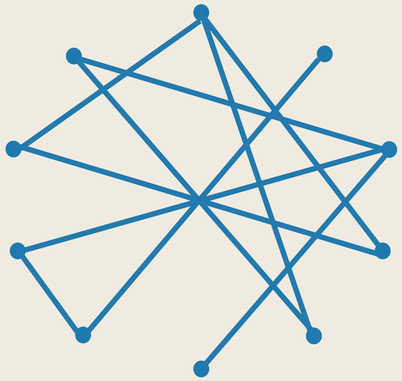
QUANTUM COMPUTATION

“SPECIAL” CLASSICAL CODES

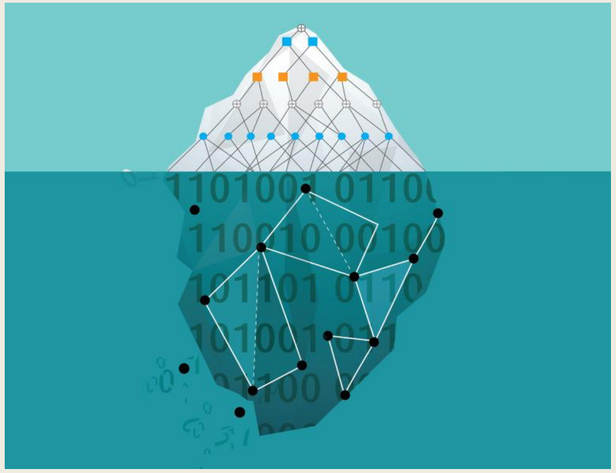




Quantum“2-D” Codes

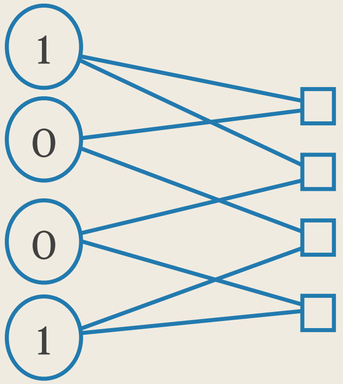


1-D Expander

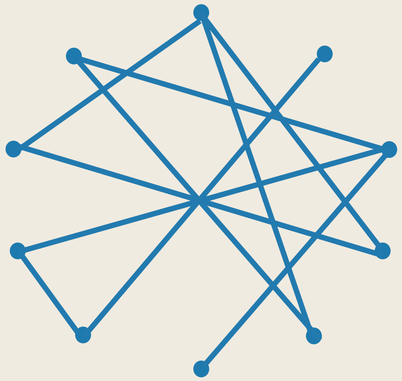


High-Dimensional Expander

High-dimensional structure is very useful



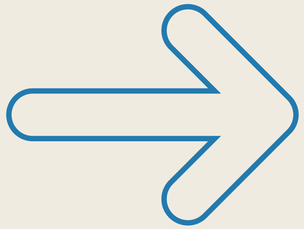
Classical 1-D Code



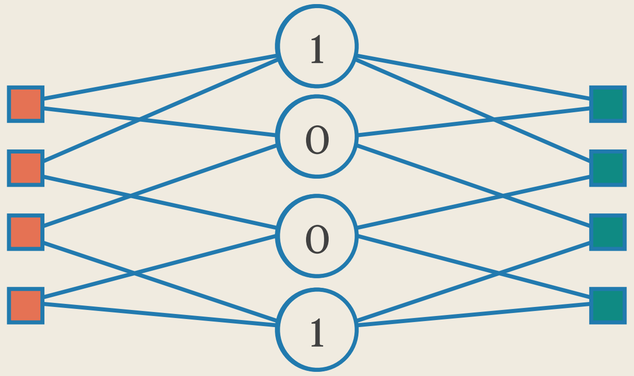
1-D Expander

QUANTUM COMPUTATION

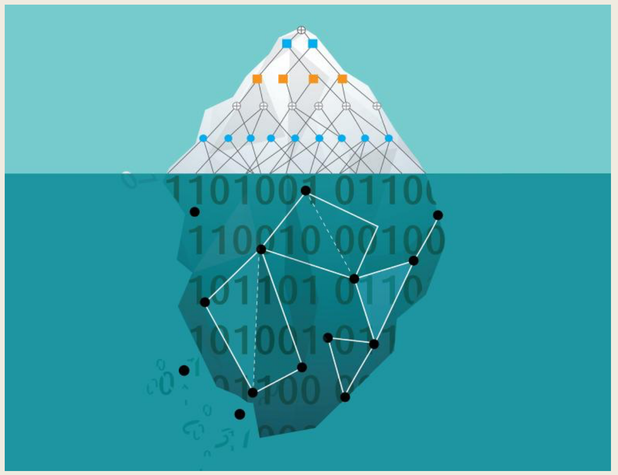
“SPECIAL” CLASSICAL CODES



FAST SAMPLING

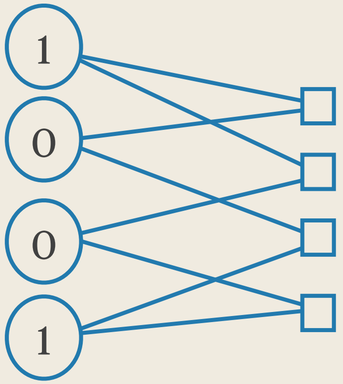


Quantum“2-D” Codes

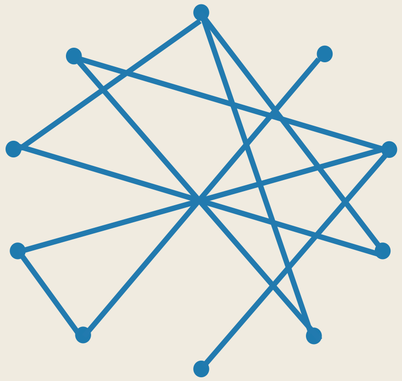


High-Dimensional Expander

High-dimensional structure is very useful



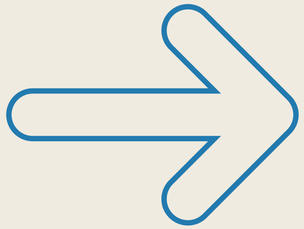
Classical 1-D Code



1-D Expander

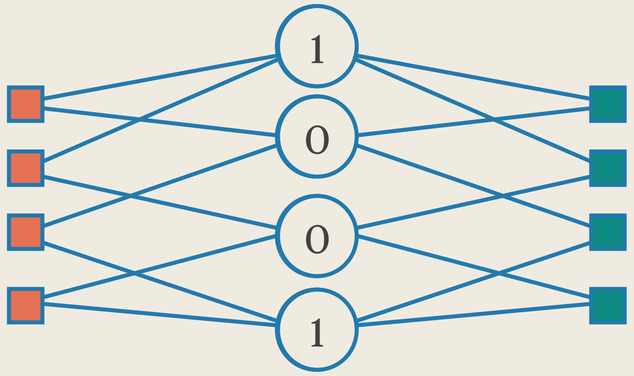
QUANTUM COMPUTATION

“SPECIAL” CLASSICAL CODES

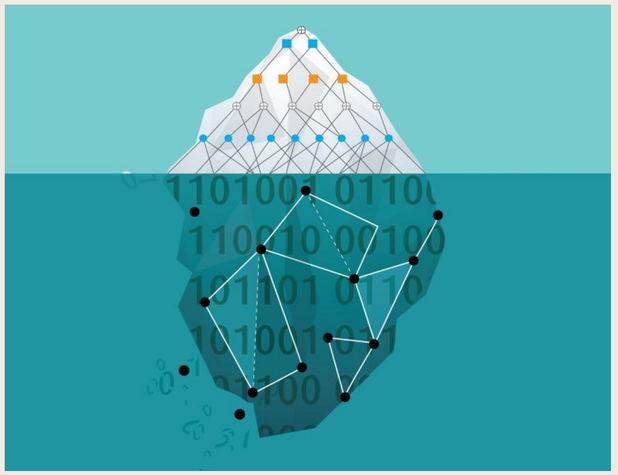


FAST SAMPLING

COMPLEXITY THEORY

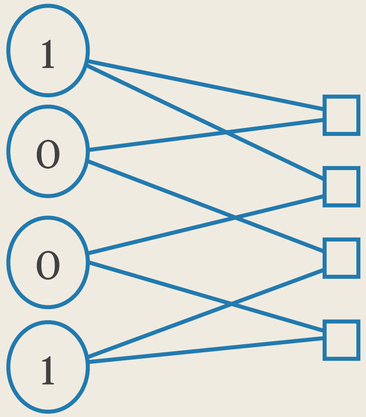


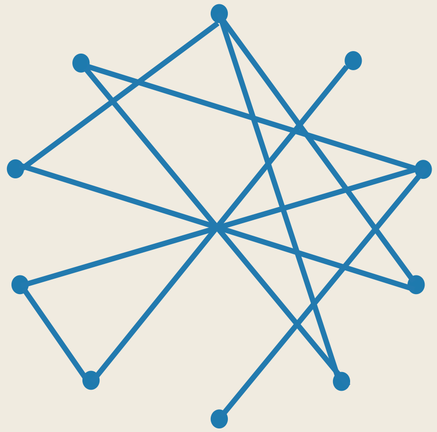
Quantum“2-D” Codes



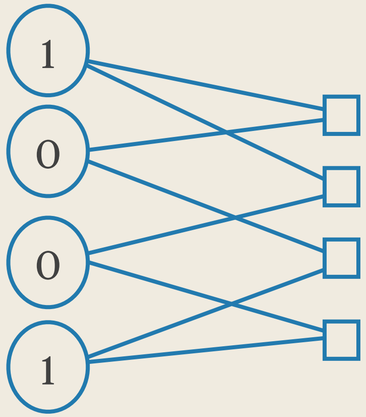
High-Dimensional Expander

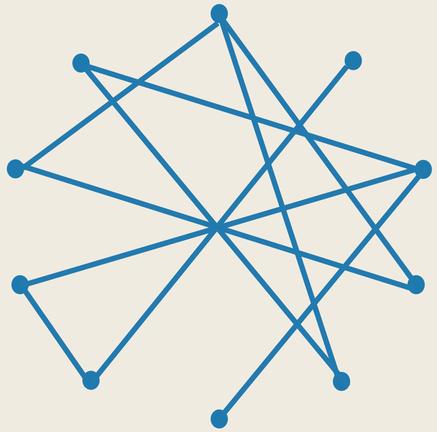
Many tools available in 1-D





Many tools available in 1-D

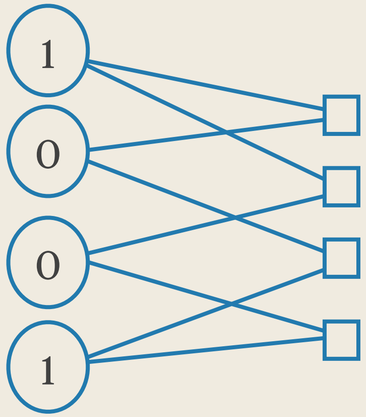


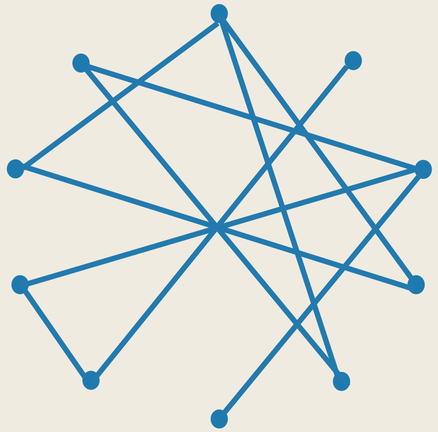


RANDOM MODELS

- Thommessen
- Gallager's Ensemble
- Random Lifts

Many tools available in 1-D





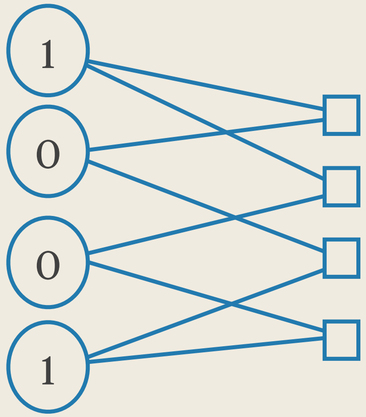
RANDOM MODELS

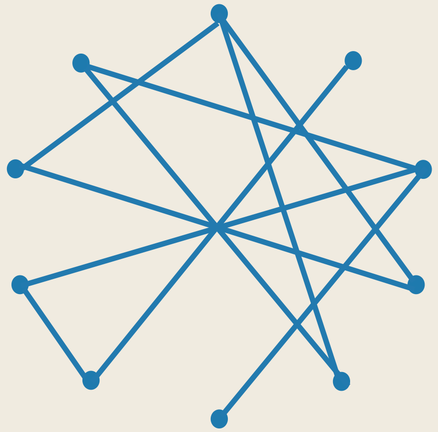
- Thommessen
- Gallager's Ensemble
- Random Lifts

AMPLIFICATION TECHNIQUES

- AEL
- Direct-Sum

Many tools available in 1-D





RANDOM MODELS

- Thommessen
- Gallager's Ensemble
- Random Lifts

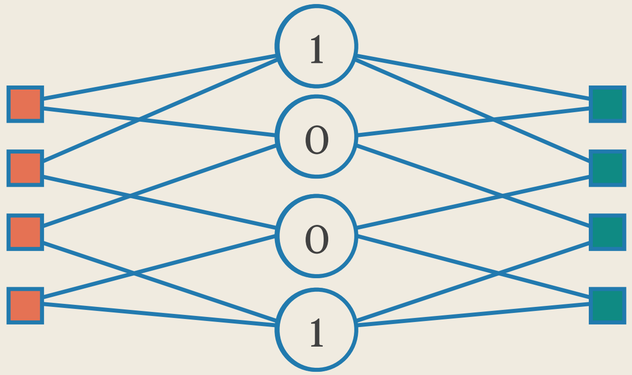
AMPLIFICATION TECHNIQUES

- AEL
- Direct-Sum

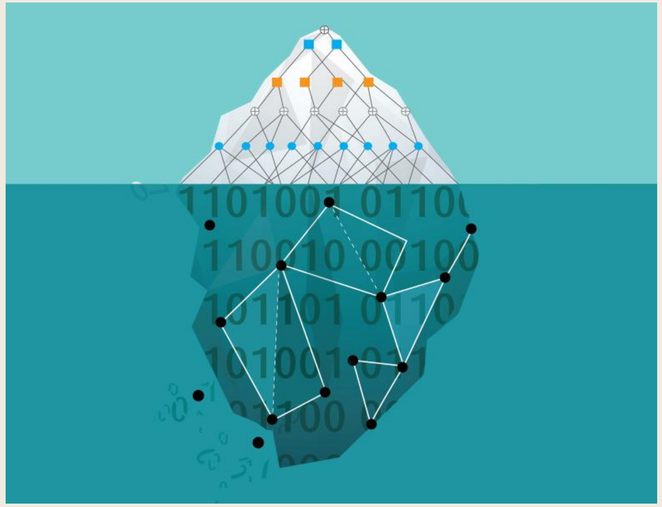
COMBINATORIAL CONSTRUCTIONS

- Derandomized Squaring
- Zig-Zag Product

Not much known in high-dimensions



Quantum"2-D" Codes



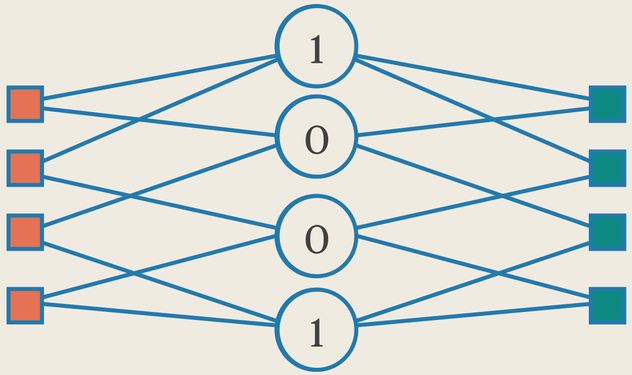
High-Dimensional Expander

RANDOM MODELS ?

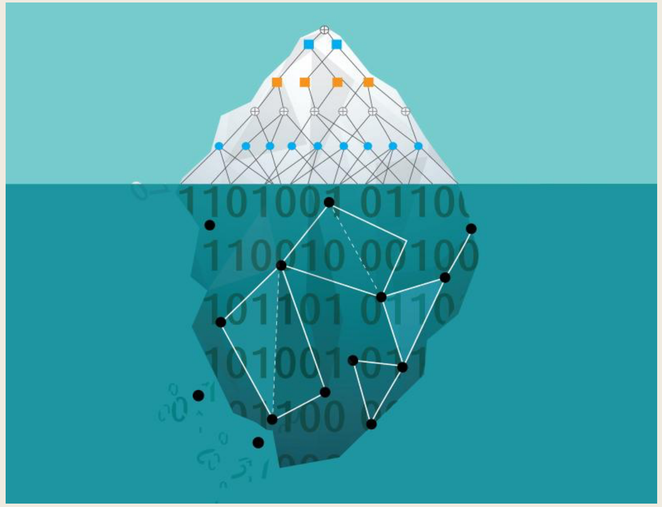
AMPLIFICATION TECHNIQUES ?

COMBINATORIAL CONSTRUCTIONS ?

Not much known in high-dimensions



Quantum"2-D" Codes



High-Dimensional Expander

RANDOM MODELS ?

AMPLIFICATION TECHNIQUES

AEL

[JMST, STOC'25, BJMST, Preprint]

COMBINATORIAL CONSTRUCTIONS ?

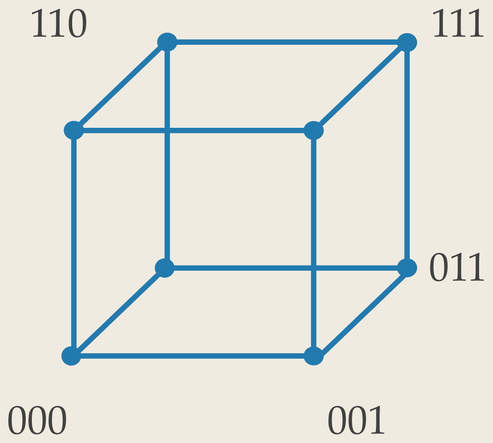
High-
Dimensional
Structure
Functions
with
Symmetry

Boolean symmetry has been very useful

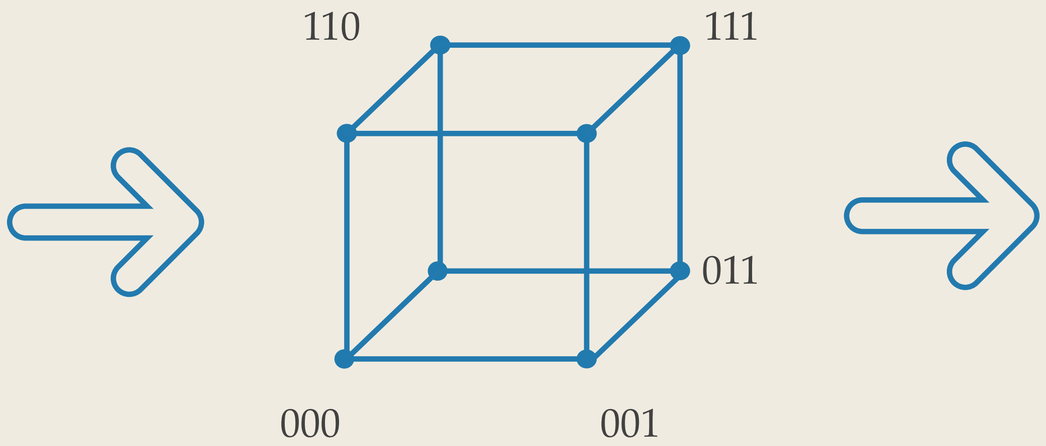
Boolean symmetry has been very useful

Boolean symmetry has been very useful

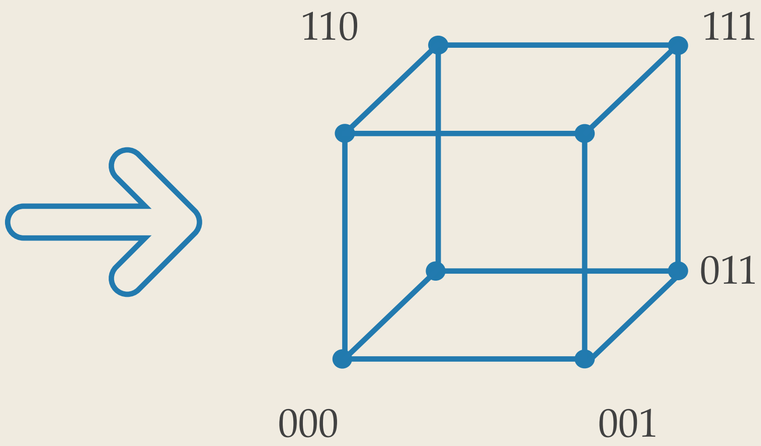
Boolean symmetry has been very useful

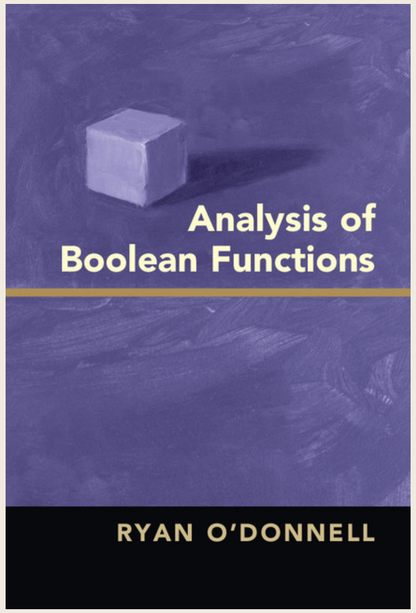


Boolean symmetry has been very useful

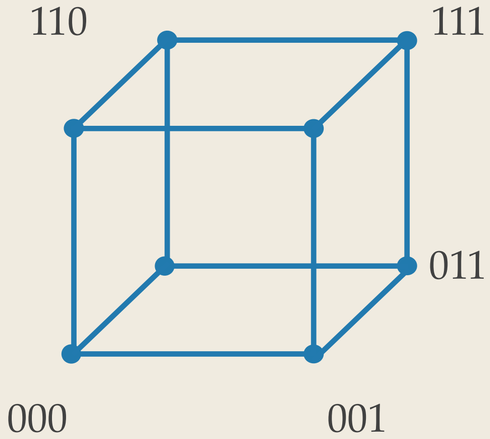


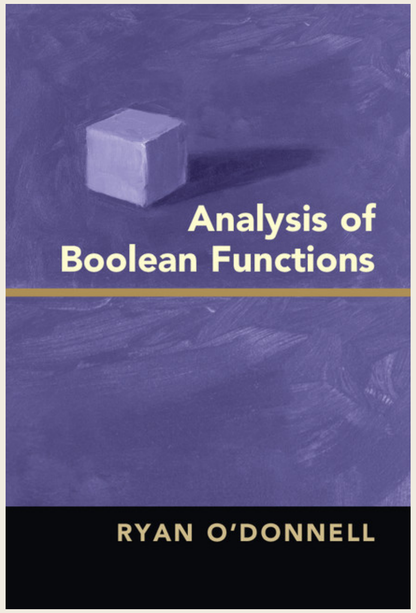
Boolean symmetry has been very useful





Boolean symmetry has been very useful



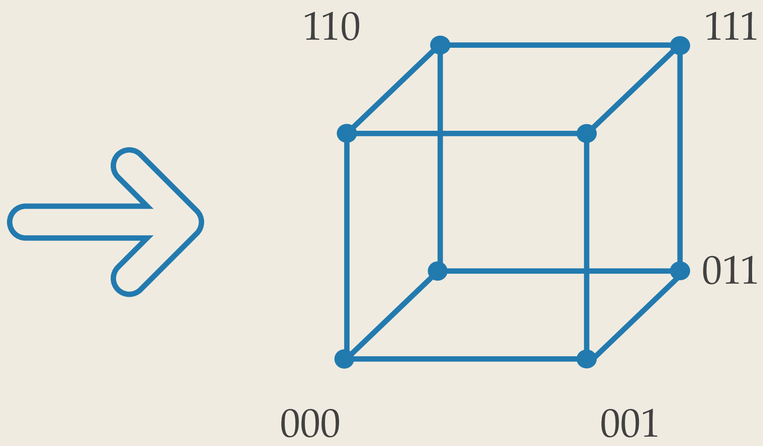


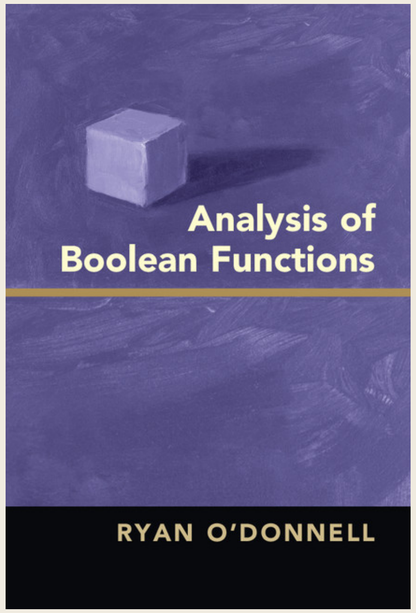
SOCIAL CHOICE THEORY

LEARNING THEORY

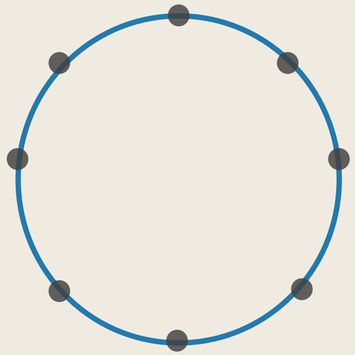
APPROXIMATION ALGORITHMS

Boolean symmetry has been very useful





?

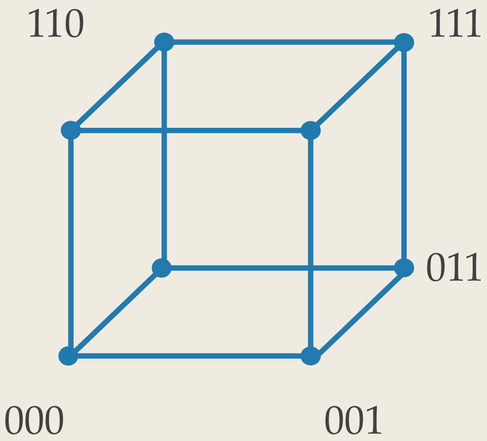


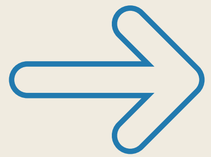
SOCIAL CHOICE THEORY

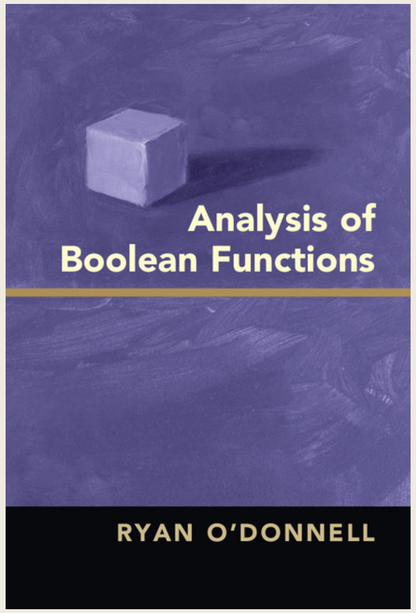
LEARNING THEORY

APPROXIMATION ALGORITHMS

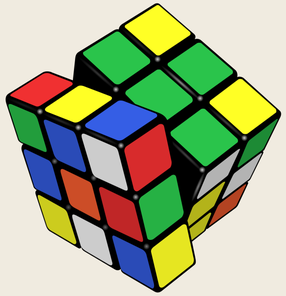
Boolean symmetry has been very useful



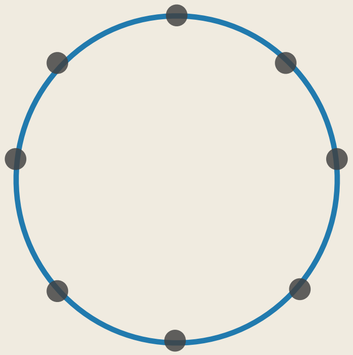




?



?

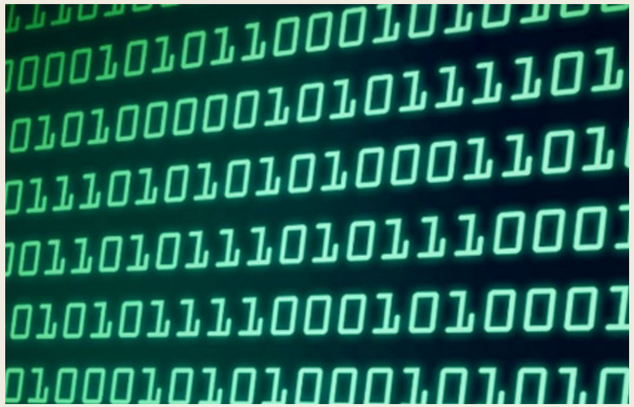


SOCIAL CHOICE THEORY

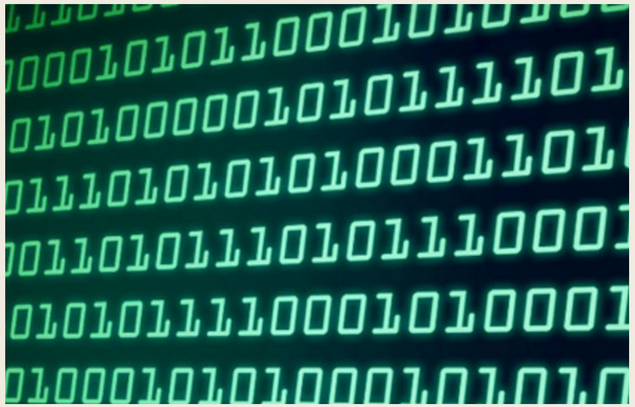
LEARNING THEORY

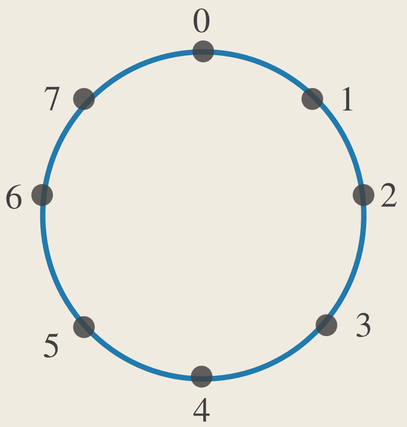
APPROXIMATION ALGORITHMS

Why respect general symmetries?



Why respect general symmetries?

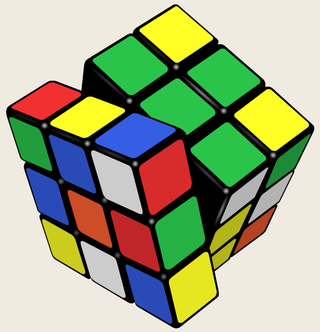


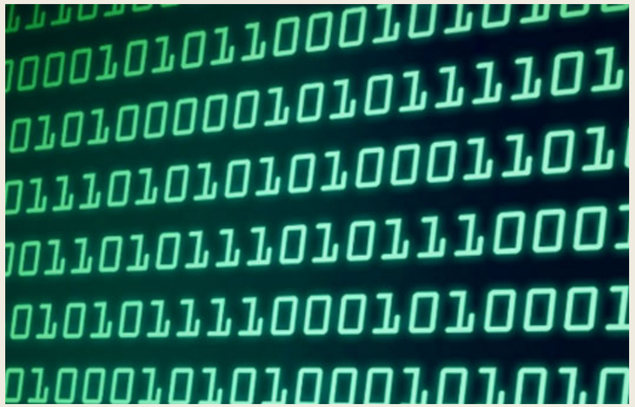


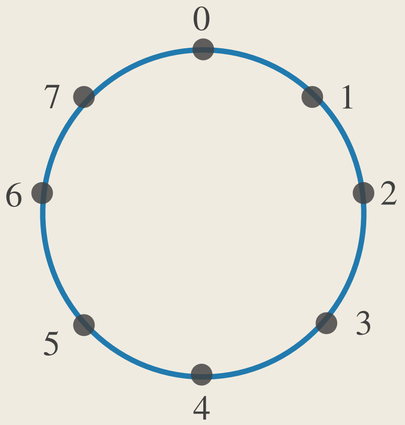
EXPONENTIALLY BETTER BOUNDS

[JMR, RANDOM'25]

Why respect general symmetries?







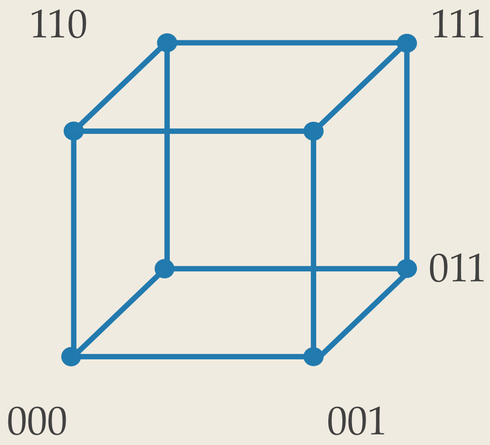
EXTREMAL COMBINATORICS

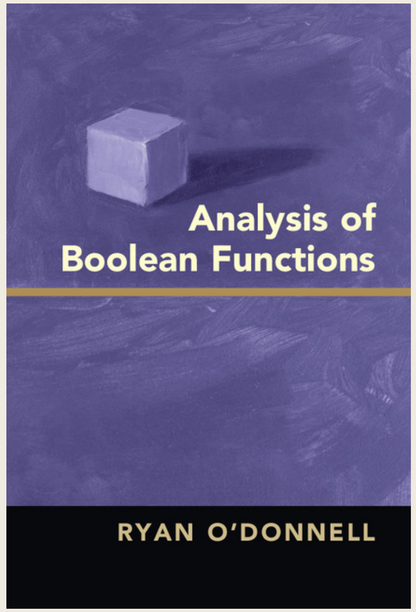
[Filmus-Kindler-Lifshitz-Minzer'22, Keevash-Lifshitz'24]

EXPONENTIALLY BETTER BOUNDS

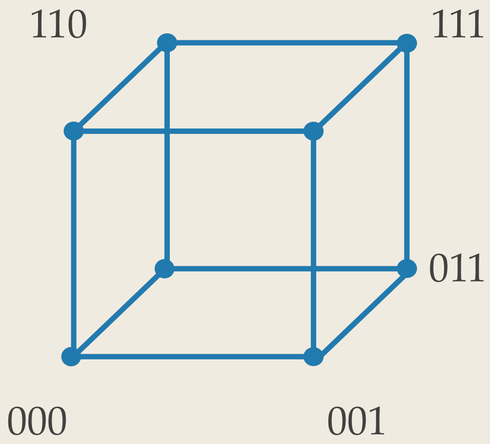
[J**M**R, RANDOM'25]

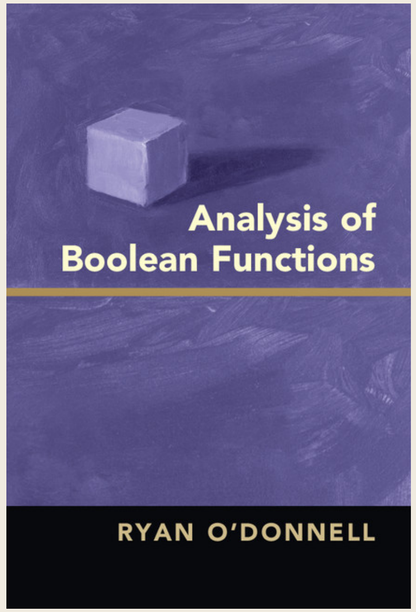
Direction 2: Functions with general symmetries

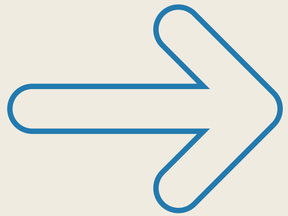




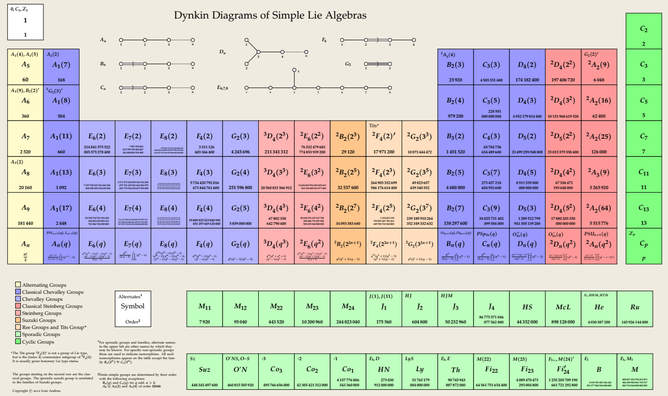
Direction 2: Functions with general symmetries







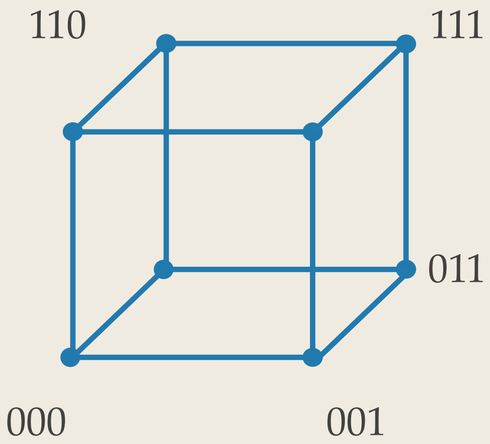
The Periodic Table Of Finite Simple Groups



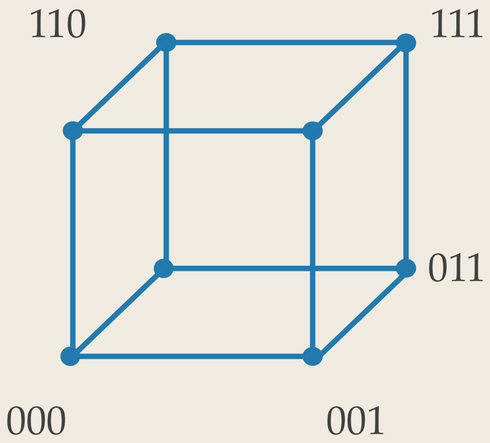
Analysis on finite groups

?

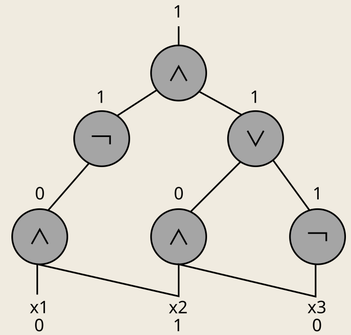
Boolean case is well-studied



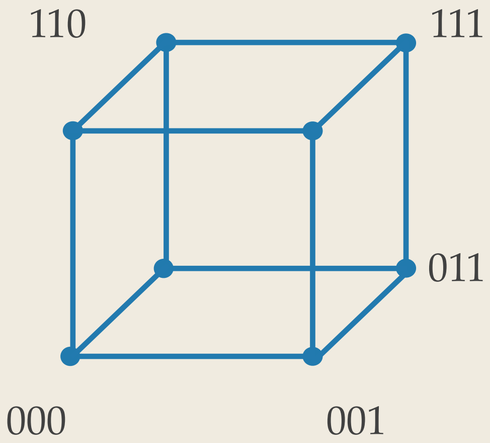
Boolean case is well-studied



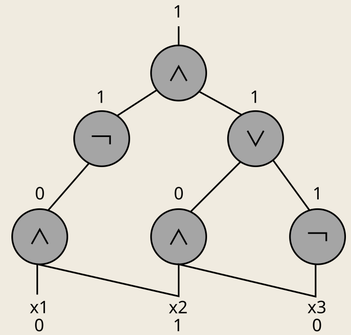
COMPUTATION MODELS



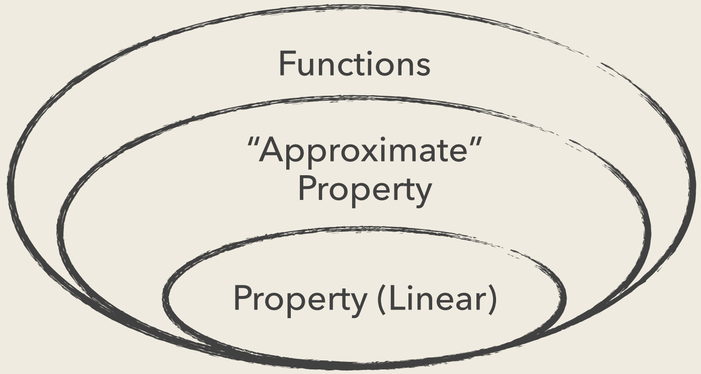
Boolean case is well-studied



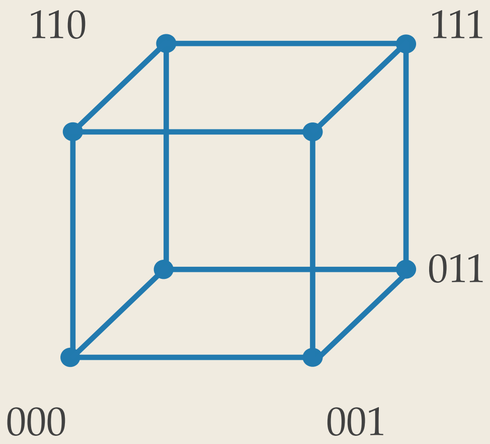
COMPUTATION MODELS



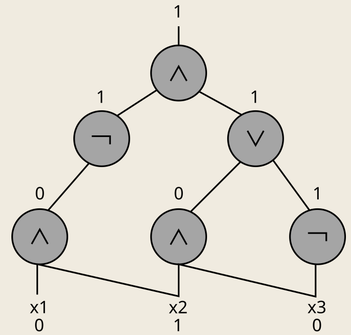
APPROXIMATE PROPERTIES



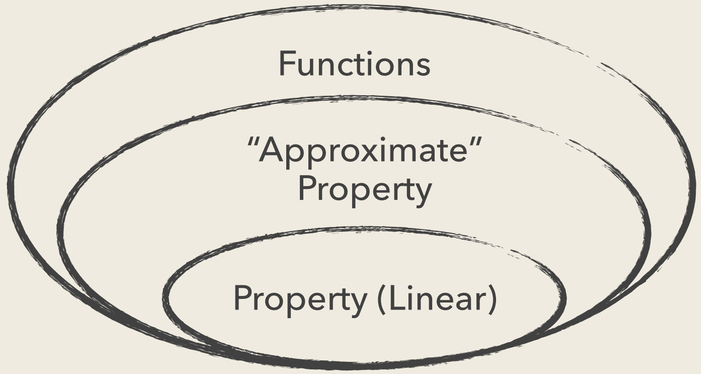
Boolean case is well-studied



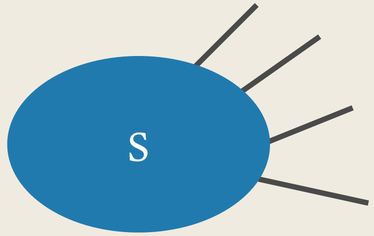
COMPUTATION MODELS



APPROXIMATE PROPERTIES



SMALL-SET EXPANSION



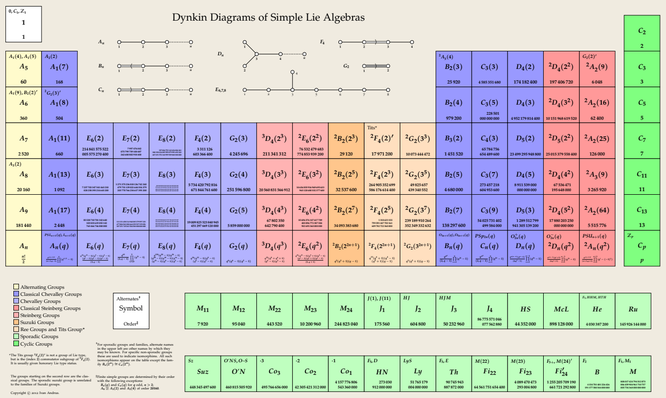
General symmetries are poorly understood

COMPUTATION MODELS ?

APPROXIMATE PROPERTIES ?

SMALL-SET EXPANSION ?

The Periodic Table Of Finite Simple Groups



Definitions ▶ Quantum Error Correction ▶ Other Pseudorandom Objects ▶ Future

General symmetries are poorly understood

COMPUTATION MODELS ?

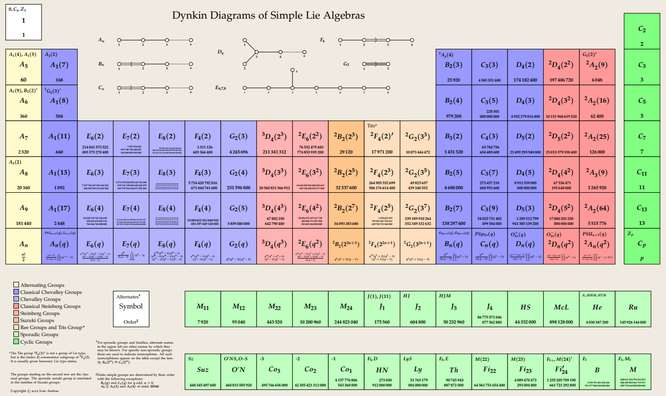
APPROXIMATE PROPERTIES

SMALL-SET EXPANSION

Permutation Groups

[Filmus-Kindler-Lifshitz-Minzer'22, Keevash-Lifshitz'24]

The Periodic Table Of Finite Simple Groups



General symmetries are poorly understood

COMPUTATION MODELS ?

APPROXIMATE PROPERTIES

Approximate Symmetry

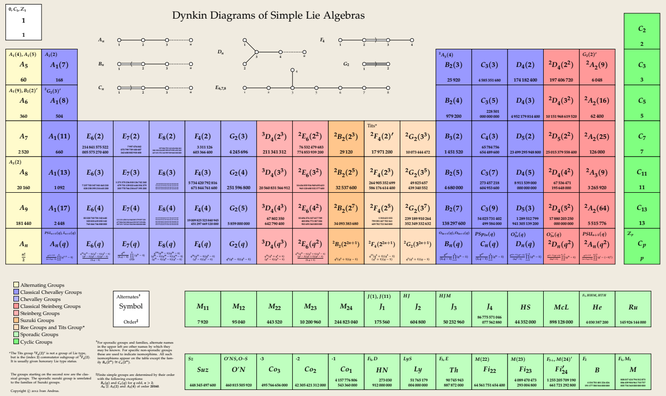
[**M**R, ITCS'26, **M**R, Preprint]

SMALL-SET EXPANSION

Permutation Groups

[Filmus-Kindler-Lifshitz-Minzer'22, Keevash-Lifshitz'24]

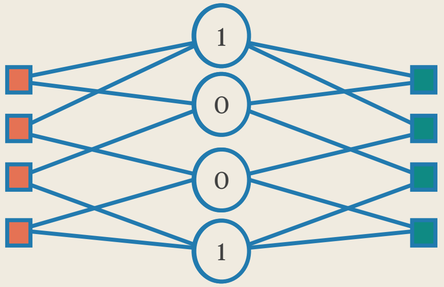
The Periodic Table Of Finite Simple Groups



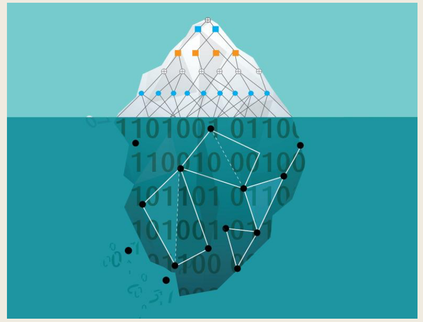
A Computational Understanding of Symmetry & Randomness

A Computational Understanding of Symmetry & Randomness

High-dimensional structure



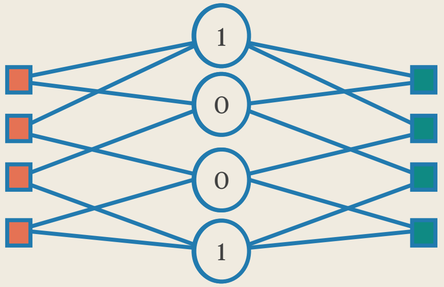
Quantum"2-D" Codes



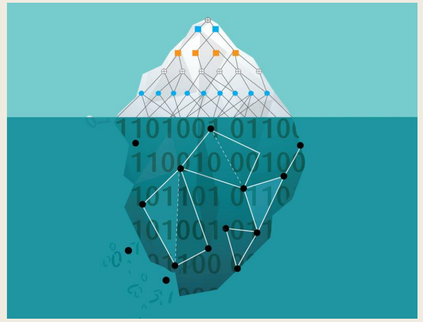
High-Dimensional Expander

A Computational Understanding of Symmetry & Randomness

High-dimensional structure

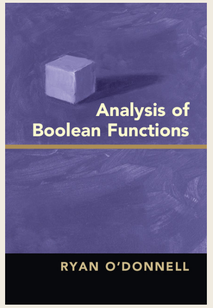


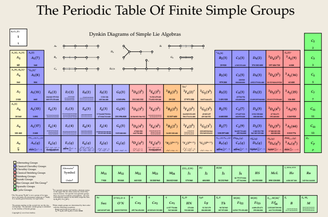
Quantum"2-D" Codes



High-Dimensional Expander

Functions with Symmetry



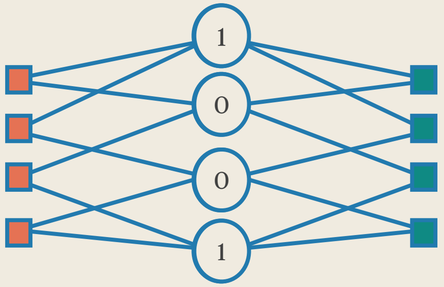


Definitions ▶ Quantum Error Correction ▶ Other Pseudorandom Objects ▶ Future

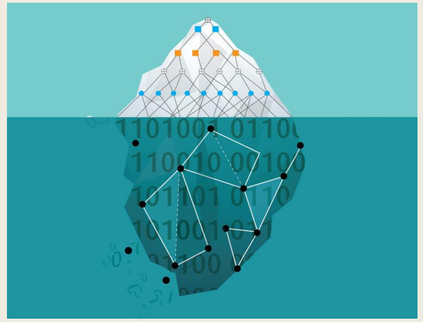
A Computational Understanding of

Symmetry & Randomness

High-dimensional structure



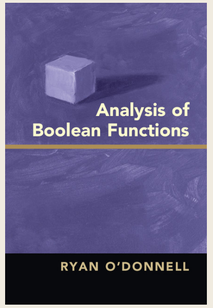
Quantum"2-D" Codes

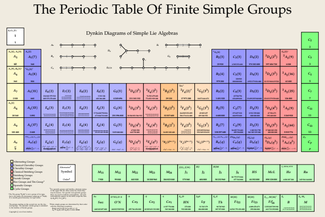


High-Dimensional Expander

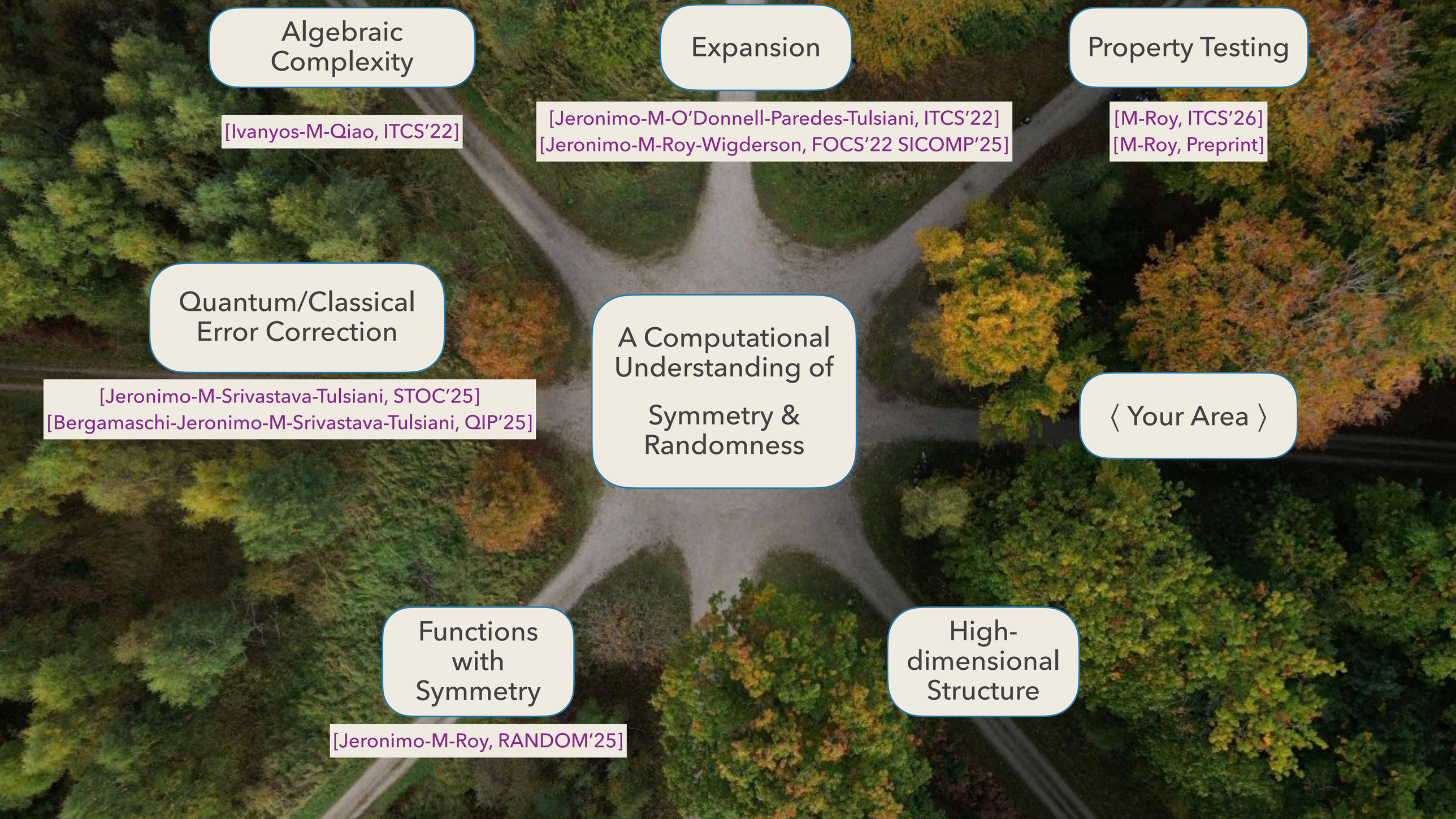
[Jeronimo–**M**–Srivastava–Tulsiani STOC'25
Bergamaschi–Jeronimo–**M**–Srivastava–Tulsiani, Preprint]

Functions with Symmetry





[Jeronimo–**M**–Roy RANDOM'25,
M-Roy, ITCS'26
M–Roy, Preprint]


Algebraic Complexity
[Ivanyos-M-Qiao, ITCS'22]
Expansion
[Jeronimo-M-O'Donnell-Paredes-Tulsiani, ITCS'22]
[Jeronimo-M-Roy-Wigderson, FOCS'22 SICOMP'25]
Property Testing
[M-Roy, ITCS'26]
[M-Roy, Preprint]
Quantum/Classical Error Correction
[Jeronimo-M-Srivastava-Tulsiani, STOC'25]
[Bergamaschi-Jeronimo-M-Srivastava-Tulsiani, QIP'25]
A Computational Understanding of
Symmetry & Randomness
⟨ Your Area ⟩
Functions with Symmetry
[Jeronimo-M-Roy, RANDOM'25]
High-dimensional Structure

Thank You!
Questions?